

Exploring consequences of rifampicin resistance mutations in *Escherichia coli*

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Abstract

Antibiotic resistance can arise through a single point mutation in the bacterial genome. These mutations are often in the gene that encodes the target of the drug. For example in the case of rifampicin, which binds to the beta subunit of RNA polymerase, resistance mutations are found within the *rpoB* gene. RNA polymerase is an essential holoenzyme responsible for the transcription of DNA to RNA, and as such mutations that alter the highly conserved sequence of its subunits can have complex pleiotropic effects on its function. Exactly how *rpoB* mutations alter transcription remains poorly understood, and while it is well established that different *rpoB* mutations can result in different fitness costs, it is likely that the transcriptome of *rpoB* mutants results in many other complex effects on bacterial phenotypes. Further, the influence of specific environmental differences on the effects of *rpoB* mutations are yet to be fully explored.

This thesis investigates how several *rpoB* mutations affect the phenotype and fitness of a model organism, *Escherichia coli*. Chapter 1 provides an introduction to antibiotic resistance more generally, discusses evolutionary considerations in the management of resistance, and outlines what is known about rifampicin resistance and *rpoB* mutations. In chapter 2, a library of *rpoB* mutants is generated and utilised to explore how their growth is altered in 3 common laboratory culture media. We find that an interaction between the specific resistance mutation and the nutrient environment shape the fitness effects of these mutants, including their sensitivity to secondary antibiotics. In chapter 3, the effect of another environmental variable—temperature—on the phenotypes and fitness of our *rpoB* mutants is explored. This chapter considers the different levels of biological organisation, from the molecular to the organismal, and identifies how temperature influences the effects of *rpoB* mutations at these different levels. Several links between effects at different levels are found, offering some mechanistic insight. Finally, chapter 4 investigates how *rpoB* mutations might influence the evolution of resistance to several secondary antibiotics with a whole-genome, deep sequencing approach. Key differences in the spectrum of secondary mutations are identified, depending on the secondary antibiotic and the specific *rpoB* mutation. Together, this thesis provides important environmental context that is largely lacking from studies characterising the complex effects of resistance mutations in essential gene, *rpoB*.

Declaration

No portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Preface: Rationale for journal format

This thesis is presented in the alternative format, otherwise known as journal format, in accordance with the regulations of the University of Manchester. The results chapters listed below (2, 3, and 4) are therefore presented as manuscripts suitable for publishing in peer reviewed journals, adhering to respective journal guidelines. Elements of the text have been reformatted to ensure a cohesive body of work; for example, figures and tables are numbered consistently throughout the thesis, with all references listed together at the end of the thesis.

Chapter 2:

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Journal: Proceedings of the Royal Society B

Author contributions: JKS conducted all experiments, data analysis, and writing with the following exceptions – MJ conducted the literature review in Chapter 2.5; CJW conducted the Cluster of Orthologous Gene analysis in Figure 2.8C; ZK, BPH and ML assisted with final review and editing of manuscript. All authors have read and approved the final version.

Chapter 3:

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Title: Exploring the effect of temperature on pleiotropic mutations in RNAP

Authors: Soley, J. K., Ozbilek, D., Walsh, C. J., Lovell, S., Howden, B. P. and Lagator, M.

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Authors: Jago, J. M.*, Soley, J. K.*, Denisov, S., Walsh, C. J., Gifford, D. R., Howden, B. P., Lagator, M. (** These authors contributed equally*)

Journal: Nature Communications

Author contributions: JKS and ML conceptualised the study. JKS and MJJ performed mutation selection experiments (Chapter 4.3.2). MJJ conducted annotation of genes and functions (Chapter 4.3.5 – 4.3.7), categorisation of mutational targets (Chapter 4.3.9), analysis of mutation dispersion (Chapter 4.3.10), and Principal Component Analysis (Chapter 4.3.11); JKS created experimental strains (Chapter 4.3.1), developed the QMS-Seq analysis pipeline and performed bioinformatic analyses (Chapter 4.3.3), conducted analysis of individual clones (Chapter 4.3.4) and global *E. coli* dataset (Chapter 4.3.8), and conducted analysis of mutation impact (Chapter 4.3.10); JKS and MJJ wrote the initial draft of the manuscript; DRG performed *breseq* analysis (Chapter 4.3.3); SD, CJW, DRG, BPH and ML assisted with final review and editing of manuscript. All authors have read and approved the final version.

Note: A revised version of this manuscript has since been published on the 17th January 2025. Further experimentation, analysis and writing was conducted by co-author *Jago, J. M.* and as such in the published manuscript the authorship was revised to reflect this. Manuscript available at doi: 10.1038/s41467-025-56050-2

1 Introduction

1.1 Antibiotic discovery and resistance

The discovery and widespread production of antibiotics in the 20th century revolutionised modern medicine, and rendered many life-threatening diseases easily treatable. The discovery of penicillin in 1928, and the so-called ‘golden age’ of antibiotic discovery 20 years later, marked a period of rapid discovery and development of novel antibiotic compounds. However, the rate of new drug discovery has since declined (Figure 1.1) (Aminov, 2010). Huge costs of production, high rates of failure, and the risk that potential drug candidates will be preserved as drugs of last resort all deter pharmaceutical companies from investing in antibiotic research and development (Renwick, Brogan and Mossialos, 2016). This slowing of the development pipeline coupled with rising levels of resistance globally leaves modern medicine to find itself in a very precarious position.

It was not long after antibiotics were first used clinically that Alexander Fleming himself warned that a failure to ensure careful use might lead to bacteria developing resistance (Fleming, 1945). Indeed, resistance to penicillin was first observed only 10 years after its discovery, and resistance to other antibiotics has been detected in even less time (Hutchings, Truman and Wilkinson, 2019). Now, some infection-causing strains are resistant to all but the very last resort antibiotics (Whiley *et al.*, 2012; World Health Organization, 2023). Particular concern surrounds the ESKAPE group of pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp.), and specific examples of resistance such as vancomycin-resistance in *Enterococcus*, methicillin- and vancomycin-resistance *S. aureus*, and carbapenem-resistant *Enterobacteriaceae* (Church and McKillip, 2021). There are a myriad of factors that have likely contributed to the increased prevalence of resistance, such as increased use of antibiotics globally with extensive use in agriculture and aquaculture (Meek, Vyas and Piddock, 2015), poor prescribing practices (King, Fleming-Dutra and Hicks, 2018), lack of novel drugs and vaccines (Hutchings, Truman and Wilkinson, 2019), and rapid environmental dissemination of resistance (Bengtsson-Palme, Kristiansson and Larsson, 2018). Improper use of these powerful antimicrobial agents can elevate the selective pressures for resistance evolution, while also creating conditions that propagate the spread of resistance from patient to patient. It has been estimated that without preventative

measures being taken, the number of annual deaths attributed to antimicrobial resistance could rise to 10 million by 2050 (O' Neill, 2014).

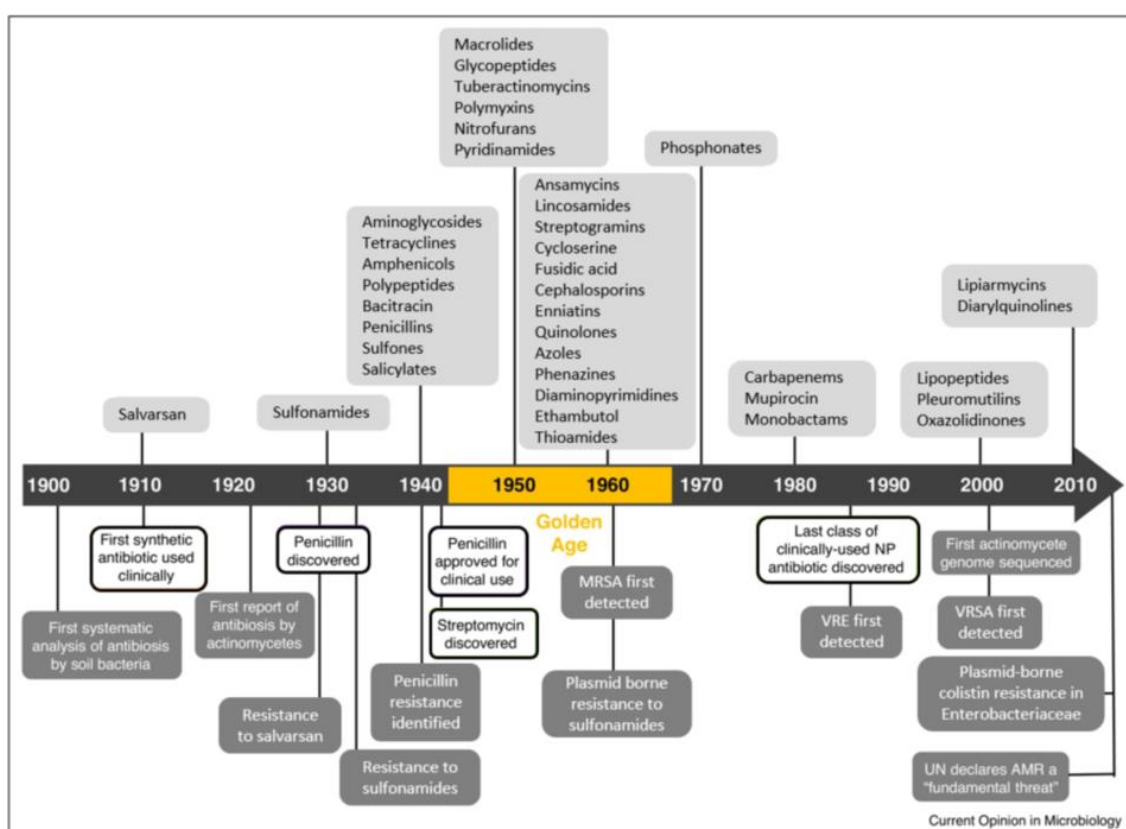


Figure 1.1 – Antibiotic discovery and resistance from 1900–2010. Above timeline: discoveries of new drug classes (light grey); below timeline: first observation of resistance (dark grey) and key dates in the history of antibiotics (white). MRSA – Methicillin-resistant *Staphylococcus aureus*; VRE – Vancomycin-resistant *Enterococci*; VRSA – Vancomycin-resistant *S. aureus*; NP – Natural product; AMR – Antimicrobial resistance. Figure adapted from (Hutchings, Truman and Wilkinson, 2019), CC BY 4.0.

1.2 Mechanisms of resistance

There are various mechanisms through which bacteria are able to develop resistance against antibiotics. There are both intrinsic and acquired mechanisms of resistance. Intrinsic resistance can be considered more of an insensitivity caused by the presence or absence of a particular functional or structural characteristic, for example the presence of the outer membrane in Gram-negative bacteria that acts as a barrier and prevents many antibiotics from entering the cell (Nikaido, 1994). Acquired resistance emerges in response to evolutionary challenge by antibiotic exposure, through spontaneous DNA mutations or the transfer of genetic information between bacteria (van Hoek *et al.*, 2011). Mechanisms can broadly be categorised into 3 groups based on how drug activity is reduced, discussed below.

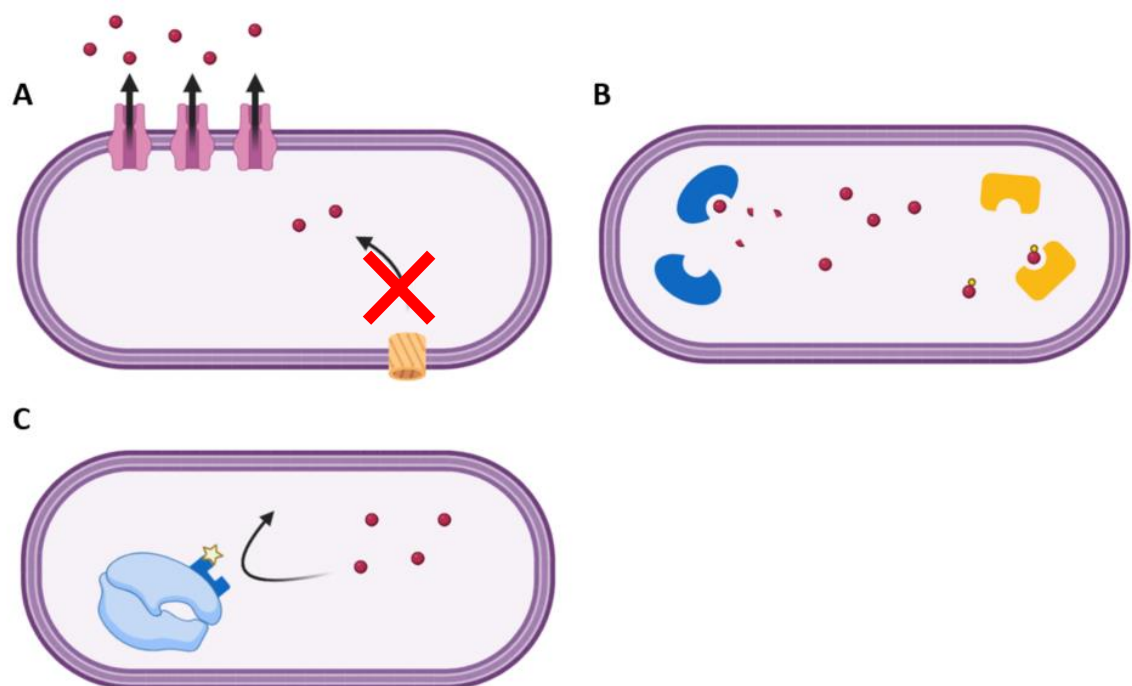


Figure 1.2 – Mechanisms of antibiotic resistance. Bacterial cells (purple) can become resistant to an antibiotic (red circles) by (A) reducing intracellular drug concentration, (B) metabolic inactivation or modification by enzymes (blue and yellow shapes), or (C) modification of the drug target site (blue) by mutation. Created with BioRender.com.

1.2.1 Reduction of intracellular drug concentration

One mechanism of resistance is a reduction of the intracellular concentration of the drug, either by decreasing drug uptake or by increasing drug export (Figure 1.2A). Membrane permeability can be actively influenced by regulation of porins – small, non-specific channels present on bacterial membranes through which antibiotics can be transported (Baroud *et al.*, 2013; Kojima and Nikaido, 2013). The influx of drug molecules can be reduced through downregulation or deletion of functional porin genes or through modifications to porin proteins (Achouak, Heulin and Pagès, 2001). Resistance can also be conferred by increasing the rate of drug efflux from the cell. The export of drug molecules is carried out by a multitude of specific and non-specific efflux pumps, the upregulation of which can contribute to resistance. These can be specific to a certain compound, such as in the case of tetracycline efflux pumps, or can confer multidrug resistance by non-specific efflux (Kosmidis *et al.*, 2012; You, Hilpert and Ward, 2013). An example of an important multidrug efflux complex in *Escherichia coli* is the TolC-AcrAB complex, implicated in the resistance of several classes of antibiotic (Hackbarth and Chambers, 1989; Poole, 2000; Nishino *et al.*, 2003).

1.2.2 Drug modification or inactivation

Another group of resistance mechanisms involves direct metabolic modification and inactivation of antibiotics (Figure 1.2B). Hydrolysing enzymes can interact with and cause the degradation of drug molecules, thus inhibiting their action. One well-known example of this mechanism is cleavage of β -lactam-containing antibiotics by β -lactamases, a common class of hydrolysing enzymes (Bush, Jacoby and Medeiros, 1995). AmpC is one such enzyme prevalent among Gram-negative strains, the expression of which can be induced in response to environmental β -lactams (Wiedemann *et al.*, 1998). Enzymes such as aminoglycoside modifying enzymes can chemically alter the structure of some antibiotics through the addition of functional groups, such as phosphate or acetyl groups (Ramirez and Tolmasky, 2010). These additions are able to sterically interfere with binding of the antibiotic to its target site (Llano-Sotelo *et al.*, 2002).

1.2.3 Target site modification

The drug target site can also be modified, causing varying levels of resistance by reducing affinity for drug binding or preventing binding completely (Figure 1.2C). Such modifications can result in either drug-specific resistance or multidrug resistance when there is overlap of drug binding sites. Drug target sites can be modified through the action of enzymes, such as RNA methyltransferases that modify ribosomes and inhibit the binding of ribosome-targeting drugs (Long *et al.*, 2006; Shen, Wang and Schwarz, 2013). Target sites can also be modified at a genetic level through mutations that either sterically block drug binding through structural alterations, or alter the binding site directly (Artsimovitch *et al.*, 2003; Babakhani, Seddon and Sears, 2014). A major example of target site mutations conferring resistance occurs against drugs that target bacterial RNA Polymerase (RNAP), and will be the mechanism of resistance focused upon in this thesis.

1.3 Fitness cost of resistance

Antibiotic resistance mechanisms often interfere with key biological processes and as such are generally associated with a reduction in bacterial fitness, known as the ‘fitness cost’ of resistance (Melnik, Wong and Kassen, 2015). For example, mutations that alter the structure of a drug target site can affect normal protein activity of essential enzymes like RNA polymerase (Gifford, Moss and MacLean, 2016). In the presence of an antibiotic, the fitness cost of resistance is offset by the fitness advantage that a resistance mutation confers as compared to the wild type. However, in an antibiotic-free environment the benefit of such a mutation is lost as bacteria are no longer subject to the selective pressure inflicted by the drug. In drug-free environments, the fitness cost has critical evolutionary consequences on the dissemination of resistance within the population – mutants with high cost resistance will be outcompeted by those with lower cost resistance, even if they confer a higher level of resistance in the presence of a drug. There are many factors that can influence the fitness cost, including the mechanism of resistance (Melnik, Wong and Kassen, 2015), the exact mutation conferring resistance (Hall, 2013), epistatic interactions (Angst and Hall, 2013), and growth environment (Gifford, Moss and MacLean, 2016). Improving our understanding of fitness effects, and the factors that influence them, is critical for predicting the long term evolutionary

trajectories of antibiotic resistance alleles and therefore how to manage the dissemination of resistance in general.

1.4 Management of resistance

As the rate of new drug development has decreased, there has been a focus on evolutionary strategies to suppress resistance evolution. Novel treatment strategies have been implemented with varying degrees of success, and sometimes with unintended evolutionary consequences (Durão, Balbontín and Gordo, 2018; Lagator, Uecker and Neve, 2021). Given the observation that countries with high levels of antibiotic use have higher levels of resistance, a simple strategy to reduce resistance is to reduce the use of antibiotics (Hsueh, Chen and Luh, 2005; Furtado *et al.*, 2010). While this has been successful in some cases (Seppälä *et al.*, 1997; Lee *et al.*, 2004), it is not always feasible and resistance to some antibiotics can emerge even in antibiotic-free environments (Rodríguez-Verdugo, Gaut and Tenaillon, 2013). Further, the evolution of compensatory mutations (which alleviate the fitness cost of resistance) can allow resistance mutations to be maintained in the population (Brandis *et al.*, 2012; Durão, Balbontín and Gordo, 2018). More complex strategies such as altering the dose and length of treatment can influence evolutionary pathways to resistance, determining the rate of resistance acquisition, minimum inhibitory concentration (MIC) of resistant strains, diversity of resistance mutations, fitness cost, and compensatory mutations – all factors critical for the maintenance of resistance in the population (Uranga *et al.*, 2016; Harmand *et al.*, 2017; Durão, Balbontín and Gordo, 2018; Wistrand-Yuen *et al.*, 2018; Lagator, Uecker and Neve, 2021).

Other strategies for suppressing resistance evolution involve the use of multiple antibiotics in various treatment regimens, such as combinatorial therapies or drug cycling (van Loon *et al.*, 2005; Tupin *et al.*, 2010). However, epistatic interactions between individual mutations can make the evolution of multidrug resistance more likely than predicted based on the independent fitness effects of those mutations. Positive epistasis, where the combined fitness cost of resistance mutations is lower than would be expected from the sum of individual mutational fitness costs, allows multi-drug resistance to be easily acquired and less likely to be reversed when antibiotic treatment is withdrawn (Angst and Hall, 2013; Borrell *et al.*, 2013; Durão *et al.*, 2015). Additionally, strains with

multidrug resistance acquire compensatory mutations more rapidly than strains resistant to a single drug, offsetting the larger fitness cost often associated with multiple resistance mutations (Moura de Sousa *et al.*, 2017). The order of drugs used in cycling regimens also has an effect on the MIC and fitness cost of resistance, further influencing the trajectory of resistance evolution (Angst and Hall, 2013; Lagator, Colegrave and Neve, 2014; Yen and Papin, 2017). Therefore, understanding the fitness effects of resistance mutations as well as interactions with other mutations and the environment is essential for predicting the evolutionary consequences, ergo the success, of management strategies.

1.5 Challenges for predicting resistance evolution

The management of resistance relies in part on our ability to predict how resistance evolves. In the past, such predictions have generally relied on a binary description of resistance – where bacteria are phenotypically characterised as either resistant or susceptible to a drug. As genomics technology continues to improve, whole genome sequencing of isolates is cheaper and more rapid than ever. As such, the importance of the exact resistance mutation in determining the phenotype and fitness (and therefore the evolutionary success) of a strain is becoming more widely appreciated (Paulander, Maisnier-Patin and Andersson, 2009; Maharjan and Ferenci, 2017; Pietsch *et al.*, 2017). Different specific resistance mutations against the same antibiotic can confer varying levels of resistance (Wistrand-Yuen *et al.*, 2018), inflict different fitness costs (Wi *et al.*, 2018), and influence sensitivity to different secondary antibiotics – a phenomenon known as collateral effects (Maltas and Wood, 2019). Critically, this means resistance mutations can variably influence the evolutionary trajectory of the population (Barrick *et al.*, 2010; Liakopoulos *et al.*, 2022). Such mutation-specific effects can make it challenging to treat infections if the genetic sequence is not known; as such, improving our knowledge of the specific effects of individual resistance mutations is crucial.

The fitness effects of resistance mutations can further vary depending on the genetic background (i.e. genotype) of the mutated strain. Epistatic interactions between resistance mutations and genetic background of different species affect the growth rate and phenotype of mutants in unpredictable ways (Vogwill, Kojadinovic and Maclean, 2016). Such complex interactions are also observed between different resistance mutations in strains that are otherwise genetically identical, with consequences for the evolution of

multidrug resistance (Trindade *et al.*, 2009; Ward, Perron and Maclean, 2009; Angst and Hall, 2013; Hoeksema *et al.*, 2019). Therefore it is important not only to characterise the fitness effects of resistance mutations in one strain (e.g. a lab strain) but to explore the effects of mutations in diverse genetic backgrounds and interactions with other resistance alleles.

Another obstacle to predicting resistance evolution is the effect of the environment and how it can alter the fitness effects of resistance mutations. Often, studies characterising resistance mutations are carried out in a single nutrient (media) environment, where the only difference is the presence or absence of antibiotic (Mariam *et al.*, 2004; Ward, Perron and Maclean, 2009; Wi *et al.*, 2018; Weinstein and Zaman, 2019). However, bacterial fitness can vary considerably in different environments (Hall, 2013; Gifford, Moss and MacLean, 2016; Leehan and Nicholson, 2022). This has direct consequences for the treatment of antibiotic resistant infections, given the environmental variations between different infection sites in the human body (Smith, Fratamico and Gunther, 2007; Ciofu *et al.*, 2015; Weiser, Ferreira and Paton, 2018; Ailloud *et al.*, 2019). More broadly, differences in the many varied environments in which bacteria can be found outside of the body could influence the evolution of resistant strains in unpredictable ways. The presence of genotype-by-environment interactions, where the fitness effects of resistant mutants with different genotypes are altered by the environment in different ways, further complicate predictions of evolution (Hall, 2013; Gifford, Moss and MacLean, 2016). Greater understanding of the influence of environment on the fitness effects of resistance mutations is needed to improve predictions of how resistance is disseminated in diverse environments.

1.6 Mission statement

In this thesis I will explore factors that influence the evolution of resistance, aiming to further our understanding of the complex relationship between resistance mutations, environment, and genetic background. I will use rifampicin resistance in *E. coli* as a model system, for several reasons. *E. coli* is an extremely well characterised organism, having been the workhorse of experimental evolution and molecular biology for decades (John E Cronan, 2014), and as such understanding the factors that influence the evolution of this organism has broad significance. While rifampicin is not generally used to treat *E.*

coli in clinical settings, there are many shared resistance mutations with organisms that are treated with rifampicin, such as *Mycobacterium tuberculosis* (Ghosh, N. and Saha, 2020; Ma *et al.*, 2021). Further, rifampicin resistance is generally conferred by single point mutations that can be easily selected and result in pleiotropic effects on bacterial phenotype and fitness (Jin and Gross, 1989; Qi, Preston and Maclean, 2014). The mechanism of action of rifampicin on RNAP and the phenotypic and fitness effects of rifampicin resistance are discussed below.

1.7 Rifampicin

1.7.1 Usage and mechanism of action

Rifampicin is a large heterocyclic antibiotic first synthesized in the 1960s through modification of naturally occurring metabolites of the species *Nocardia mediterranei* in an effort to produce a more bioavailable rifamycin-class antibiotic (Figure 1.3) (Maggi *et al.*, 1966). It has activity against both Gram-positive and Gram-negative bacterial species and is used clinically as a treatment against various bacterial infections including leprosy, legionnaires' disease and endocarditis (Peard *et al.*, 1970; Rees, Pearson and Waters, 1970; Dournon *et al.*, 1990). However, owing to particularly effective activity against mycobacteria species, its primary use is as a treatment against *M. tuberculosis* infections (Grobbelaar *et al.*, 2019). The crystal structure of RNAP in complex with rifampicin reveals that rifampicin binds to a region deep within the DNA/RNA channel of the RNAP β subunit, where it sterically prevents the addition of nucleotides onto nascent RNA transcript 2-3 nucleotides long (Campbell *et al.*, 2001). Through this mechanism, it acts as an inhibitor of transcription initiation.

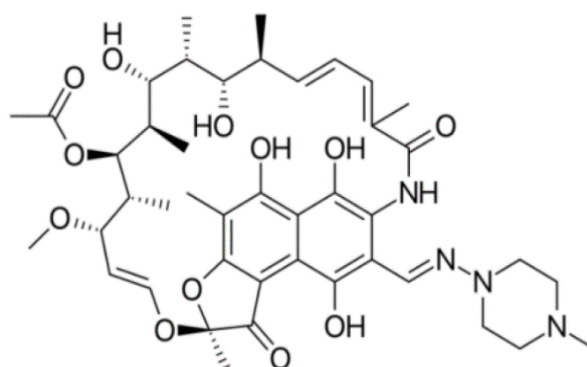


Figure 1.3 – Heterocyclic structure of the antibiotic rifampicin.

1.7.2 Bacterial RNAP and transcription

RNAP is a large, highly conserved protein of around 400 kDa that transcribes DNA into mRNA, and is therefore essential for the growth and survival of cells (Sweetser, Nonet and Young, 1987; Zhang *et al.*, 1999). The core structure of this enzyme is formed from 5 constituent subunits ($\alpha 2\beta\beta'\omega$), which can associate with an additional subunit σ to form a holoenzyme (Hermann, 2019). The two largest subunits (β and β') form a 'crab claw' structure with a cleft in which there are several discrete channels – the primary, secondary and RNA exit channel. (Zhang *et al.*, 1999). Respectively, these channels accommodate DNA and DNA-RNA hybrid structures, permit nucleotide entry, and carry out separation of RNA-DNA hybrid strands (Zhang *et al.*, 1999; Korzheva *et al.*, 2000). These three functions loosely define the three stages of transcription – initiation, elongation, and termination.

To initiate transcription of a target gene, RNAP must first bind to a promoter. RNAP is able to recognise different promoters in a sequence-specific manner through association with one of various σ factors (Lonetto, Gribskov and Gross, 1992; Dombroski, 1997; Wösten, 1998). Further transcriptional control is possible through variations in the strength of this initial binding, in part determined by the sequences of two upstream promoter elements known as the -35 and -10 hexamers (Harley and Reynolds, 1987; Murakami *et al.*, 2002). Once bound to the promoter, RNAP facilitates the unwinding and separation of a short stretch of the two DNA strands around the -10 region in an Mg^{2+} dependent manner (Suh, Ross and Record, 1993; Chen and Hermann, 1997). The separated strands of DNA form a 'transcription bubble', the site of RNA synthesis (Zuo and Steitz, 2015; Duchi *et al.*, 2018). Successive short abortive RNA transcripts are formed as the RNAP fails to dissociate from the promoter, until the nascent RNA transcript reaches 11 nucleotides and RNAP is able to break free from the promoter (Revyakin *et al.*, 2006; Saecker, Record and DeHaseth, 2011).

During the elongation phase, RNAP is very stable as it progresses along the DNA template, protecting the transcription bubble as well as 8-9 base pairs of RNA-DNA hybrid and ~5 nucleotides of nascent single-stranded RNA (Nudler *et al.*, 1997; Korzheva *et al.*, 2000). Single-stranded DNA upstream of the transcription site is reannealed to form double-stranded DNA before it leaves the enzyme so as to prevent degradation by nucleases (Nudler *et al.*, 1997). Elongation continues until RNAP reaches a termination

sequence, either intrinsic or Rho-dependent (Gusarov and Nudler, 1999; Peters *et al.*, 2009; Ray-Soni, Bellecourt and Landick, 2016). For intrinsic termination, transcription of the termination sequence produces an RNA with a G/C-rich hairpin loop that disrupts RNAP to terminate transcription (Gusarov and Nudler, 1999). For Rho-dependent termination, a recognition sequence must be present that allows the translocation of Rho and formation of the Rho-RNAP complex, the presence of which results in clamp unloading and breakdown of the elongation complex (Lau and Roberts, 1985; Epshtein *et al.*, 2010).

There are several characteristics of bacterial RNAP that make it an attractive target for pharmaceutical inhibition: (i) RNAP is essential for cell viability; (ii) RNAP genes are highly conserved among bacteria, allowing broad-spectrum activity; (iii) Bacterial RNAP is genetically distant enough from eukaryotic RNAPs to permit selectivity (Sweetser, Nonet and Young, 1987; Ma, Yang and Lewis, 2016).

1.7.3 Resistance to rifampicin

The effectiveness of rifampicin as a clinical treatment is limited as mutations conferring resistance are common and diverse, and can arise spontaneously in bacterial populations (Jin and Gross, 1988). Early research identified specific mutational sites that conferred varying levels of resistance to rifampicin in *E. coli* (Ovchinnikov *et al.*, 1983). Unsurprisingly, these sites occurred exclusively in the *rpoB* gene, encoding the target of rifampicin – the RNAP β subunit. Additional mutation sites in *rpoB* have since been characterised, largely confined to three distinct clusters now known as the Rifampicin Resistance Determining Region (RRDR) – a region that forms the site of rifampicin binding upon folding of the protein (Jin and Gross, 1988; Williams *et al.*, 1994; Garibyan *et al.*, 2003). Rifampicin resistance mutation sites are highly conserved among bacterial species, and several loci predominate the mutational landscape: residues at position 516, 526 and 531 of the β subunit (*E. coli* numbering) are most frequently substituted in clinical resistant isolates (Goldstein, 2014). This cluster is within the RRDR, and upon folding of the RNAP protein is part of the active centre at which nucleotide addition occurs (Severinov *et al.*, 1995).

1.7.4 Consequences of *rpoB* mutations

1.7.4.1 Fitness cost

Heavily conserved regions of RNAP genes are likely critical to its function, and as such mutations in these regions can affect the phenotype of mutant strains. Indeed as with many other resistance mutations, *rpoB* mutations can inflict a fitness cost (Jin and Gross, 1988; Reynolds, 2000; Qi, Preston and Maclean, 2014), and mutations with a high impact on structure such as deletions and duplications have an especially high fitness cost (Brandis and Hughes, 2018). However some rifampicin resistance mutations have a negligible effect on fitness in the absence of the antibiotic, while several mutants even appear to have improved fitness as compared to the wildtype (Reynolds, 2000; Garibyan *et al.*, 2003; Durão *et al.*, 2015). The magnitude (or even the presence) of fitness cost is therefore highly dependent on the specific resistance allele. Notably, even mutations that affect the same residue can have differences in fitness cost depending on the nature of the substitution (Gifford, Moss and MacLean, 2016). Epistatic interactions can further influence the fitness cost of *rpoB* mutations, either reducing or increasing the cost depending on the genetic background of the strain or on the presence of a specific secondary resistance mutation (Durão *et al.*, 2015; Trauner *et al.*, 2021).

1.7.4.2 Transcription

The exact mechanism through which *rpoB* mutations inflict a fitness cost is unclear, however it is likely underpinned by alterations to transcription. Different *rpoB* mutations have been shown to interfere with the critical processes of transcription including the stability of initiation complexes and the rate of elongation (Zhou and Jin, 1998; Svetlov *et al.*, 2007). As such, these mutations can alter the expression levels of genes – notably in a mutation-specific way, where the number and nature of differentially expressed genes are dependent on the loci and identity of the mutated amino acid (Qi, Preston and Maclean, 2014; Meenakshi and Munavar, 2018; Trauner *et al.*, 2021). Transcriptional alteration can result in pleiotropic phenotypic effects that depend on which specific genes are differentially expressed. Altered transcription of a key regulator could have knock on effects on the expression of many genes, potentially interfering with diverse biological systems. If these biological systems are targets of other antibiotics, such interference could alter the level of sensitivity to secondary drugs. For example, a specific *rpoB*

mutation (though not one that caused rifampicin resistance) results in increased cell wall thickness and altered cell surface charge – phenotypic changes hypothesised to cause daptomycin and vancomycin resistance (Cui *et al.*, 2010). Several ciprofloxacin-selected *rpoB* mutations (that also decreased the susceptibility to rifampicin) increase the expression of multidrug efflux pump *mdtK*, concomitantly increasing resistance to multiple classes of antibiotics (Pietsch *et al.*, 2017). Therefore, rifampicin resistance mutations in *rpoB* could influence the sensitivity to secondary antibiotics in a similar way, especially given such mutations have been shown to influence collateral effects (Waller *et al.*, 2023).

1.7.4.3 Environment-dependent effects

Not only are the fitness effects of *rpoB* mutations highly genotype-dependent, but they can also be influenced by the bacterial growth environment. Several mutants display a temperature sensitivity, failing to grow either at low temperatures, high temperatures, or both (Jin and Gross, 1989). Variations in carbon source also alter the fitness effects of *rpoB* mutants, potentially altering the competitive fitness of the resistant strain compared to the wildtype (Hall, 2013; Song *et al.*, 2014; Gifford, Moss and MacLean, 2016). This has several important consequences: firstly, for the characterisation of *rpoB* mutations and their effect on phenotype and fitness, as comparisons between different studies is made challenging if growth conditions aren't consistent; secondly, for the dissemination of rifampicin resistant strains in clinical settings, as their survival and thus evolutionary outcomes are dependent on the growth environment.

1.8 Thesis outline

Broadly, this thesis highlights how a single point mutation—especially in an essential gene such as *rpoB*—can have complex effects on bacterial phenotype and fitness, and as such can have significant and unpredictable effects on the evolutionary trajectories of mutant strains. These complex effects caused by different genotypes are further compounded by interactions with the environment – for example nutrient environment, temperature and secondary antibiotics. In chapter 2, I explore how different mutations in *rpoB* affect the fitness of *E. coli* and the transcription of genes by RNA polymerase across

several different media environments commonly used in laboratory research. The effects of temperature on RNAP function, and more generally on the relationship between genotype, phenotype and fitness of *rpoB* mutants, is investigated in chapter 3. Finally, the different evolutionary trajectories of *rpoB* mutants evolving secondary resistance are uncovered in chapter 4 with a novel technique involving whole genome metagenomic sequencing of a heterogenous population. Together, this thesis demonstrates how different mutations in an essential gene cause unpredictable changes to bacterial phenotype; that these changes are highly dependent on the growth environment in which they are measured; that such phenotypic changes lead to measurable changes to bacterial fitness; and finally that these single point mutation variations between strains can result in differences in their downstream evolutionary outcomes.

2 Pervasive Genotype-by-Environment interactions shape the fitness effects of antibiotic resistance mutations

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2.1 Abstract

The fitness effects of antibiotic resistance mutations are a major driver of resistance evolution. While the nutrient environment affects bacterial fitness, experimental studies of resistance typically measure fitness of mutants in a single environment only. We explored how the nutrient environment affected the fitness effects of rifampicin-resistant *rpoB* mutations in *Escherichia coli* under several conditions critical for the emergence and spread of resistance—the presence of primary or secondary antibiotic, or the absence of any antibiotic. Pervasive Genotype-by-Environment (GxE) interactions determined fitness in all experimental conditions, with rank order of fitness in the presence and absence of antibiotics being strongly dependent on the nutrient environment. GxE interactions also affected the magnitude and direction of collateral effects of secondary antibiotics, in some cases so drastically that a mutant that was highly sensitive in one nutrient environment exhibited cross-resistance to the same antibiotic in another. It is likely that the mutant-specific impact of *rpoB* mutations on the global transcriptome underpins the observed GxE interactions. The pervasive, mutant-specific GxE interactions highlight the importance of doing what is rarely done when studying the evolution and spread of resistance in experimental and clinical work: assessing fitness of antibiotic-resistant mutants across a range of relevant environments.

2.2 Introduction

The rise of antimicrobial resistance in recent decades represents an immense global health and economic risk, particularly in developing nations, with stark estimates of the cost to human life (O' Neill, 2014). Efforts to combat the rise of resistance have largely focussed on the development of new drugs. Yet, whilst the last century enjoyed unprecedented levels of new drug discovery, the pipeline has dried up in recent years (Spellberg *et al.*, 2004). The dramatic slowdown in the development of new drugs has resulted in the need for novel approaches to tackle resistance as well as preserving the usefulness of existing drugs (Qadri, Shah and Mir, 2021). Improving our understanding of the evolutionary processes underpinning resistance is essential for prolonging the lifetime of existing drugs and, hence, essential for slowing down the spread of antimicrobial resistance. This includes understanding the factors that impact how resistance emerges, what effect mutations have on phenotype and fitness, and how they influence the evolutionary fate of a population.

Bacteria can evolve resistance through spontaneous DNA mutations, often single point mutations, that inhibit drug binding or action (Woodford and Ellington, 2007). While in many cases there are multiple possible mutations that can confer resistance to an antibiotic, not all mutations confer the same level of resistance. Therefore, some mutants will have lower fitness than others in the presence of an antibiotic (sometimes referred to as 'survival' or just 'resistance') (Garibyan *et al.*, 2003). Some resistance mutations also impose *collateral effects*, by altering fitness in the presence of secondary antibiotics to which the population has not been previously exposed (Lázár *et al.*, 2013; Roemhild and Andersson, 2021). Furthermore, resistance mutations are often associated with a reduction in fitness in an antibiotic-free environment (commonly referred to as the *fitness cost of resistance*) (Andersson *et al.*, 2014), which can arise as a consequence of the resistance mutation altering the structure or function of an essential protein (Petersen, Aarestrup and Olsen, 2009; Qi, Preston and Maclean, 2014). For example, resistance to rifampicin often carries a fitness cost as the resistance-conferring mutations occur in the *rpoB* gene, which encodes the β subunit of RNA polymerase (Qi, Preston and Maclean, 2014; Song *et al.*, 2014). The likelihood of acquiring and maintaining multi-drug resistance critically depends on the fitness effects of resistance mutations in the presence and the absence of the primary antibiotic, as well as the associated collateral effects in the

presence of secondary antibiotics (Perron *et al.*, 2012; Imamovic and Sommer, 2013; Lagator, Colegrave and Neve, 2014).

The fitness effects of mutations, therefore, depend not only on the genotype but also on the environment (Hall, 2013; Gifford, Moss and MacLean, 2016). Broadly speaking, when attempting to understand the key factors that affect the emergence and spread of drug resistance, it is critical to characterise fitness in several environments: (i) presence of the primary antibiotic to which resistance has originally evolved; (ii) absence of any antibiotic, as would occur when the primary treatment is withdrawn and as measured by the fitness cost of resistance; (iii) presence of a secondary antibiotic used to treat the resistant infection, captured by the collateral effects; and (iv) low antibiotic concentrations, as concentrations vary drastically between different niches in the body and the environment (Kümmerer, 2009; Dartois, 2014). The fitness associated with a mutation can vary considerably between these environments (Gifford, Moss and MacLean, 2016; Allen, Pfrunder-Cardozo and Hall, 2021).

However, the presence and absence of an antibiotic is not the only environmental factor that determines the fitness effect of a mutation. The fitness effects of a resistance mutation can depend on the nutrient composition of the environment, resulting in a Genotype-by-Environment (GxE) interaction (Hall, 2013; Gifford, Moss and MacLean, 2016). The existence of such GxE interactions means that, at a given concentration of an antibiotic, one mutant might outperform another in one nutrient environment, while exhibiting lower fitness in a different environment (Martin and Lenormand, 2015). GxE interactions are difficult to predict, especially when mutations have pleiotropic effects that affect many phenotypes (Jin and Gross, 1989; Qi, Preston and Maclean, 2014; Pietsch *et al.*, 2017; Ostrer, Yinduo and Khodursky, 2021). The complexity of environments in which bacteria live means that the effect of GxE interactions on fitness can play a key role in determining the emergence and spread of antimicrobial resistance. And yet, the majority of studies measuring fitness of resistance mutations tend to do so in a single nutrient environment, only varying whether an antibiotic is present or not (Jin and Gross, 1989; Mariam *et al.*, 2004; Ward, Perron and Maclean, 2009).

We investigated how GxE interactions (in this study, we consider the environment to be the nutrient composition of the growth media) affect bacterial fitness in the presence and absence of the primary antibiotic, as well as the collateral effects of secondary antibiotics.

We focused on 11 genetically identical mutants that differ only with respect to a single point mutation in the *rpoB* gene (coding for the β subunit of RNA polymerase), which confers resistance to the antibiotic rifampicin. We identified complex GxE interactions that make fitness effects of resistance mutations and the collateral effects of other antibiotics strongly dependent on the nutrient environment.

2.3 Materials and methods

2.3.1 Bacterial strain and culture conditions

E. coli strain BW25113 $\Delta tolC$ was used in all experiments in this study to avoid selecting for resistance mutations targeting efflux systems. This strain is part of the Keio collection (Baba *et al.*, 2006). It should be noted that similar effects of environment and genotype on resistance profiles have been reported in strains with a functioning *tolC* (Allen, Pfrunder-Cardozo and Hall, 2021), as well as in other species (Liakopoulos *et al.*, 2022). Strains were stored as 15% glycerol stocks at -80°C. Liquid cultures were grown at 37°C with shaking at 200 RPM. Agar plates were grown at 37°C without shaking. Culture media used were Lysogeny Broth (LB), Mueller-Hinton (MH), and M9 supplemented with 2% glucose and 0.2% casamino acids. These were selected as commonly used growth media and include 2 complex media (LB and MH) and 1 defined media (M9).

2.3.2 Antibiotics

All antibiotics, their solvents and the range of concentrations tested can be found in Table 2.1.

Table 2.1 – Antibiotics and their solvents used in this study.

Antibiotic	Solvent	Concentration range tested (µg/mL)
<i>Rifampicin</i>	<i>DMSO</i>	-
<i>Chloramphenicol</i>	<i>70% Ethanol</i>	<i>MH: 0.25 – 8; LB: 0.25 – 8; M9: 0.5 – 16</i>
<i>Ciprofloxacin</i>	<i>1% Acetic acid</i>	<i>MH: 0.0005 – 0.016; LB: 0.001 – 0.032; M9: 0.0005 – 0.016</i>
<i>D-cycloserine</i>	<i>Water</i>	<i>MH: 10 – 60; LB: 10 – 60; M9: 4 – 128</i>
<i>Streptomycin</i>	<i>Water</i>	<i>MH: 0.25 – 8; LB: 1.5 – 15; M9: 2 – 64</i>
<i>Tetracycline</i>	<i>50% Ethanol</i>	<i>MH: 0.031 – 1; LB: 0.25 – 8; M9: 0.063 – 2</i>

2.3.3 Rifampicin-resistant mutant selection

rpoB mutants were generated with a fluctuation assay-based protocol in which mutations were first generated during growth in a non-selective environment before plating on selective agar plates to isolate resistant colonies. With this method the likelihood of generating more than a single mutation is very low given the mutation rate in *E. coli* ($\sim 1 \times 10^{-3}$) (Lee *et al.*, 2012). Firstly, overnight cultures were prepared by inoculating 1 mL MH from frozen stocks. The following morning, cultures were diluted 1 in 100 in 1 mL fresh MH media and grown for 24 hours. Evolved cultures were spread on MH agar plates containing 20 µg/mL rifampicin and grown for a further 24 hours. Colonies were picked and grown in liquid MH overnight for storage. 11 unique *rpoB* mutations were generated, identified through Sanger sequencing of the *rpoB* gene. We carried out Illumina whole genome sequencing on these strains to confirm the presence of only a single mutation (MicrobesNG performed genome sequencing then adapter trimming with Trimmomatic v 0.30 (Bolger, Lohse and Usadel, 2014), with a sliding window quality

cut-off of Q15). Trimmed reads have been deposited as FASTQ files to the US National Centre for Biotechnology Information (NCBI) Sequence Read Archive database under Bioproject accession number PRJNA988908.

2.3.4 Fitness measurements

Fitness measurements in Figure 2.2 were obtained using a CLARIOstar Plus microplate reader. Overnight cultures were prepared in respective media (LB, MH or M9) from frozen stocks, before being diluted 1 in 1000 in the same media, with or without rifampicin at 8 µg/mL. We used this concentration of rifampicin after measuring it as the MIC of the WT in MH using standard log₂ dose response experiments (see *Chapter 2.3.5 – Determining Minimum Inhibitory Concentrations (MICs)*). While some growth of the WT was observed in both LB and M9 at 8 µg/mL of rifampicin, suggesting that in these environments the MIC is higher, we wanted to keep the concentration of antibiotic consistent when comparing growth of resistant mutants so did not change it between environments. OD₆₀₀ was measured every 10 minutes for 24 hours at 37°C with shaking. 7 biological replicates were taken for each condition. Growth curve analysis was carried out using R and the package *growthcurver* (Sprouffske and Wagner, 2016). We used the empirical area under the curve between 0 and 24 hours as our measurement of fitness, and values were normalised against the fitness of the WT in antibiotic-free MH. To test whether there was a significant change in rank order between nutrient environments in Figure 2.3A and Figure 2.3B, we performed an aligned ranks transformation ANOVA using the *art* function in the R package *ARTool* (Wobbrock *et al.*, 2011). In this model we used AUC as the response variable, with nutrient environment, mutant (genotype) and the interaction between the two as response variables. The WT was excluded from the analysis in 2A as its growth was greatly diminished in the presence of rifampicin. Tukey’s pairwise comparisons were added to the analysis in Figure 2.3B to determine which mutants had significantly different fitness from one another. The linear regression in Figure 2.3C and Figure 2.7 was performed with the *lm* function in R. Welch’s two sample t-tests were used to test for significance between the fitness of specific pairs of mutants. To investigate the interaction between mutant, nutrient environment and the presence of rifampicin, a mixed effect model was fitted using the *aov* function with nested error terms of nutrient environment, rifampicin concentration and replicate.

Fitness measurements in Figure 2.5 were obtained using CLARIOstar Omega microplate reader with stacker. Overnight cultures were prepared as above, then diluted 1 in 1000 in media supplemented with a sub-MIC concentration of antibiotic. The sub-MIC concentration used for each antibiotic/media combination was determined by obtaining the MIC of every mutant in every antibiotic/media combination through standard dose-response experiments (see *Chapter 2.3.5 – Determining Minimum Inhibitory Concentrations (MICs)*), and then halving the lowest MIC of all mutants in that environment. For example, the lowest MIC of ciprofloxacin in MH media was 0.001 $\mu\text{g/mL}$ (shared across several mutants), therefore the sub-MIC concentration of ciprofloxacin used for all mutants in MH was 0.0005 $\mu\text{g/mL}$. OD₆₀₀ was measured every 30 minutes for 24 hours at 37°C with shaking before each reading. 4 biological replicates were measured for each condition. As described above, we used area under the curve as fitness and measurements were normalised to the fitness of the WT in antibiotic-free MH. To determine whether the fitness of a mutant at sub-MIC concentrations could be predicted from its MIC (Figure 2.6A), we fitted a linear mixed effect model using the *lme* function from the R package *nlme*, with fitness (area under the curve) as the response variable, MIC as the fixed effect, and mutation (genotype) and replicate as random effects. To test whether there was a significant change in rank order of fitness between nutrient environments in the presence of sub-MIC concentrations of secondary antibiotics (Figure 2.6B), we performed an aligned ranks transformation ANOVA as previously described.

2.3.5 Determining Minimum Inhibitory Concentrations (MICs)

Overnight cultures were prepared in respective media then diluted 1 in 100 in the same media supplemented with antibiotic at a range of concentrations (Table 2.1), prepared by serial dilution. Three biological replicates were tested. OD₆₀₀ measurements were taken before and after 24 hours of incubation. Blank values were subtracted from the final measurements, then MIC was determined as the lowest concentration at which at least 2 out of 3 replicates failed to reach an OD₆₀₀ greater than 0.1. In more than 80% of cases this concentration was the same in all 3 replicates. The MIC of each mutant was normalised to the MIC of the WT in each respective antibiotic-media combination.

2.3.6 RNA-Seq

We conducted RNA-Seq analysis on three *rpoB* mutants to investigate their effect on the transcriptome (phenotype). We selected two strains with a mutation in the same residue (Q513L and Q513R) and a random third mutant (H526Y). RNA-Seq was carried out in two nutrient environments (either LB or M9) as the environments with the highest and lowest growth rates (averaged across all mutants). RNA was extracted using *QIAGEN* RNAprotect and RNEasy kits, according to manufacturer's instructions. Briefly, $\sim 7.5 \times 10^8$ RNAprotect stabilised cells harvested at $OD_{600} = 0.7$ were digested by incubating with 200 μ L TE (*QIAGEN*) containing 15 mg/mL Lysozyme (*Thermo Fisher Scientific*) and 10 μ L Proteinase K (*QIAGEN*) for 10 minutes at room temperature. RNA was then purified using RNEasy spin columns. Samples were sent to *Azenta Life Sciences* (Germany) for quality control, rRNA depletion, library preparation and sequencing with the Standard RNA-Seq service using an Illumina NovaSeq platform and 150 bp paired end reads. Raw reads have been deposited in the NCBI Sequence Read Archive database under Bioproject accession number PRJNA988908.

Sequencing adapters and low-quality regions were trimmed using *Trim Galore* (v.0.6.7) (Krueger *et al.*, 2023) using default parameters and reads shorter than 50bp after trimming were discarded. rRNA reads were removed with *BBDuk* (v.38.18) (<https://sourceforge.net/projects/bbmap>) using the bundled rRNA k-mers database. Remaining mRNA reads were aligned to the reference genome (CP009273) using *Bowtie2* (v.2.2.5) (Langmead and Salzberg, 2012) and count tables were constructed from the resulting SAM files using *featureCounts* (v.2.0.1) (Liao, Smyth and Shi, 2014) – see *Supplementary stats and data*[†]. *DEGUST* (Powell, 2015) was used to perform differential expression analysis using the voom/limma method with a false discovery rate threshold of $p < 0.05$ and absolute log fold change of > 1 , and to generate multidimensional scaling (MDS) plots. To determine COG categories for differentially expressed genes, coding sequences of the reference genome were downloaded from RefSeq and functionally annotated using the online instance of *eggNOG-mapper* (<http://eggnog-mapper.embl.de/>) with default parameters and Diamond as the underlying alignment algorithm (Huerta-Cepas *et al.*, 2019; Buchfink, Reuter and Drost, 2021; Cantalapiedra *et al.*, 2021).

[†] Can be found at <https://doi.org/10.6084/m9.figshare.c.6764136>

2.3.7 Antimicrobial susceptibility testing literature review

The literature survey examining antimicrobial susceptibility testing in multiple environments was conducted using PubMed's database. The search terms were "antibiotic resistant", "isolate", and "Escherichia coli"; filters were used to exclude review publications and select only those published in 2019, providing a current assessment without bias from COVID-19. The first 300 of 1,351 results were annotated according to the distinct environments in which they measured antimicrobial susceptibility or minimum inhibitory concentration. In cases where the utilised growth medium was unstated, it was assumed that measurements were taken only in one environment.

2.4 Results

2.4.1 Genotype-by-Environment (GxE) interactions alter rank order of fitness in the primary antibiotic environment

One of the key factors that determine the evolutionary success of a newly acquired resistance mutation is its fitness effects in the presence of the selective (primary) antibiotic. To explore the relationship between nutrient environment and fitness in the presence of the primary antibiotic, we generated 11 rifampicin-resistant strains of *E. coli* using a fluctuation assay-based approach (see *Chapter 2.3 – Materials and Methods*). Each strain was genetically identical to the wildtype (WT) (BW25113, carrying a *tolC* deletion to prevent the activity of TolC-dependent efflux pumps) with the exception of a single substitution mutation in *rpoB* (Figure 2.1A). All these mutations (except V146F) are within the rifampicin resistance determining region of the *rpoB* gene, and all (including V146F) lie close to the rifampicin binding site (Figure 2.1B). We measured fitness (as the area under the curve (AUC) of growth curves; Figure 2.2) of all 12 strains (11 *rpoB* mutants and the WT) in three common laboratory nutrient environments: Mueller-Hinton (MH) broth, Lysogeny broth (LB), and M9 minimal media with 2% glucose (M9), in the presence of 8 µg/mL of the primary antibiotic rifampicin (Figure 2.1C). This concentration of rifampicin was selected as we determined it to be the minimum inhibitory concentration of the WT strain measured in MH. We used AUC as a measure of fitness as it is a well-used growth metric that integrates various features of

the bacterial growth curve, such as growth rate and lag phase (Dunai *et al.*, 2019; Shea *et al.*, 2020; Arrieta-Ortiz *et al.*, 2023; Gifford *et al.*, 2023).

The rank order of all mutants varied considerably between environments, indicating that the genotypes interact significantly with nutrient environment (Aligned ranks transformation ANOVA: $F_{20, 198} = 27.93$, $p < 0.0001$) (Figure 2.3A). The effect of nutrient environment can be so dominant that one mutant (Q513P) exhibited significant levels of resistance in MH and LB but showed very little growth in M9. Such prominent GxE interactions, which alter the rank order of mutant fitness in the presence of rifampicin between environments, can result in one mutant (for example, R529H) having the highest fitness in one environment (LB), but exhibiting low relative fitness in another (M9).

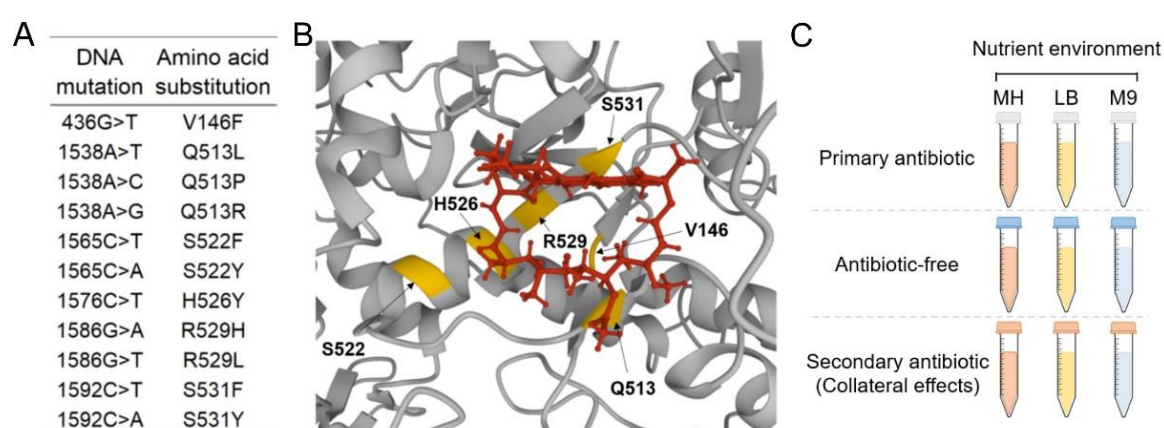


Figure 2.1 – Investigating Genotype-by-Environment interactions of *rpoB* mutations. (A) *rpoB* mutants (and the respective amino acid substitutions) generated in this study. (B) Location of the affected residues (yellow) and their proximity to a rifampicin molecule (red) in the 3D structure of RNA polymerase (grey; structure PDB ID: 4KMU) (Berman *et al.*, 2000; Molodtsov *et al.*, 2013; Sehnal *et al.*, 2021). (C) Schematic outlining the environments investigated herein. In this instance the primary antibiotic was rifampicin, and the secondary antibiotic was either chloramphenicol, ciprofloxacin, D-cycloserine, streptomycin, or tetracycline. MH – Mueller Hinton media, LB – Lysogeny Broth, M9 – M9 media supplemented with 2% glucose.

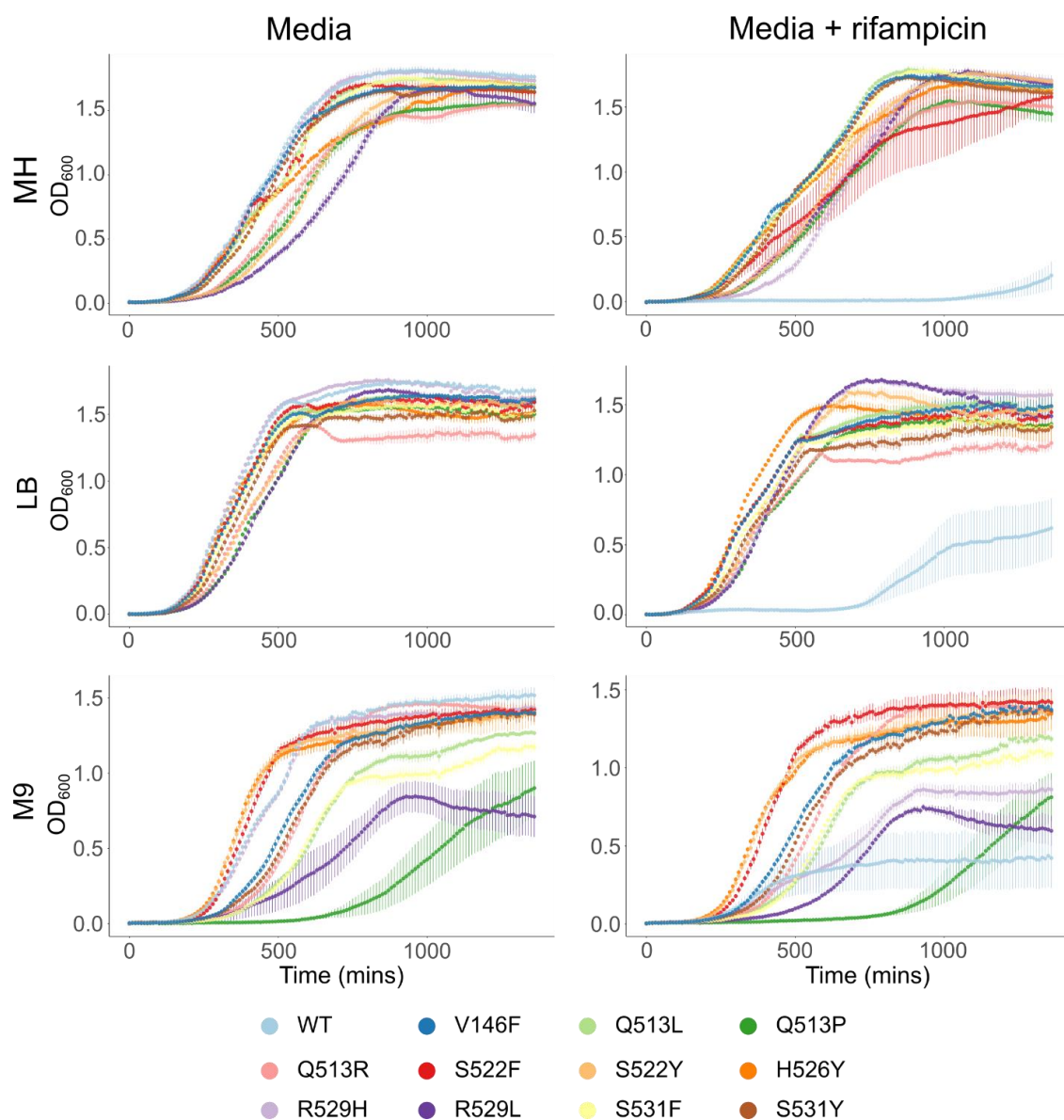


Figure 2.2 – Growth curves of *rpoB* mutants in 3 nutrient environments, with and without rifampicin (8 µg/mL). Data are mean of 7 biological replicates; error bars are standard error of the mean.

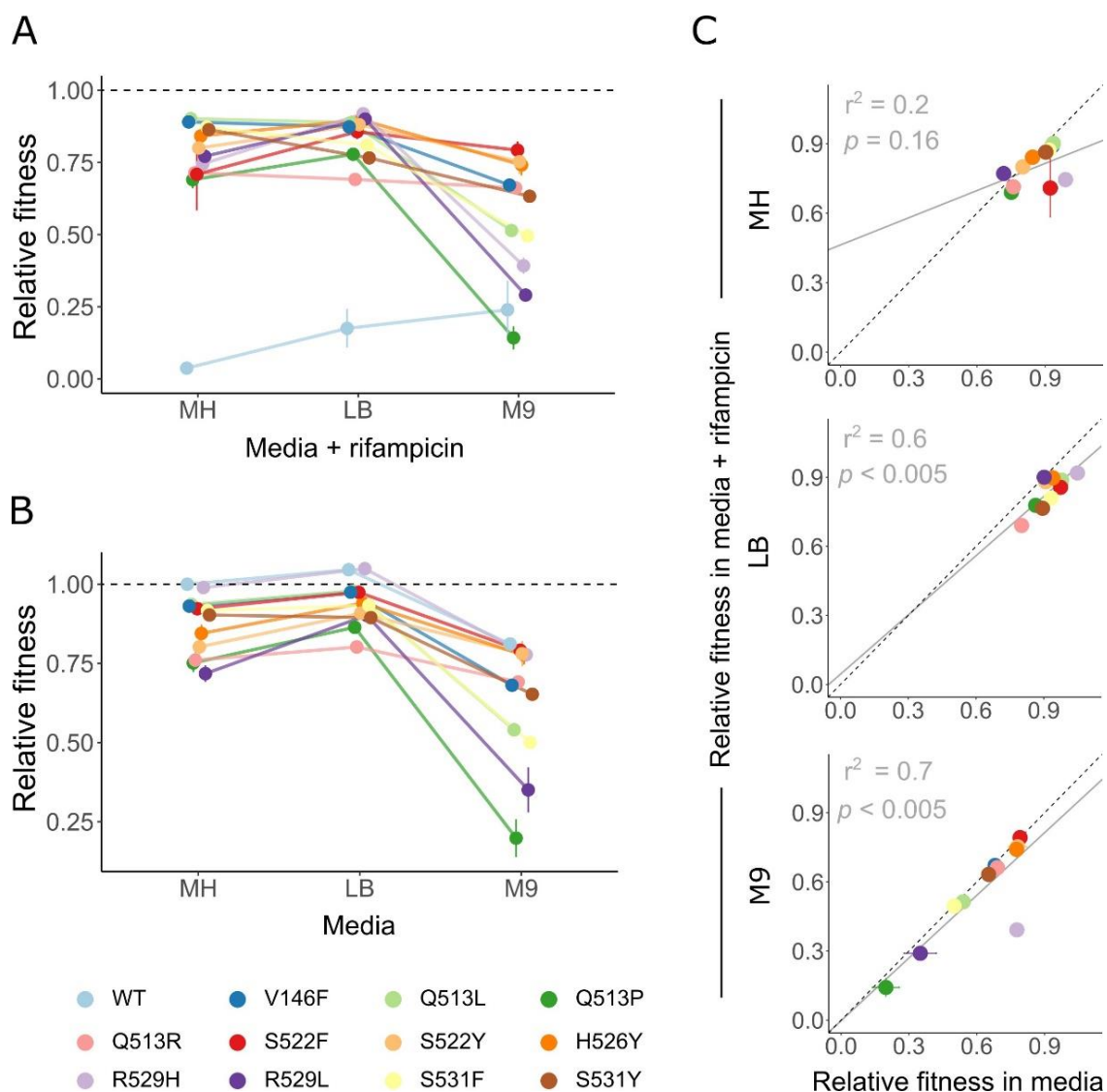


Figure 2.3 – GxE interactions of *rpoB* mutations influence fitness in the presence and absence of rifampicin. (A) Relative fitness of mutants and the WT grown in 3 nutrient environments supplemented with 8 $\mu\text{g/mL}$ rifampicin (determined as the minimum inhibitory concentration of the WT in MH media; whilst some growth of the WT was observed in LB and M9 with this concentration, it was maintained in all 3 environments for consistency). (B) Relative fitness of mutants and the WT grown in 3 antibiotic-free nutrient environments. (C) Relative fitness of mutants in each antibiotic-free nutrient environment against relative fitness in media supplemented with rifampicin. Results of linear regression analysis are shown in grey (solid line); dotted line shows perfect correlation. WT was excluded from regression analysis as its fitness was unsurprisingly lower in the presence of rifampicin. Points in all plots show mean of 7 biological replicates; error bars show standard error of the mean. Relative fitness values are area under the curve (AUC) values normalised to the AUC of the WT in antibiotic-free MH.

2.4.2 GxE interactions alter rank order of fitness in the absence of antibiotics

Once a bacterial infection has evolved resistance to the initial treatment, the use of that first antibiotic will likely be withdrawn (Hanretty and Gallagher, 2018). Strains carrying primary resistance mutations may then occupy an antibiotic-free environment as the drug is cleared from the body. To explore GxE interactions in the absence of antibiotics, we measured fitness of all 12 strains in the 3 nutrient environments. Around half of the mutants exhibited a significant fitness cost compared to the WT (MH: 5; LB: 5; M9: 6; for all pairwise comparisons see *Supplementary stats and data*[†]). No mutant had higher fitness than the WT in the absence of antibiotics. Interestingly, we observed greater diversity of fitness amongst *rpoB* mutants when grown in M9 than when tested in MH or LB media, both in the presence and the absence of the primary antibiotic (Standard deviations – MH: 172.5; LB: 124.3; M9: 337.3; MH + Rif: 211.3; LB + Rif: 144.5; M9 + Rif: 345.5) (Figure 2.3). Why the variation in fitness between *rpoB* mutants might be more constrained in some environments than in others is unclear.

GxE interactions affected the rank order of mutant fitness between nutrient environments in the absence of antibiotics (Aligned ranks transformation ANOVA: $F_{22, 216} = 23.42$, $p < 0.0001$) (Figure 2.3B). The observation that the rank order of fitness depends on the nutrient environment suggests that the maintenance of resistance mutations is contingent on both the specific mutation and the growth conditions. For example, consider the mutants Q513R and S531F. In MH media, the fitness of Q513R (mean = 1251.9, SEM = 43.4) was significantly lower than the fitness of S531F (mean = 1510, SEM = 17.5) (Tukey pairwise comparison: $p < 0.005$), suggesting that if they were to coexist in a population S531F would outcompete Q513R. However, in M9 the opposite was true (Q513R: mean = 1138.5, SEM = 11.03; S531F: mean = 823.6, SEM = 31.6; Tukey pairwise comparison: $p < 0.0001$), suggesting that in different nutrient environments the population dynamics of the primary resistant mutants are shifted. The ability to predict the long-term evolutionary success of these two mutants is further complicated by the fact that their rank order of fitness in the presence of the primary antibiotic rifampicin is also dependent on the nutrient environment (Figure 2.3A).

[†] Can be found at <https://doi.org/10.6084/m9.figshare.c.6764136>

2.4.3 Nutrient environment alters the relationship between fitness in the presence and the absence of rifampicin

In some instances, the fitness of resistant mutants in the presence of the primary antibiotic is correlated with the fitness in its absence (Vogwill *et al.*, 2012; Wen *et al.*, 2021). To determine whether fitness in the absence of rifampicin was a significant predictor of fitness in the presence of the drug, we fitted a linear regression model on the mean fitness of 11 mutants, in each of the 3 environments (the WT was excluded from the analysis as its fitness in rifampicin is unsurprisingly diminished – see *Chapter 2.3 – Materials and Methods*). In LB and M9, the fitness of *rpoB* mutants in the absence and presence of rifampicin was positively correlated, while no such relationship was identified in MH (Figure 2.3C). Therefore, fitness in the presence of rifampicin can be predicted based on mutant fitness in the absence of the drug only under some nutrient conditions.

Interestingly, whether the addition of rifampicin significantly altered mutant fitness was also environment dependent, as we identified a significant interaction between genotype, nutrient environment, and the presence/absence of rifampicin (Mixed effect model: $F_{20, 393} = 2.77$, $p < 0.001$). This means that, for example, the presence of rifampicin did not significantly alter the fitness of the mutant S531F in M9 (Welch Two Sample t-test: $t_{11.06} = 0.2$, $p = 0.85$), while in MH and LB its fitness was significantly different in the presence and absence of rifampicin (MH: $t_{11.95} = 2.85$, $p < 0.05$; LB: $t_{9.38} = 7.83$, $p < 0.001$). The existence of a relationship between fitness in the absence and presence of antibiotic in two of the three media environments further demonstrates the complex GxE interactions that shape the evolutionary fate of rifampicin resistance mutations.

2.4.4 GxE interactions shape collateral effects of secondary antibiotics

The fitness effects of a resistance mutation in the presence of the primary antibiotic critically determine the likelihood of that mutation being fixed during treatment, while its persistence in the population following treatment is to a larger extent determined by its fitness in the absence of antibiotics. We have shown that GxE interactions play a large role in determining the fitness effects of mutations in both conditions. However, when considering the emergence of multi-drug resistance, another factor plays a major role: the collateral effects of resistance mutations – *i.e.*, their fitness effects in the presence of a

secondary antibiotic (Imamovic and Sommer, 2013; Gjini and Wood, 2021). To test if GxE interactions shape collateral effects, we measured the minimum inhibitory concentration (MIC) of all 11 mutants to 5 secondary antibiotics with differing mechanisms of action to rifampicin (chloramphenicol, ciprofloxacin, D-cycloserine, streptomycin, and tetracycline) in the 3 nutrient environments. This includes drugs that have been trialled in combinatorial therapy with rifampicin (Vouking, Tamo and Tadenfok, 2013; van Straalen *et al.*, 2021), as a treatment for rifampicin-resistant tuberculosis (Li *et al.*, 2019), or in the case of ciprofloxacin, have a similar mode of action to the anti-tubercular drug moxifloxacin (Naidoo *et al.*, 2017).

Collateral effects were very common in rifampicin resistance mutants, with mutants having different MIC compared to the WT strain in 55% of the tested environment/antibiotic combinations (Figure 2.4). Interestingly, even mutations at the same residue but with a different amino acid substitution exhibited different collateral effects. For example, a substitution to leucine at position 513 (Q513L) did not increase the sensitivity to streptomycin, while the proline substitution did (Q513P). Whether the collateral effects were more likely to result in increased sensitivity (lower MIC than WT) or cross-resistance (higher MIC than WT) depended on the antibiotic and the mutant (Figure 2.4A,B). For example, while cross-resistance was the most common collateral effect observed against D-cycloserine (14/19), no mutants had cross-resistance to tetracycline or chloramphenicol, instead exhibiting only varying degrees of collateral sensitivity. The likelihood of observing collateral effects also varied between antibiotics (Figure 2.4B).

In addition to the antibiotic and the mutant itself, the nutrient environment also had an impact on collateral effects (Figure 2.4A,C). This Genotype-by-Environment-by-Antibiotic interaction means that the strength of cross-resistance or sensitivity, and in some cases even the direction of the collateral effect, is dependent not only on which secondary antibiotic is used but also on the nutrient environment. One of the more dramatic examples is S522F, which exhibited cross-resistance to ciprofloxacin in LB but increased sensitivity in MH. The same mutant exhibited cross-resistance to streptomycin in MH but collateral sensitivity in LB. The nutrient environment also impacted the total number of cross-resistance interactions - in LB only 1 such interaction was found (S522F against ciprofloxacin), whilst in M9 there were 11 examples of cross-resistance (Figure 2.4C). Meanwhile, mutants grown in MH exhibited the highest number of collateral

sensitivity interactions – 28, whilst in M9 only 20 such interactions were observed. These findings demonstrate the environment-dependent variability in not only the strength but also the direction of collateral effects.

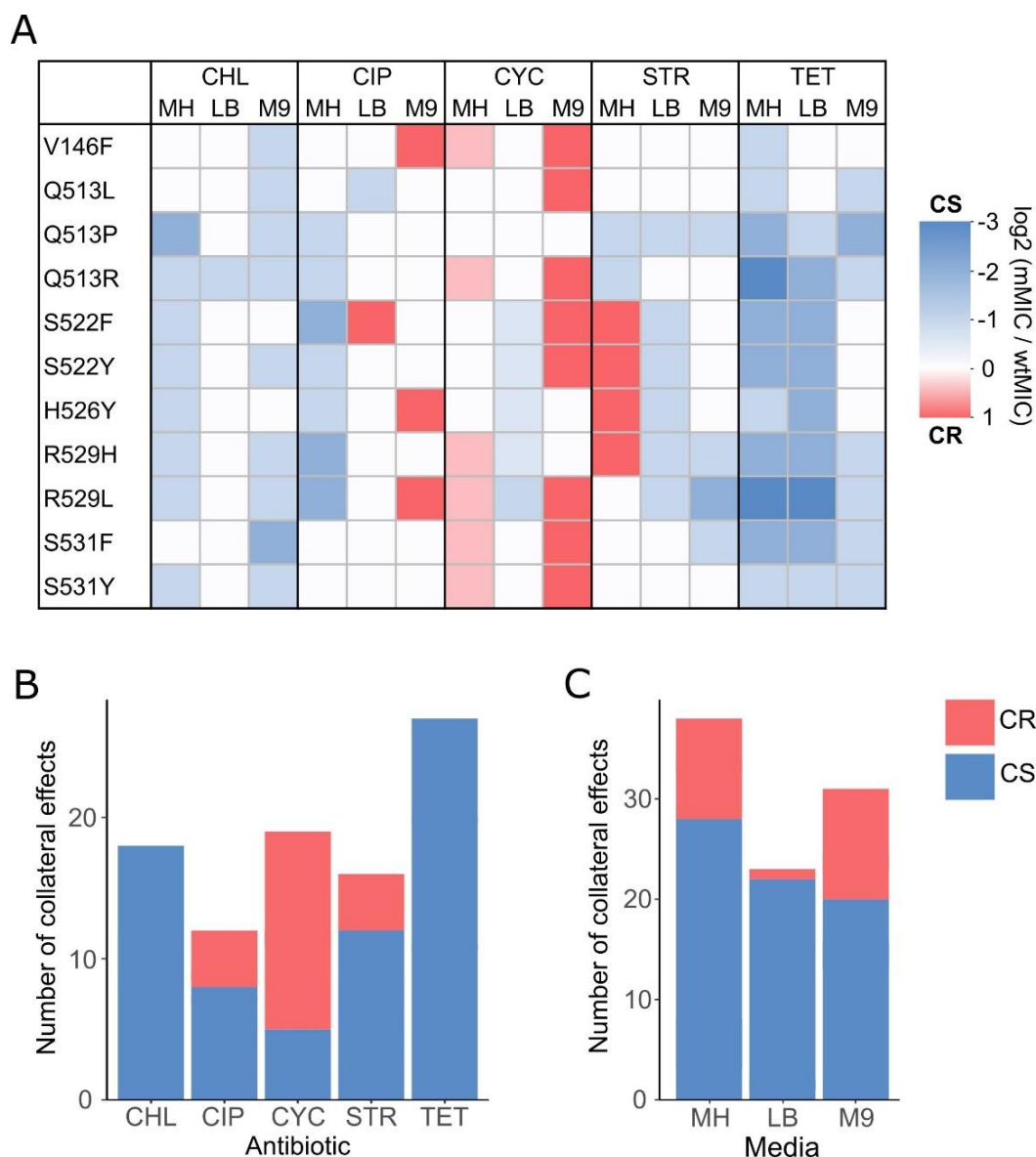


Figure 2.4 – Pervasive collateral effects of *rpoB* mutants are genotype- and environment-dependent. (A) Collateral sensitivity (CS; blue) and cross resistance (CR; red) of *rpoB* mutants against 5 secondary antibiotics (CHL: chloramphenicol, CIP: ciprofloxacin, CYC: D-cycloserine, STR: streptomycin, TET: tetracycline), in 3 nutrient environments. Strength of collateral effects are determined as the log base 2 of the MIC of the mutant divided by the MIC of the WT in each respective nutrient environment/antibiotic combination. (B) Total number and direction of collateral effects against each secondary antibiotic, and (C) in each nutrient environment.

2.4.5 Genotype-by-Environment-by-Antibiotic interactions shape fitness at sub-MIC concentrations of secondary antibiotics

Differences in collateral effects (Figure 2.4) identified certain concentrations of secondary antibiotics that are permissive to the growth of some mutants, but not others. GxE effects on collateral sensitivity ensure that the same concentration of antibiotic might not be sufficient to control the bacterial population as the nutrient environment changes. Another level of complexity arises from the fact that concentrations of antibiotics vary dramatically during the course of treatment (Markus, Amparo and Hartmut, 2004), potentially leaving bacteria in the presence of sub-inhibitory (sub-MIC) concentrations of secondary antibiotics. This led us to question whether the differences in collateral effects (Figure 2.4) correlate with fitness at low concentrations of secondary antibiotics, thus influencing the evolution of mutants when the concentration is below the MIC. Is the fitness of rifampicin-resistant mutants in the presence of sub-MIC concentrations of secondary antibiotics similarly influenced by GxE interactions?

To explore these questions, we measured the fitness of the mutants in all 3 nutrient environments in the presence of sub-MIC concentrations of 3 antibiotics – ciprofloxacin, D-cycloserine, and tetracycline. We selected half the concentration of the lowest MIC of any mutant (measured in each nutrient environment separately) as our sub-MIC.

We found inconsistent and unpredictable fitness of *rpoB* mutants in sub-MIC doses of secondary antibiotics (Figure 2.5; Figure 2.6A). Contrary to our hypothesis, we identified a significant relationship between MIC and fitness in sub-MIC of the same antibiotic in only two tested conditions (Linear mixed effect model: MH + Ciprofloxacin: $p < 0.005$; M9 + Tetracycline: $p < 0.05$). As such, the MIC of a mutant to a given secondary antibiotic generally did not correlate to its growth at sub-MIC concentrations of that antibiotic. Furthermore, the rank order of mutant fitness depended strongly on GxE interactions in sub-MIC concentrations of secondary antibiotics (Aligned ranks transformation ANOVA: Ciprofloxacin: $F_{22,108} = 3.43$, $p < 0.001$; D-Cycloserine: $F_{22,108} = 1.81$, $p < 0.05$; Tetracycline: $F_{22,108} = 4.16$, $p < 0.001$) (Figure 2.6B). However, growth in the presence of sub-MIC of a secondary antibiotic was significantly correlated with growth in the absence of antibiotics in 7/9 tested conditions (Figure 2.7). In other words, fitness at sub-MIC concentrations of secondary antibiotic was not significantly related to the mutant tolerance of that antibiotic (*i.e.*, its MIC), but was to its fitness in the absence

of antibiotics. It is important to note that these correlations may be different if sub-MIC concentrations used were determined independently for each strain, however in typical applications of antibiotics low concentrations are unlikely to be strain specific.

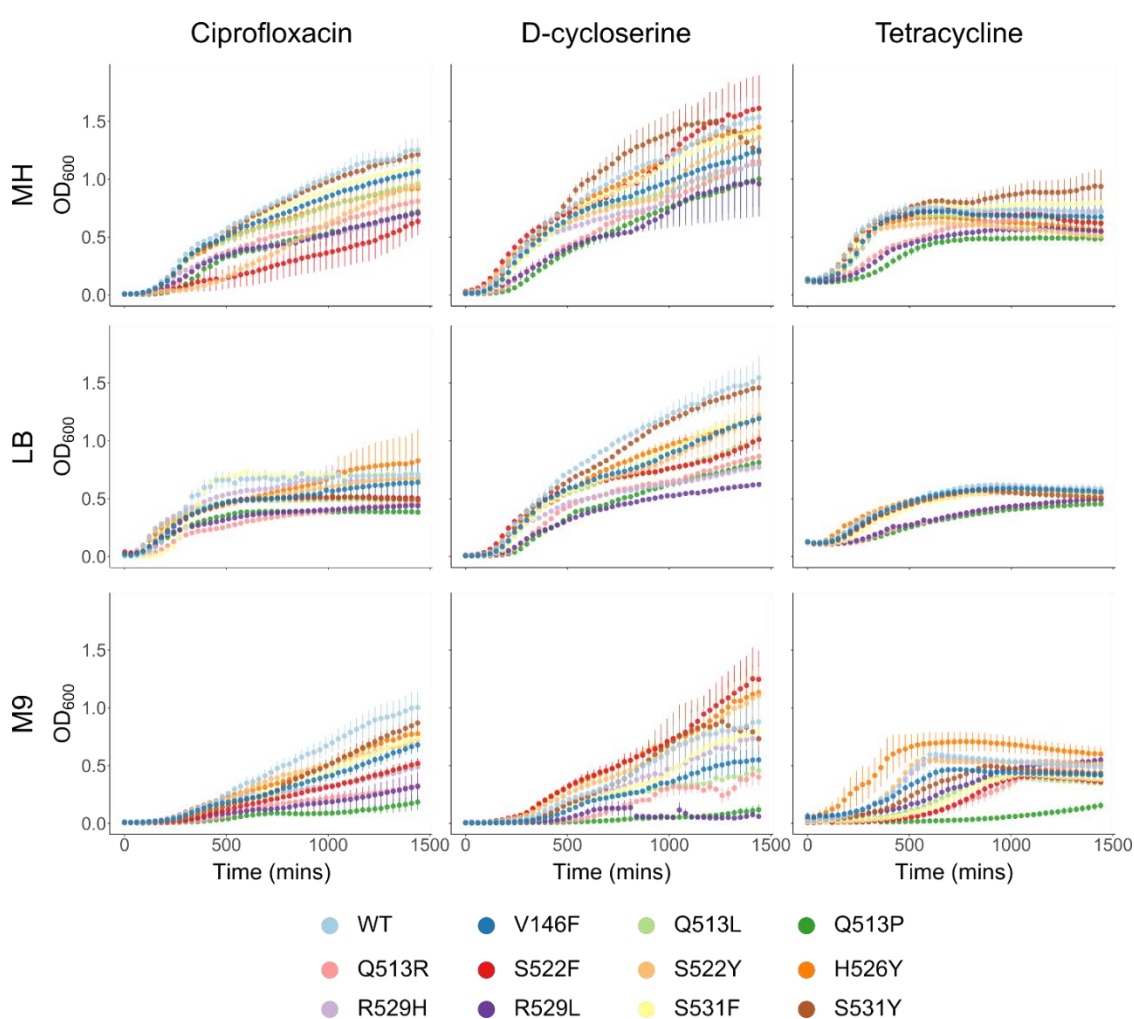


Figure 2.5 – Growth curves of *rpoB* mutants in 3 nutrient environments, in the presence of 3 secondary antibiotics at sub-MIC concentrations. Data are mean of 4 biological replicates; error bars are standard error of the mean.

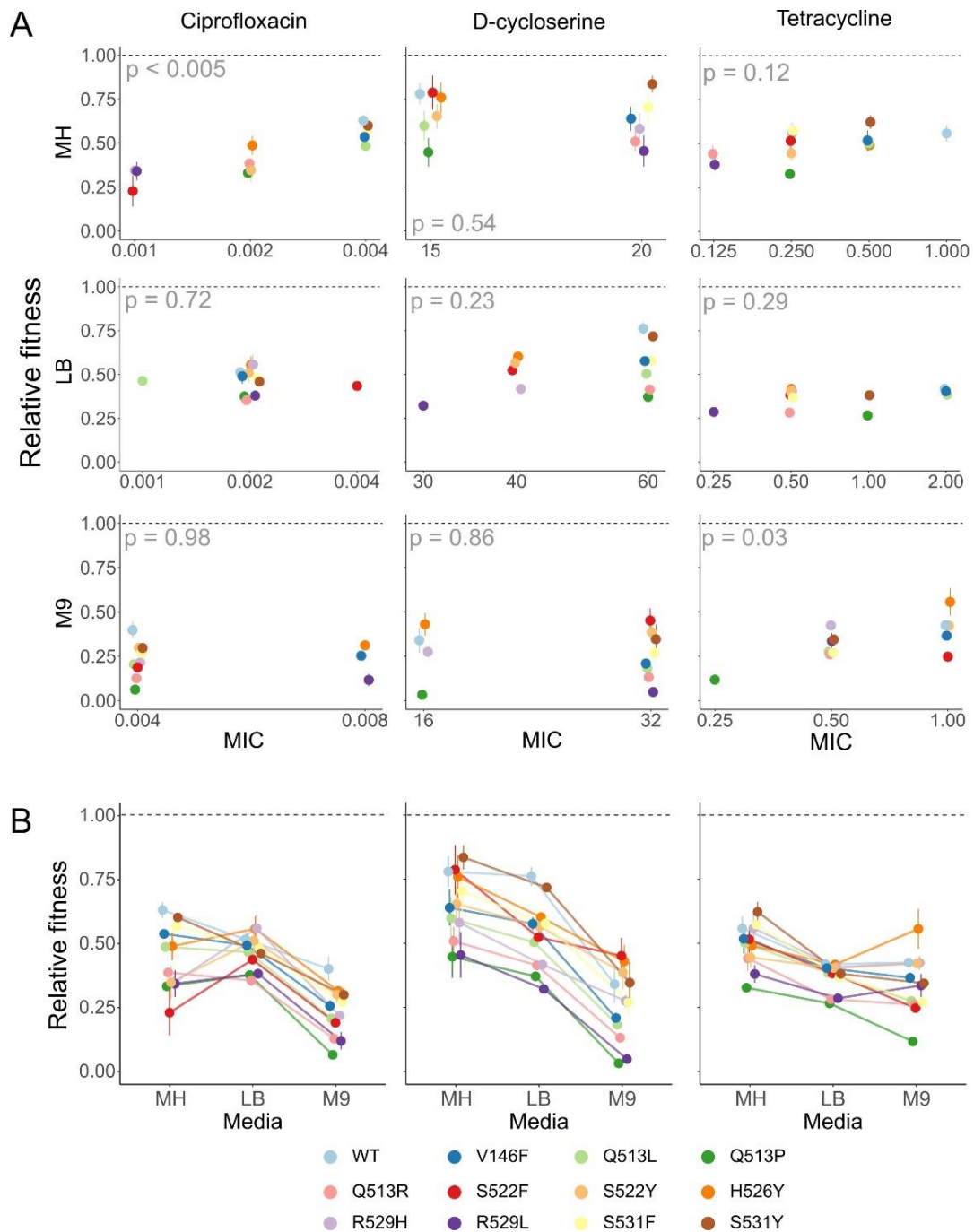


Figure 2.6 – GxE interactions shape fitness of *rpoB* mutants at sub-MIC concentrations of secondary antibiotics. (A) Relative fitness of mutants grown at sub-MIC concentrations of secondary antibiotics ciprofloxacin (left), D-cycloserine (middle) and tetracycline (right), in the 3 nutrient environments, plotted against the MIC of each mutant in the corresponding nutrient environment-antibiotic combination. P-values from a linear mixed-effects model are shown in grey. (B) Relative fitness of mutants in sub-MIC concentrations of secondary antibiotics in each nutrient environment, plotted to highlight GxE interactions. Points are mean of 4 biological replicates; error bars are standard error of the mean. Fitness values are area under the curve (AUC) values normalised to the AUC of the WT in antibiotic-free MH.

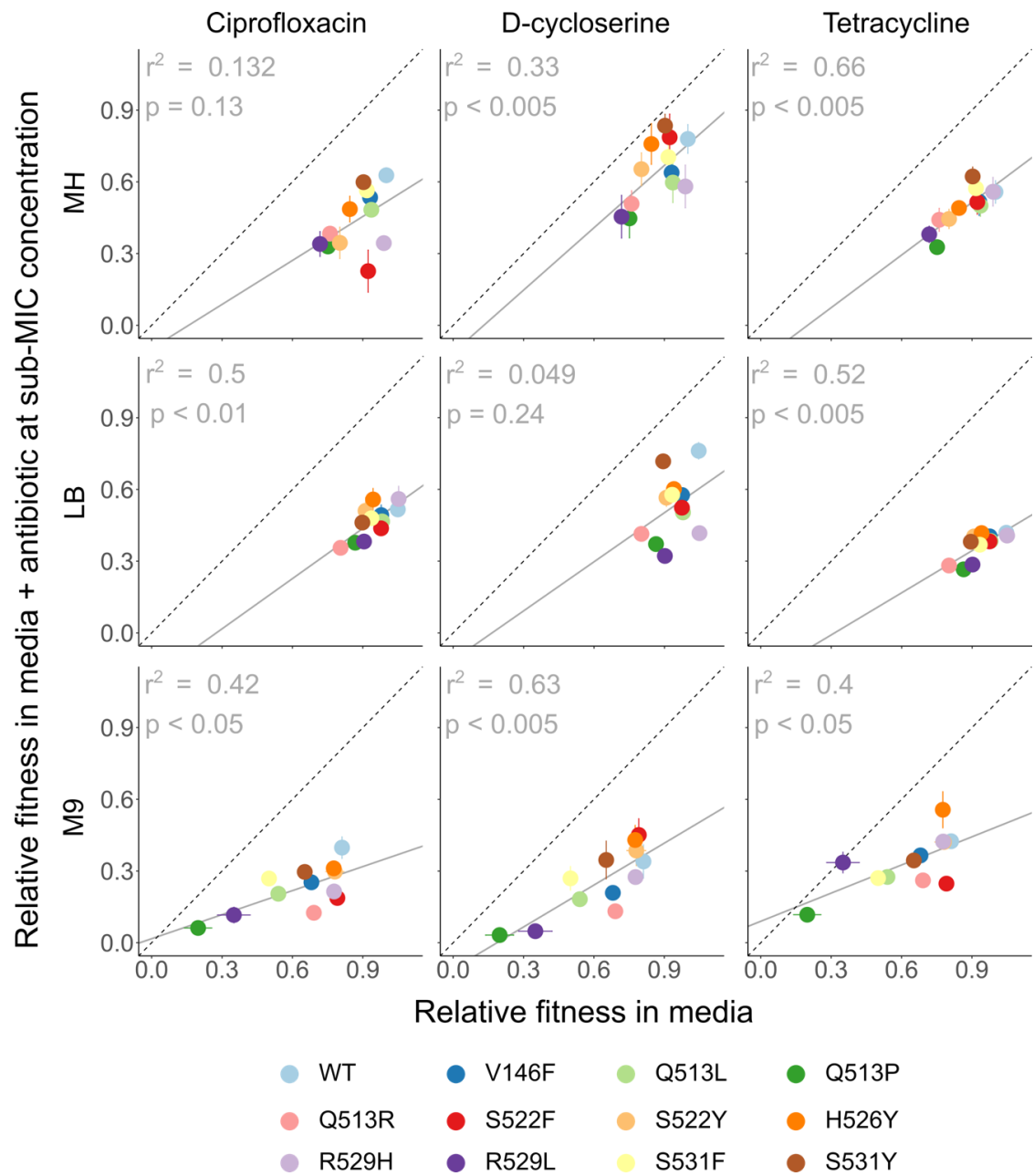


Figure 2.7 – Relative fitness of mutants in 3 nutrient environments plotted against relative fitness in media containing sub-MIC concentrations of secondary antibiotic. Points are mean of 5 biological replicates; error bars are standard error of the mean. Results of linear regression are shown in grey text and solid line, dotted line shows perfect correlation. Fitness values are area under the curve (AUC) values normalised to the AUC of the WT in antibiotic-free MH.

2.4.6 Pervasive pleiotropy as a possible source of complex Genotype-by-

Environment interactions

In this study, we identified complex ways in which GxE interactions shape fitness of rifampicin-resistant mutants in the presence (Figure 2.3A) and the absence of the primary drug (Figure 2.3B), as well as their collateral effects (Figure 2.4) and fitness in the presence of secondary antibiotics (Figure 2.6). Why does fitness of these mutants and their rank order depend so highly and so unpredictably on the nutrient and antibiotic environment?

Like most known rifampicin resistance mutations (Garibyan *et al.*, 2003), the 11 used in this study were point mutations in the *rpoB* gene coding for the β subunit of RNA polymerase. Mutations in *rpoB* can alter global gene expression profiles, leading to some or even many genes being over- or under-expressed (Utrilla *et al.*, 2016; González-González *et al.*, 2017; Meenakshi and Munavar, 2018; Trauner *et al.*, 2021). We explored whether such changes to global transcription levels were caused by the mutations we studied and if they were dependent on the nutrient environment, using three mutants (Q513L, Q513R, and H526Y). We measured their transcriptome using *RNA-Seq* in LB and M9 and compared it to the expression levels observed in the WT strain in each respective media.

The effects of rifampicin resistance mutations on the global transcriptome were often highly pleiotropic, significantly altering the expression levels of as many as several hundred genes (Figure 2.8A). The sets of genes that were over- and under-expressed in each environment were mutant-specific, meaning that each rifampicin resistance mutation resulted in a unique transcriptional response (Figure 2.8B). Similarly, the transcription profile of each mutant was different between nutrient environments (Figure 2.8A). Together, these results show that the pleiotropic effects of rifampicin resistance mutations on gene expression levels are often large, mutant-specific, and affected by the environment.

In some instances, changes to the expression levels of one or more specific genes can impact bacterial fitness in a given environment, by altering a key phenotype (Utrilla *et al.*, 2016). In general, we did not find such straightforward relationships between fitness measurements and individual key genes being over- or under-expressed. In one specific case (Q513R grown in LB) we found increased expression of the outer membrane porin

gene *ompF*. *ompF* deletion has been implicated in increased resistance to a range of antibiotics including D-cycloserine, tetracycline, and chloramphenicol (Choi and Lee, 2019). Increasing the uptake of these antibiotics through upregulation of *ompF* could explain the increased sensitivity of this mutant to chloramphenicol and tetracycline (Figure 2.4A). However, we observed no increased sensitivity to D-cycloserine, another substrate of this porin channel. Further, we found increased expression of *ompF* repressor *ompR* in H526Y grown in LB, but no concurrent decreases in sensitivity compared to the wildtype. We did not identify over- or under-expression of specific resistance genes to any of the secondary antibiotics used in this study. As such, the highly pleiotropic effects of *rpoB* mutations on transcriptomic profiles prevented us from drawing direct explanatory links as to how Genotype-by-Environment-by-Antibiotic interactions shape bacterial fitness. Instead, it is more likely that the dependence of fitness effects on nutrient composition stems from the cumulative effect of altering expression levels of many genes, especially as many of the genes with altered expression levels are involved in metabolism (Figure 2.8C).

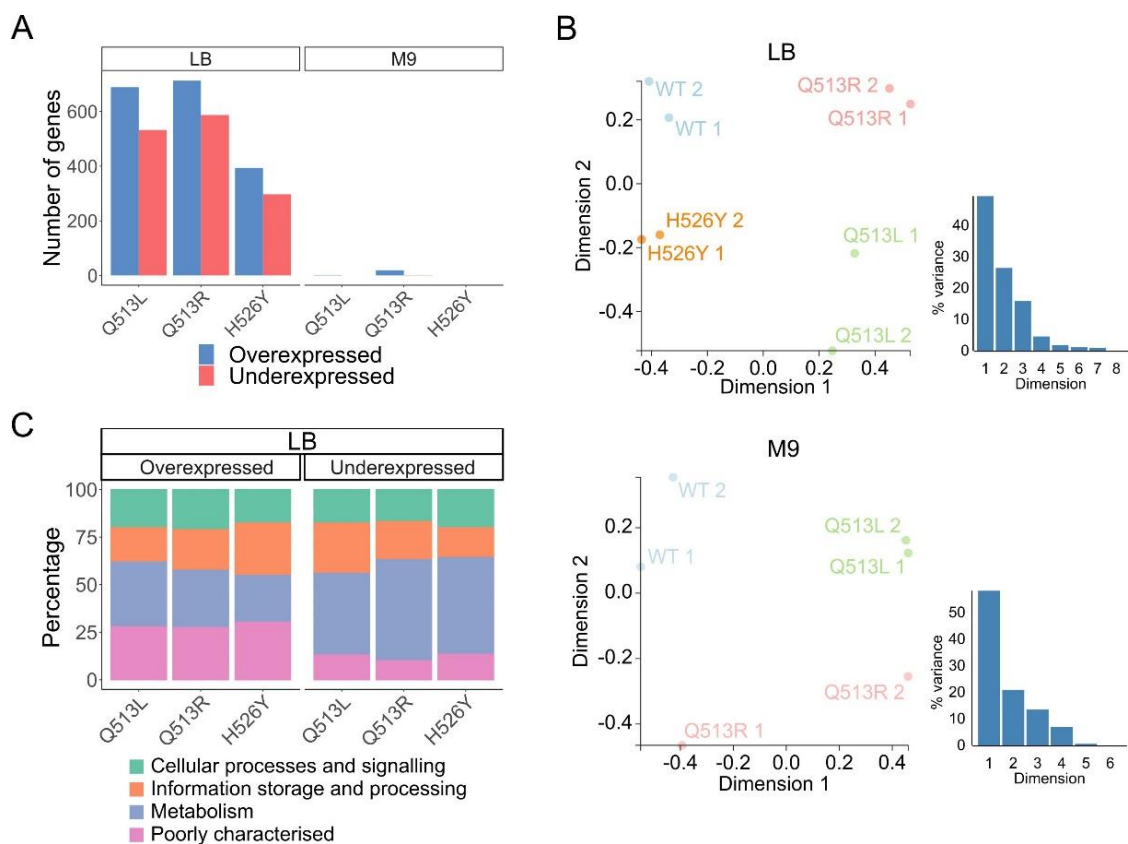


Figure 2.8 – Pleiotropic effects of *rpoB* mutations on the transcriptome. (A) Number of differentially expressed genes of 3 *rpoB* mutants grown in LB (left) or M9 (middle), as compared to the WT grown in the same media. Threshold for differentially expressed genes was a false discovery rate of $p < 0.05$ and absolute $\log_2$ fold change of > 1 . (B) Multidimensional scaling (MDS) plots of normalised expression (CPM) values of all genes in both LB and M9. This plot visualises the distance between samples based on the similarity of their gene expression profiles. H526Y in M9 is not shown as our analysis determined there were no differentially expressed genes. (C) Percentage of differentially expressed genes in LB divided into Clusters of Orthologous Genes (COG) categories. We do not provide this analysis for M9 as the number of differentially expressed genes was very low.

2.5 Discussion

The global challenge of antimicrobial resistance necessitates novel strategies in addition to ongoing efforts to develop new drugs (Ventola, 2015). Understanding the evolutionary forces that shape the emergence and spread of resistance, especially multi-drug resistance, underpins such key strategies. While much work has been put into describing resistance evolution and the dynamics of resistant populations (Furusawa, Horinouchi and Maeda, 2018), much less is known about how those dynamics depend on Genotype-by-Environment (GxE) interactions (Gifford, Moss and MacLean, 2016; Maharjan and Ferenci, 2017; Clarke *et al.*, 2020). In this study, we identified complex GxE interactions shaping the fitness effects of 11 highly pleiotropic *rpoB* mutations in 3 common laboratory nutrient environments, in the presence and absence of the primary antibiotic rifampicin or a secondary antibiotic with a different mechanism of action. The extent of fitness variation across environments was remarkably large. Perhaps most strikingly, we found that GxE interactions could alter the rank order of mutant fitness, while also affecting the direction and the strength of collateral effects against secondary antibiotics.

The fitness of a mutant in the presence of the selective antibiotic and in its absence plays a role in determining the evolutionary fate of a novel resistance mutation. As such, GxE interactions that alter mutant fitness between environments, and especially those that alter the rank order of fitness between mutants, result in more complex evolutionary dynamics. As one mutant can have a selective advantage (relative to other mutants) in one environment but a disadvantage in another, understanding and especially predicting the dynamics of resistance evolution becomes contingent on knowing fitness effects across a range of environments, not just in one. Though it has been previously observed that fitness in the presence or the absence of antibiotics can vary between environments (Martin and Lenormand, 2015; Gifford, Moss and MacLean, 2016; Maharjan and Ferenci, 2017; Lin *et al.*, 2018), this factor is often omitted in studies characterising resistance mutations and their costs (Mariam *et al.*, 2004; Ward, Perron and Maclean, 2009; Melnyk, Wong and Kassen, 2015; Pietsch *et al.*, 2017; Card, Jordan and Lenski, 2021). Indeed, surveying a sample of 300 papers published in 2019 that performed antimicrobial susceptibility testing on *E. coli* isolates showed that only 1.5% assessed resistance in multiple nutrient environments. While a further 4% partially considered environmental effects by observing viability in both liquid and agar cultures, it's clear that the vast majority of resistance characterisation occurs in only one nutrient context. For example, in a recent

study examining the extensive collateral effects of a large number of antibiotic-resistant *M. tuberculosis* strains (including several *rpoB* mutants), all MICs were tested in only a single nutrient environment (Waller *et al.*, 2023). Furthermore, MH media is often used as a ‘gold standard’ for MIC determinations owing to its stable pH and low concentrations of antibiotic inhibitors, and yet measurements of fitness and MIC in MH do not correlate to those made in LB or M9. Our study highlights how pervasive GxE interactions can be in shaping bacterial fitness, further emphasizing the need to experimentally measure fitness of resistant mutants across a range of environments. Nearly every surveyed paper referenced susceptibility testing guidelines published either by CLSI or EUCAST to describe their methodology; as neither yet mentions the advantages of examining multiple growth environments, updating these guidelines could provide an avenue for the rapid adoption of more rigorous screening protocols.

Environmental variation influencing resistance profiles in such an unpredictable manner is particularly pertinent in clinical settings, where bacterial infections can colonise a variety of microenvironments and ecological niches around the human body (Ciofu *et al.*, 2015; Weiser, Ferreira and Paton, 2018; Ailloud *et al.*, 2019). Exploiting collateral sensitivity between antibiotics has been proposed as a novel therapeutic strategy that could slow down resistance evolution by exposing strains carrying resistance to one drug to a second antibiotic with an associated increase in sensitivity (Imamovic and Sommer, 2013; Aulin *et al.*, 2021). However, complex GxE interactions that alter the direction of collateral effects between nutrient environments (Allen, Pfrunder-Cardozo and Hall, 2021), as well as different resistance mutations exhibiting different collateral sensitivity profiles (Barbosa *et al.*, 2017; Nichol *et al.*, 2019; Liakopoulos *et al.*, 2022; Waller *et al.*, 2023), pose a difficulty for implementing such treatment strategies. Whilst we tested lab media environments that are artificial and not necessarily representative of physiological growth environments, it is likely that such GxE interactions occur in more realistic environments as well. *E. coli* is implicated in a wide range of infection sites including the urinary tract, bloodstream, and gastrointestinal tract (Smith, Fratamico and Gunther, 2007), all of which have unique microenvironments; our data suggest that resistant mutants may not necessarily exhibit the same collateral effects to secondary antibiotics in each of these environments. For example, data from fosfomycin-resistant *E. coli* strains (with a variety of different resistance alleles) show that variation of the microenvironment even within the same bodily site can alter the MIC of strains in a genotype-dependent

manner (Martín-Gutiérrez *et al.*, 2017). Together with such previous studies, our data suggest that complex GxE effects on collateral effects might be common and complicate the design of resistance-proof therapeutic strategies.

The effects of rifampicin resistance mutations on global transcription levels likely underpin the pervasive GxE interactions we observed (Qi, Preston and Maclean, 2014; González-González *et al.*, 2017; Pietsch *et al.*, 2017; Meenakshi and Munavar, 2018). Indeed, the altered expression of 15 – 20 genes has been used to predict the levels of resistance to secondary stressors in evolved populations of *E. coli*, suggesting that altered expression of a few genes may be responsible for determining collateral effects (Horinouchi *et al.*, 2017). Which exact genes are responsible for influencing the sensitivity to secondary drugs may depend on the antibiotic's mechanism of action or pharmacological properties. Specific mechanisms that might give rise to collateral effects include diverse cellular processes such as cell membrane charge, cell wall thickness, metabolic networks, and SOS response (Cui *et al.*, 2010; Roemhild, Linkevicius and Andersson, 2020). This diversity, in turn, makes predictions of collateral effects challenging – difficulties which are exacerbated by the fact that different *rpoB* mutations affect the transcriptome in different (mutant-specific) ways (Qi, Preston and Maclean, 2014; Pietsch *et al.*, 2017). Which mechanisms of resistance are associated with highly pleiotropic effects, and whether resistance that is not associated with pleiotropic effects on transcription can also give rise to complex GxE interactions on fitness and collateral effects, remains to be explored (Allen, Pfrunder-Cardozo and Hall, 2021).

When studying the factors that shape antibiotic resistance evolution, most studies rely on characterising the fitness of resistance mutations under a range of antibiotic and genetic conditions. However, much of these data are generated from experiments in a single laboratory medium. Similarly, resistance in clinical isolates is typically determined through growth on a single nutrient medium in the presence of a fixed dose of an antibiotic (Khan, Siddiqui and Park, 2019). Our findings demonstrate pervasive GxE interactions that result in large variability in fitness of rifampicin resistance mutants across different nutrient environments and their ability to tolerate secondary antibiotics. These effects, while likely linked to the pleiotropic effects of mutations in *rpoB* on global transcription, were unpredictable. Together with previous studies that identified pervasive GxE effects on fitness of resistance mutants (Hall, 2013; Gifford, Moss and MacLean, 2016; Clarke *et al.*, 2020; Allen, Pfrunder-Cardozo and Hall, 2021), our work points to the need for

caution when extrapolating fitness estimates across environments and identifies the need to re-evaluate how we assess and evaluate resistance in experimental and clinical settings.

3 Exploring the effect of temperature on pleiotropic mutations in RNAP

The contents of this chapter have been compiled as a manuscript in preparation for publication.

3.1 Abstract

Despite occurring in a highly conserved molecule critical to an essential processes of life, RNA polymerase mutations are common in bacteria where they give rise to antimicrobial resistance and elicit complex pleiotropic effects across different levels of organisation. The mutational effects at each of these levels—sequence, molecular, functional, transcriptomic, and organismal—are also modulated by environment. While the effects at individual levels have been studied previously, we take some of the first steps in connecting the effects of mutations between different levels of biological organization to ask how they are affected by an environmental factor – temperature. The transcriptomic response to temperature change was highly genotype specific: no differentially expressed genes upon temperature change were shared between all mutants. These transcriptomic differences could, in part, be explained by the differences in the effects of mutations on the molecular level, as they had different distance to sigma factors and template DNA. The transcriptomic differences drove the effects of mutations on organismal level, affecting key traits such as growth rate and biofilm formation. We argue that understanding how the effects of mutations at one level of biological organization propagate to other levels is critical for predicting how mutations alter organismal fitness, especially in targets with highly pleiotropic consequences.

3.2 Introduction

RNA polymerase (RNAP) is essential for all life, as its function is to transcribe DNA into RNA. In bacteria, one sole RNAP is responsible for the transcription of all genes (Darst, 2001). Because of its essentiality, bacterial RNAP is a desirable target for novel antimicrobials, especially in an era of increasing antibiotic resistance and slowing rates of antibiotic development (Hutchings, Truman and Wilkinson, 2019). However, resistance to RNAP-targeting antibiotics can arise through spontaneous mutations in subunits that constitute RNAP. For example, the drug rifampicin (which is commonly used in the treatment of tuberculosis) can be rendered almost completely ineffective through one of many single substitution mutations in the *rpoB* gene encoding the beta subunit of RNAP (Jin and Gross, 1988; Goldstein, 2014). Studying the consequences of such mutations is important not only for improving our ability to predict and control the evolution of resistance to antibiotics like rifampicin, but also to further our understanding of the structure and function of one of the most important and well conserved molecules, RNAP.

Mutations in genes that constitute RNAP can have widespread pleiotropic effects on bacterial phenotypes and fitness (Qi, Preston and Maclean, 2014; Trauner *et al.*, 2021). The most direct consequence of RNAP mutations, which lies at the heart of their pleiotropy, is the differential expression of a large number of genes. Differential gene expression as a consequence of RNAP mutations can interfere with diverse biological systems such as the production of antibiotics and metabolites, decreased motility, salt tolerance, and host immune responses (Perkins and Nicholson, 2008; Tanaka *et al.*, 2013; Cai *et al.*, 2017; Cutugno *et al.*, 2020; Bobba *et al.*, 2023). The complex consequences of single point mutations, like those in *rpoB*, occur at several levels of biological organisation – at the molecular level (i.e. modifying protein structure), at the cellular level (RNAP function and transcriptomic dysregulation), and at the organismal level through changes to fitness (Figure 3.1) (Bershtein *et al.*, 2015; Utrilla *et al.*, 2016). While many of the consequences of *rpoB* mutations have been previously studied, they are typically studied independently of each other. As such, we lack the understanding of how changes at one level of biological organisation impact changes at other levels, preventing predictions of mutational effects and their consequences for evolution.

Understanding the relationship between different levels of biological organisation impacted by mutations in RNAP (Figure 3.1) is further complicated as the consequences of *rpoB* mutations can also be dependent on the environment (Song *et al.*, 2014; Gifford, Moss and MacLean, 2016; Hinz *et al.*, 2024). For example, mutations conferring a low fitness cost in one environment can be more costly in another and vice versa (Soley *et al.*, 2023). Environmental variations such as nutrient content, temperature, and presence of antibiotics, could occur within a patient during infection and so are likely to play a role in the evolution of clinical strains. While previous work has explored how the fitness of *rpoB* mutants is altered in response to environmental fluctuations, the mechanisms underlying these responses remain poorly understood (Yubero and Poyatos, 2021; Choudhury *et al.*, 2023).

Here, we explore how one environmental variable, temperature, affects the consequences of 6 *rpoB* mutations in *Escherichia coli*. A range of mutations were selected that included different amino acid sites, a pair of mutations affecting the same amino acid site, and one mutation outside of the rifampicin resistance determining region. We characterise mutational effects at the different levels of biological organisation at two different temperatures: 37°C, considered the optimum growth temperature for *E. coli*, and 42°C, to explore how heat stress influences the effects of *rpoB* mutations, attempting to find explanatory links between the levels of biological organisation. In doing so, we also provide a template on how to approach studying the consequences of mutations with complex, pleiotropic effects – by dissecting mechanistically how and why mutations alter molecular function, phenotypes, and fitness. Such comprehensive understanding of mutational effects across levels of biological organization lends itself to generalizing beyond the specifics of a given system and improve the predictability of evolution at a larger scale.

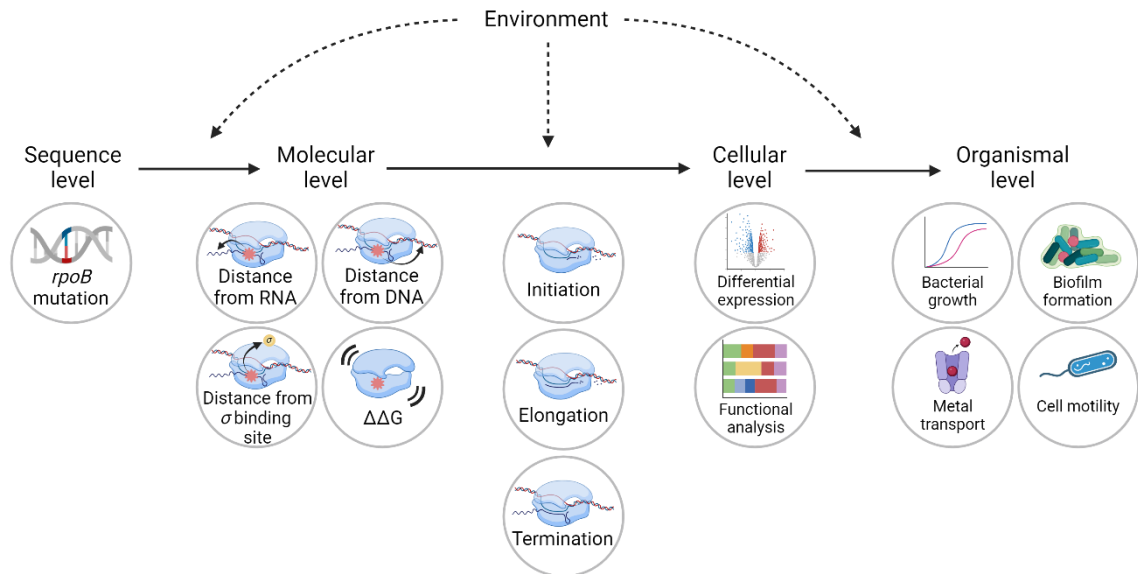


Figure 3.1 – A holistic approach to understanding the effects of mutations with complex pleiotropic effects across different levels of biological organization. We illustrate and study some of the major consequences of *rpoB* mutations. Created with BioRender.com.

3.3 Materials and methods

3.3.1 Bacterial strains

All mutants used in this study were evolved from an ancestral WT strain - *E. coli* BW25113 $\Delta tolC$ from the Keio collection (Baba *et al.*, 2006). The *tolC* deletion was used to ensure the selection of *rpoB* mutations and avoid resistance through efflux pump activity as many efflux mechanisms require TolC to function. *rpoB* mutants were selected with a fluctuation assay-based approach and the presence of a single mutation was validated with WGS as described in (Soley *et al.*, 2023).

3.3.2 Predictions of molecular level effects

Protein structures used for computational calculations of molecular level effects were downloaded from the Protein DataBank (PDB; RCSB.org) (Berman *et al.*, 2000). The PDB accession ID of all structures used can be found in Table 3.1. To estimate distances from each mutation to key functional sites within RNAP, PDB structures were imported to R with the *bio3d* package (Grant *et al.*, 2006) then distances were calculated from C-

alpha of the mutated residue to every C-alpha on the relevant site or subunit (RpoC, sigma binding site, template/non-template DNA strand, nascent RNA strand). Distances were sorted and the shortest distance for each site was used. Gibbs energy of protein folding values ($\Delta\Delta G$) for mutated RpoB were predicted in FoldX (Schymkowitz *et al.*, 2005) using the *BuildModel* command. A one-way ANOVA was performed to compare the effect of different *rpoB* mutations on each molecular characteristic.

Table 3.1 – Protein structures used for predicting molecular effects of *rpoB* mutations.

PDB ID	Structure notes	Where structure is used	Reference
4KMU	In complex with rifampicin	Figure 3.2A	(Molodtsov <i>et al.</i> , 2013)
6UU9	Sigma-S, initiation complex, 8-nt nascent RNA	Figure 3.2B Figure 3.2C (DNA, RNA, β' , σ binding site) Figure 3.4E	(Zuo <i>et al.</i> , 2020)
6PMJ	Sigma-28, initiation complex, 3-nt nascent RNA	Figure 3.2C (DNA, β' , σ binding site) Figure 3.4D,E	(Shi <i>et al.</i> , 2020)
8HKC	Sigma-32, initiation complex	Figure 3.2C (DNA, β' , σ binding site, $\Delta\Delta G$)	(Lu <i>et al.</i> , 2023)
5IPL	Sigma-S, initiation complex, 4-nt nascent RNA	Figure 3.2C (DNA, RNA, β' , σ binding site) Figure 3.4D,E	(Liu, Zuo and Steitz, 2016)
6GFW	Sigma-54, initiation complex, 4-nt nascent RNA	Figure 3.2C (DNA, β' , sigma binding site)	(Glyde <i>et al.</i> , 2018)
6JBQ	Sigma-24, initiation complex, 5-nt nascent RNA	Figure 3.2C (DNA, β' , sigma binding site)	(Fang <i>et al.</i> , 2019)
6N62	Sigma-70	Figure 3.2C (RNA, β' , σ binding site, $\Delta\Delta G$)	(Braffman <i>et al.</i> , 2019)
7MKP	RNAP core enzyme	Figure 3.2C ($\Delta\Delta G$)	(Qayyum <i>et al.</i> , 2021)
6ALH	Elongation complex	Figure 3.2C ($\Delta\Delta G$)	(Kang <i>et al.</i> , 2017)
6C9Y	Sigma-70	Figure 3.2C ($\Delta\Delta G$)	(Narayanan <i>et al.</i> , 2018)

When calculating correlations between molecular and transcriptomic effects, we used PDB structure 6UU9 for distances to RNA as it had the longest RNA chain; distances to RpoC were predicted from the same structure as there was little variation between structures (Figure 3.2). Distances to sigma factor binding site, and both template and non-template DNA strands were used from multiple PDB structures and tested separately for correlations, as these distances varied between structures (Figure 3.2). When testing correlations with $\Delta\Delta G$, the mean $\Delta\Delta G$ of all structures was used. Correlations were calculated in R with the *lm* command, using default settings. All interactions with an adjusted $r^2 \geq 0.5$ were plotted, however no interactions were statistically significant after Bonferroni correction.

3.3.3 Measurement of transcriptomic effects

We performed RNA-Seq analysis on 6 *rpoB* mutants and the wildtype at two temperatures (37°C and 42°C). RNA was extracted as in (Soley *et al.*, 2023), then samples were sent to Azenta Life Sciences (Germany) for quality control, rRNA depletion, library preparation and sequencing with the Standard RNA-Seq service (Illumina NovaSeq, 150 bp paired end reads). Raw sequencing reads were trimmed with *Trim Galore* (v0.6.7) (<https://github.com/FelixKrueger/TrimGalore>) using default parameters to remove sequencing adapters and low quality regions. Reads shorter than 50 base pairs after trimming were then discarded. Any remaining rRNA reads were removed with *BBDuk* (v38.18) (<https://sourceforge.net/projects/bbmap>) using the bundled rRNA k-mers database. Trimmed mRNA reads were aligned to the reference genome (CP009273) using *Bowtie2* (v2.2.5) (Langmead and Salzberg, 2012), then transcript count tables were constructed using *featureCounts* (v2.0.1) (Liao, Smyth and Shi, 2014). Untransformed counts were adjusted for batch effects based on negative binomial regression with the *ComBat-seq* tool available in R/Bioconductor package *sva* (v3.48.0) (Zhang, Parmigiani and Johnson, 2020). Differential expression analysis was performed with *DESeq2* (v1.40.1) (Love, Huber and Anders, 2014) to all genes with rowSums ≥ 10 in the adjusted count table. An interaction term between mutation and temperature was included in the DESeq design formula to allow comparisons of each mutant at 42°C versus 37°C. Significantly differentially expressed genes were defined as genes with a log-fold change > 1 at an adjusted P value for false discovery rate < 0.05 . PCA plot in Figure 3.3 was

constructed with the *plotPCA* function in *DESeq2*, using *rlog* transformed data for variance stabilising. UpSet plots in Figure 3.4 were created with *UpSetR* (Lex *et al.*, 2014; Conway, Lex and Gehlenborg, 2017) with intersections of 5 or more genes. Functional analysis of differentially expressed genes was performed using Gene Set Enrichment Analysis (GSEA) with KEGG terms (Kyoto Encyclopaedia of Genes and Genomes, (Kanehisa *et al.*, 2023)), using the R/Bioconductor package *clusterProfiler* (v4.8.2) (Wu *et al.*, 2021).

3.3.4 Measurements of organismal level effects

3.3.4.1 Growth curves

Growth curves were obtained using a CLARIOstar Plus microplate reader. Strains were grown overnight at 37°C in LB inoculated from frozen stocks, then diluted 1:1000 into fresh media in microtiter plates. Plates were incubated in the microplate reader for 24 hours with shaking, either at 37°C or at 42°C, and OD₆₀₀ was measured every 10 minutes. 12 biological replicates were taken for each mutant in each condition. Growth curves were analysed in R with the package *growthcurver* (Sprouffske and Wagner, 2016) to calculate the empirical area under the curve between 0 and 24 hours.

3.3.4.2 Biofilm formation

Strains were grown overnight in LB media, then diluted the following morning 1:100 into fresh media. 100 µL of diluted culture was added per well into 96 well plates, one for growth at 37°C and one for growth at 42°C, with 6 replicates per sample. Plates were incubated at the relevant temperature for 4 hours without shaking. After incubation, plates were tipped to remove cells, then washed gently twice by gentle submersion in water. 125 µL of a 0.1% crystal violet solution was added to each well, before incubation at room temperature for 15 minutes. Plates were then washed 4 times with water as before then dried upside down overnight. To solubilize the crystal violet, 125 µL of 30% acetic acid was added to each well and plates were incubated at room temperature for 15 minutes.

Solubilized crystal violet was transferred to a new flat bottom microtiter plate, then absorbance at 550nm was quantified in a CLARIOstar plus microplate reader.

3.3.4.3 ABC transporter activity

Strains were grown overnight at 37°C in LB, then diluted the following morning 1:100 into 20 mL fresh media and grown to stationary phase at either 37°C or 42°C with shaking (200 RPM), with 3 biological replicates per sample. 2 mL of culture was taken for serial dilution and CFU counts. The remaining culture was centrifuged (3660 RPM, 40 minutes) and pellets resuspended in 10 mL TE buffer, then repeated for a total of 3 washes. After final wash, pellets were resuspended in 10 mL mQH₂O, spun again (3660 RPM, 10 minutes), and supernatant was removed before drying pellets in 80°C oven overnight (with tube lids removed). The following morning, 0.5 mL 70% Nitric acid was added to each tube, lids were replaced with parafilm to reduce evaporation, and samples were incubated overnight at 37°C with shaking (200 RPM) to dissolve pellets. On the final day, samples were diluted in a final volume of 10 mL mQH₂O and filter sterilised, then sent for analysis by Inductively Coupled Plasma Mass Spectrometry (ICP-MS) at the Manchester Analytical Geochemistry Unit for detection of nickel parts per billion (ppb).

For each organismal level effect, the propagation of error was calculated for the difference between 42°C and 37°C (σ_f), where σ is standard deviation:

$$\sigma_f = \sqrt{\sigma_{37^\circ\text{C}}^2 + \sigma_{42^\circ\text{C}}^2}$$

One-way ANOVA was then performed to test the significance between different mutants (Figure 3.6B,C,D; bottom plots).

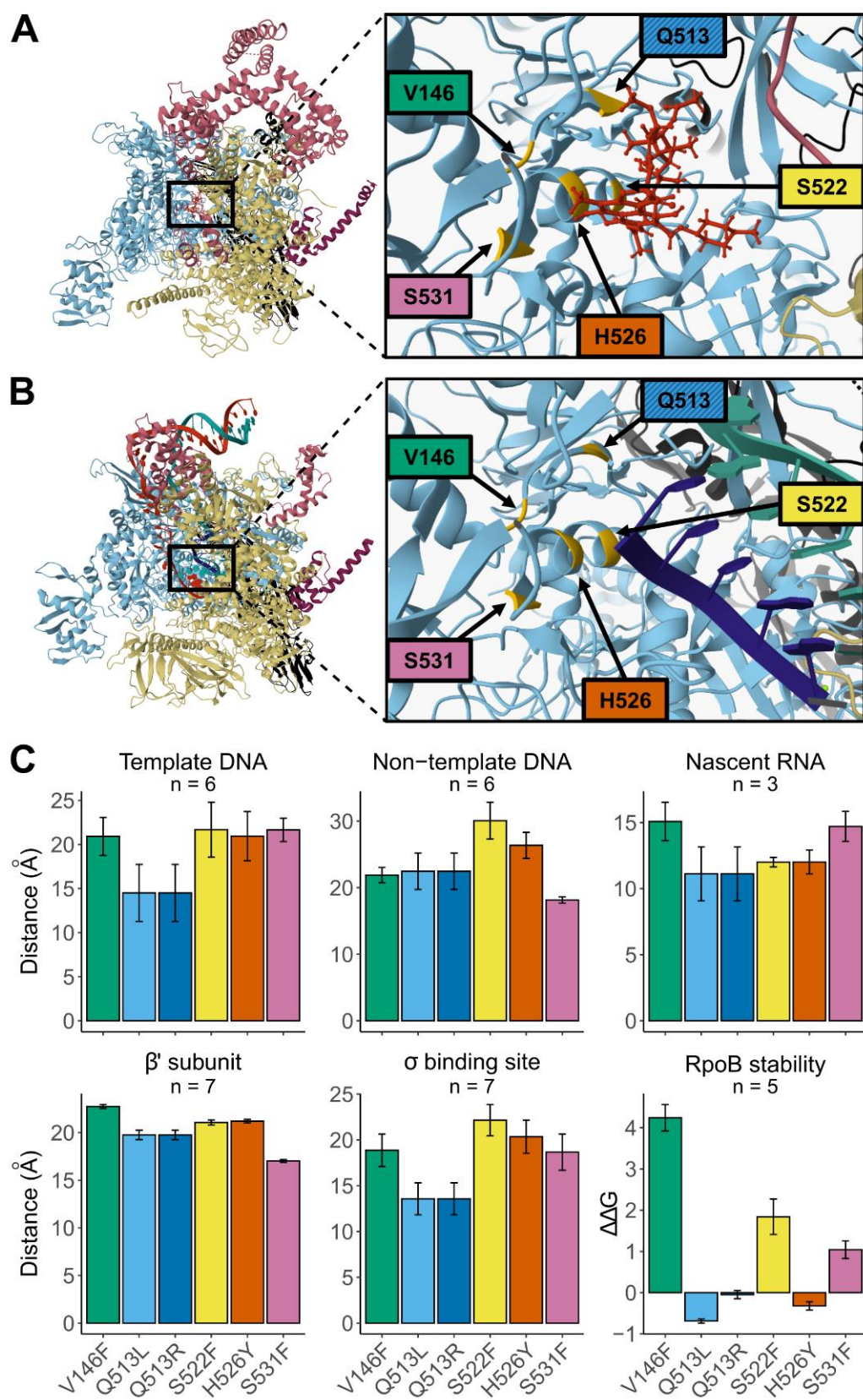
3.4 Results

To investigate how temperature influences the effects of *rpoB* mutations, we selected 6 mutants each with a single point mutation in *rpoB* conferring resistance to rifampicin, resulting in the following amino acid substitutions: V146F, Q513L, Q513R, S522F, H526Y, and S531F (Figure 3.2A,B). Strains were generated with a fluctuation assay-based approach and the presence of a single mutation was confirmed with whole genome sequencing, as previously described in (Soley *et al.*, 2023). Genomes of the six strains were otherwise identical to each other and that of the wildtype.

3.4.1 Linking molecular to cellular effects

We first predicted computationally how each *rpoB* mutation affected structural characteristics of RNAP complex, including the distance of each mutated residue from several regions critical for transcription, and how each mutation altered the structural stability of RpoB. RNAP structures were downloaded from the RCSB Protein DataBank for predictions of molecular level effects of mutations (see *Chapter 3.3.2 – Predictions of molecular level effects*). In RNAPs quaternary structure, mutated residues sit within the RNA exit channel, at the site of rifampicin binding (Figure 3.2A,B). We calculated the shortest distance between each mutation and β' subunit (RpoC), σ factor binding site, nascent RNA, and DNA (both template and non-template strand) in several different PDB structures of RNAP (Table 3.1). We relied on multiple different RNAP structures to capture the complex bound to different σ factors and in different structural conformations during initiation and elongation. The distance between mutated residues and the non-template strand of DNA, β' , and σ factor binding site, differed significantly between *rpoB* mutants ($F_{5,30} = 3.64$, $p = 0.01$ (non-template DNA); $F_{5,36} = 34.14$, $p < 0.001$ (RpoC); $F_{5,36} = 3.94$, $p < 0.01$ (σ binding site)) (Figure 3.2C). We found no significant difference in the distance to template DNA, or to RNA, between mutants ($F_{5,30} = 1.66$, $p = 0.17$ (template DNA); $F_{5,12} = 1.48$, $p = 0.27$ (RNA)) (Figure 3.2C). We hypothesise that mutations closer to these key sites are likely to exert a larger effect on RNAP function and therefore have a greater influence on fitness – a hypothesis we will test in relation to transcriptomic and organismal level effects of *rpoB* mutations in subsequent sections.

Figure 3.2 – Molecular characteristics of *rpoB* mutations. (A) Proximity of mutated residues (yellow) in RpoB (cyan) to rifampicin (red) in RNA polymerase quaternary structure (light yellow: β' subunit, black: α subunit, purple: ω subunit, pink: σ factor; PDB ID: 4KMU). (B) Same as (A), but showing proximity to DNA and RNA during transcription (dark blue: nascent RNA, teal: template DNA strand, red: non-template DNA strand; PDB ID: 6UU9). (C) Distances between mutations and several structural regions critical for transcription, and predicted stability of mutant RpoB (bottom right). Data are means of predictions from different PDB structures (see *Chapter 3.3 – Materials and methods*), error bars show standard error of the mean. Colours correspond to mutant labels in A/B.



Another molecular phenotype that may influence the function of RNAP is protein stability. We estimated the stability of RpoB in each mutant by calculating $\Delta\Delta G$ values: comparing the change in Gibbs free energy between the unfolded and folded state of the wildtype protein with the equivalent change in energy of the mutated protein. Positive $\Delta\Delta G$ values indicate the mutation causes destabilisation of the protein, whilst negative values suggest the mutation is stabilising. We found that $\Delta\Delta G$ varied significantly between mutants ($F_{5,24} = 9.49$, $p < 0.001$); 3 mutants (V146F, S522F, S531F) appear to destabilise RpoB whilst 3 mutants (Q513L, Q513R, H526Y) had negative $\Delta\Delta G$ values (Figure 3.2C). Q513R was predicted to have a negligible impact on protein stability, with a $\Delta\Delta G$ very close to 0.

Our molecular predictions suggest that different *rpoB* mutations result in significant differences in key molecular phenotypes of the β subunit and RNAP. While our aim is to specifically investigate how temperature influences these mutation effects, predicting the effects of temperature on specific molecular phenotypes remains challenging. Therefore, as we investigate the link between the molecular and the transcriptomic/organismal consequences of *rpoB* mutations, we will assume that any temperature effect on molecular phenotypes is comparable between mutants and that the rank order we observed (Figure 3.2) is maintained.

To explore the cellular level effects of *rpoB* mutations, we performed transcriptomic profiling of each mutant at 37°C and 42°C using RNAseq. This allowed us to make several comparisons – the *mutation* effect when grown at 37°C and 42°C (i.e., differentially expressed genes (DEGs) in each mutant when compared to the wildtype at each temperature), and the *temperature* effect of each mutant (e.g., DEGs in V146F grown at 42°C versus in V146F grown at 37°C). The mutation effect when grown at either 37°C or 42°C was highly genotype-dependent; in other words, the number of genes differentially expressed varied depending on the specific mutation (Figure 3.3A,B). On average, more genes were downregulated at 37°C than 42°C, with the inverse true for upregulated genes (upregulated genes: mean at 37°C = 41; mean at 42°C = 44; downregulated genes: mean at 37°C = 84; mean at 42°C = 63; total DEGs: mean at 37°C = 122; mean at 42°C = 105). Principal component analysis of RNA-Seq counts revealed strong separation of samples by temperature with principal components 1 and 2 (Figure 3.3C).

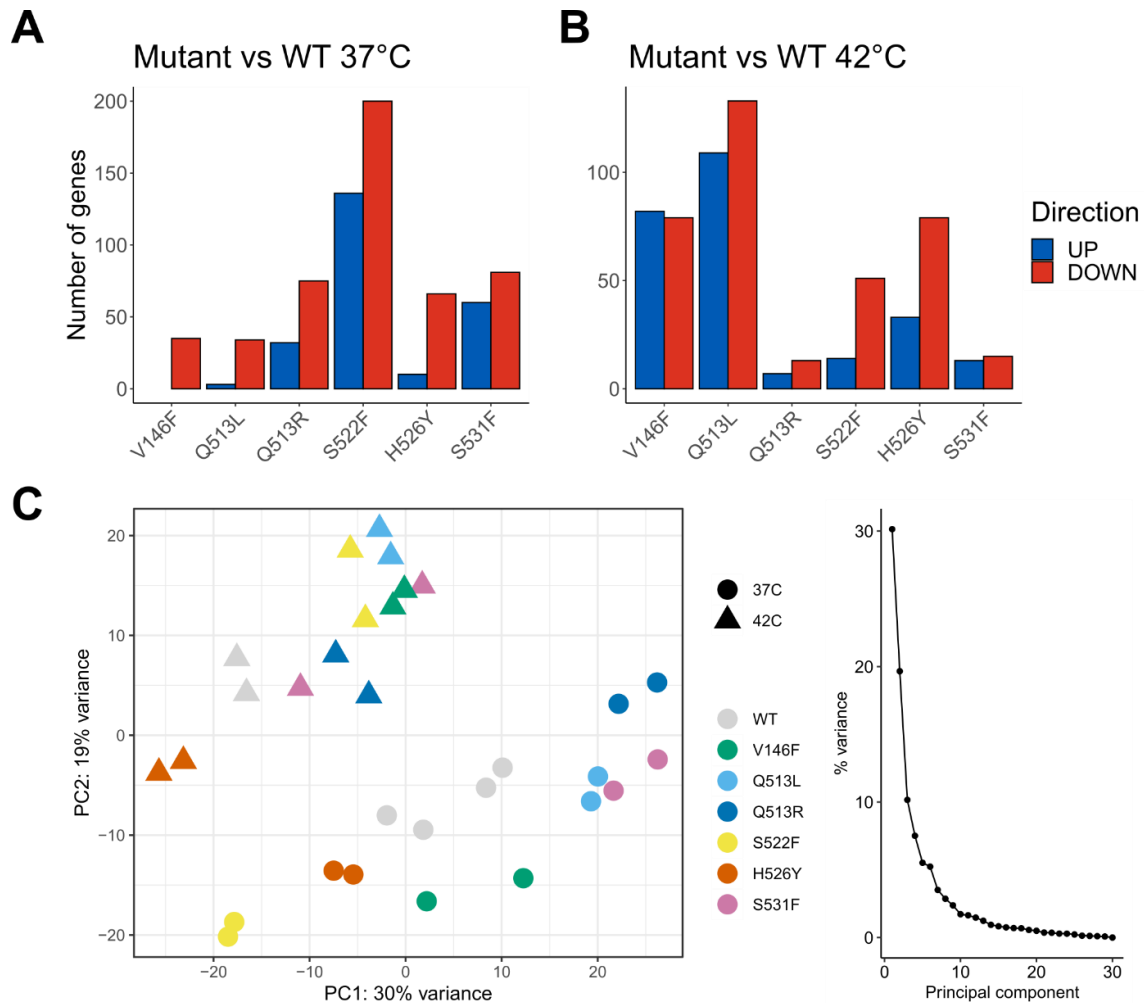


Figure 3.3 – Cellular-level phenotypic effects of *rpoB* mutations are temperature-dependent. (A) Number of differentially expressed genes in each *rpoB* mutant growing at 37°C. (B) Number of differentially expressed genes in each *rpoB* mutant growing at 42°C. (C) Principal component analysis (PCA) of regularised log transformed normalised counts, showing Principal Component 1 vs 2 (left) that differentiate by mutant and temperature; scree plot showing percent variance of each principal component (right). Differentially expressed genes defined as genes with log-fold change > 1 at an adjusted P value for false discovery rate < 0.05 (see Chapter 3.3.3 – Measurement of transcriptomic effects).

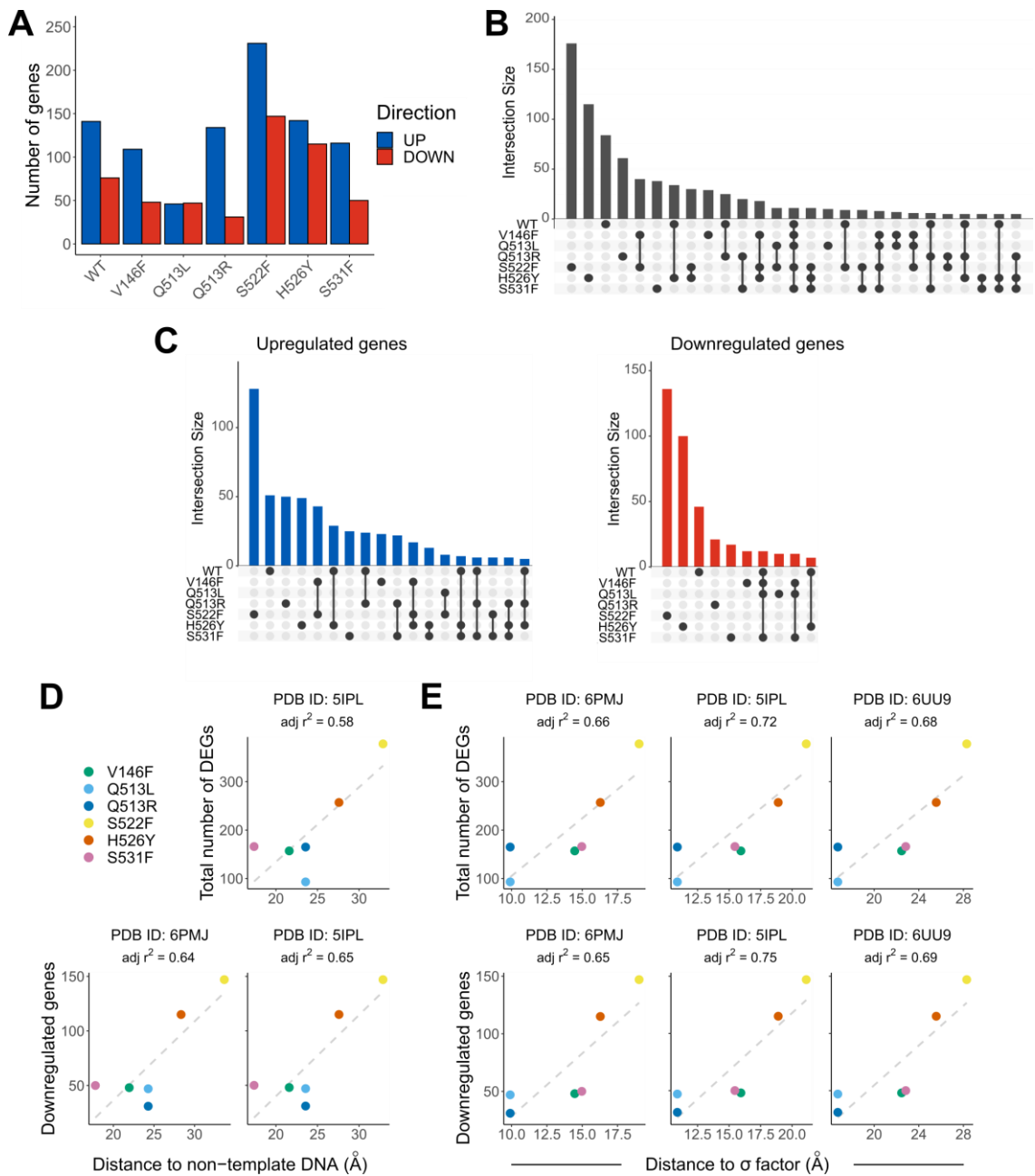


Figure 3.4 – Cellular-level phenotypic response of *rpoB* mutants to change in temperature is genotype-dependent. (A) Number of differentially expressed genes (DEGs) in each mutant and the wildtype when grown at 42°C versus 37°C. (B) UpSet plot showing the number of unique and overlapping DEGs. Intersections with less than 5 genes removed to improve legibility. (C) Same as (B) but separated by upregulated (top) and downregulated (bottom) genes. (D) Correlations between total DEGs or number of downregulated genes and the distance between each mutated residue and the non-template strand of DNA. (E) Correlations between total DEGs or number of downregulated genes and the distance between each mutated residue and the sigma binding site. Grey lines in D and E show results of linear regression analysis; only those interactions with an adjusted $r^2 \geq 0.5$ are shown.

Given the effect of nutrient environment on transcription in *rpoB* mutants (Soley *et al.*, 2023), we were particularly interested in how temperature influences the transcriptomic effects of a mutation. Thus, we next focused on the temperature effect of each mutant. The number of genes differentially expressed at 42°C vs 37°C was also dependent on the *rpoB* mutation: the total number of DEGs ranged from 93 (Q513L) to 378 (S522F) (Figure 3.4). In all mutants except Q513L, more genes were upregulated than downregulated at 42°C vs 37°C (upregulated mean = 131; downregulated mean = 73.4). Interestingly, a wide number of genes were dysregulated across all six mutants, with the total of 567 unique upregulated and 401 unique downregulated genes. In addition to the number of genes, we found that the specific genes that were differentially expressed were also genotype-dependent (Figure 3.4B,C) – the majority of genes were differentially expressed in only 1 mutant (328/567 upregulated genes; 342/401 downregulated genes). This meant that only a minority of genes were affected by more than one mutation, with no genes differentially expressed in all mutants (Figure 3.4C). This is surprising because we expected a portion of the differentially expressed genes to be shared across all mutants and the wildtype as a general response to temperature change (Li *et al.*, 2018). Instead, our data suggests that *rpoB* mutations substantially alter the transcriptomic response to temperature change in a genotype-specific manner.

We asked if the predicted molecular consequences of *rpoB* mutations affected their transcriptomic effects by correlating the changes to molecular and transcriptomic levels of organization. We found no molecular-cellular relationships at 37°C nor at 42°C, indicating that the effects of mutations on transcriptomes at a given temperature were not driven by any of the molecular changes we predicted. When examining the temperature effect on DEGs, we observed a relationship between the number of differentially expressed genes (when compared for each mutant between 37°C and 42°C) and the distance to non-template DNA strand (Figure 3.4D) and the sigma factor binding site (Figure 3.4E) in several PDB structures used. This correlation was driven by downregulation, as the number of downregulated genes correlated with the two molecular phenotypes, unlike the number of upregulated genes (Figure 3.4D,E). We did not identify a relationship between the number of DEGs (neither the total number nor the number of upregulated/downregulated) and the distance between the mutation and β' , nor the distance between the mutation and the nascent RNA chain. Interestingly, we found no correlation between the stability of RpoB ($\Delta\Delta G$) and the number of differentially

expressed genes. As such, our data may hint to a key interaction between the distance of a mutation to sigma factors and DNA, and how the transcriptomic impact of that mutation is modulated by temperature. However, as none of the observed correlations were statistically significant after Bonferroni correction, the exact mechanism of interaction between the molecular and transcriptomic effects of these *rpoB* mutations remains unclear.

3.4.2 Linking cellular to organismal effects

To identify the effects of temperature on organismal level effects of mutations in *rpoB*, we first measured the fitness of mutants at 37°C and 42°C. Fitness was measured as the area under the curve (AUC) of growth curves, as is typical for measurements of bacterial fitness (see *Chapter 3.3.4.1 – Growth curves*). For the majority of *rpoB* mutants and the wildtype, there were slight increases in fitness when grown at 42°C compared to 37°C (Figure 3.5), while two mutants (Q513R and S531F) exhibited a larger increase in growth at the higher temperature.

We next sought explanatory links between these fitness effects and changes at the transcriptomic level, asking whether the altered transcriptional profiles of mutant strains lead to changes in fitness. A search for correlations as before appears to show a negative correlation between the number of DEGs and the AUCs, but this was not a significant relationship (Figure 3.5D). However, this approach ignores the context of the function of DEGs, as genes are not functionally independent. To contextualise genes with regards to their function, we performed a functional analysis of transcriptomic data using Gene Set Enrichment Analysis (GSEA). GSEA is a well-established tool in which genes are ranked (here by log fold change in expression) before testing for enrichment of user-defined gene sets at the top or bottom of the list. We used the Kyoto Encyclopaedia of Genes and Genomes (KEGG) database for high-level function classification of genes.

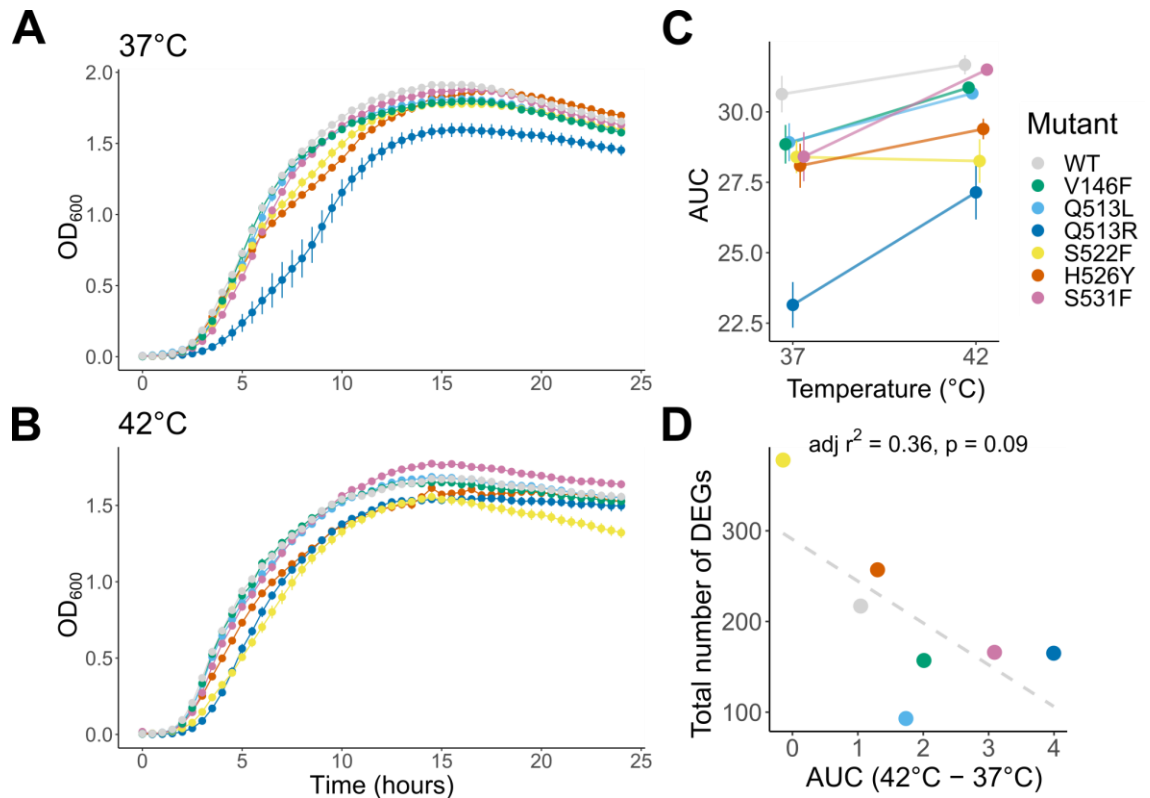


Figure 3.5 – Organismal fitness of *rpoB* mutants at two different temperatures. (A) Growth curves at 37°C. (B) Growth curves at 42°C. (C) Scatter plot showing area under the curve (AUC) of each mutant at both temperatures. Data are mean of 12 biological replicates, error bars are standard error of the mean. (D) Correlations between the total DEGs at 42°C vs 37°C and the AUC difference between temperatures. Grey line shows results of linear regression analysis.

Through the grouping of genes into functional categories, we found altered expression of groups of genes involved in specific aspects of bacterial behaviour that are linked to fitness in a context-dependent manner, including metabolism, chemotaxis, membrane transport, flagellar assembly, and biofilm formation (Figure 3.6A). Interestingly, the total number of DEGs did not strongly influence the number of altered organismal functions ($r(5) = -0.077$, $p = 0.87$), indicating that the impact of temperature on the organismal consequences of *rpoB* mutations might be genotype-specific. In other words, different *rpoB* mutations might have a specific and consistent response to changes in temperature.

To test this idea, we measured several organismal-level responses *in vitro* to determine whether differential expression of gene sets reflected on organismal fitness. First, we returned to fitness data from growth curves (Figure 3.5) and found that the same 2 mutants with the greatest increase in growth when comparing 42°C to 37°C (Q513R and S531F) were also the only ones with a significant overexpression of genes in metabolic pathways (Figure 3.6B). The difference between growth at 42°C and 37°C varied significantly between mutants ($F_{6, 165} = 4.59$, $p < 0.001$). While many factors impact bacterial growth, our finding points to a key role of metabolic genes in dictating the consequences of changing temperature.

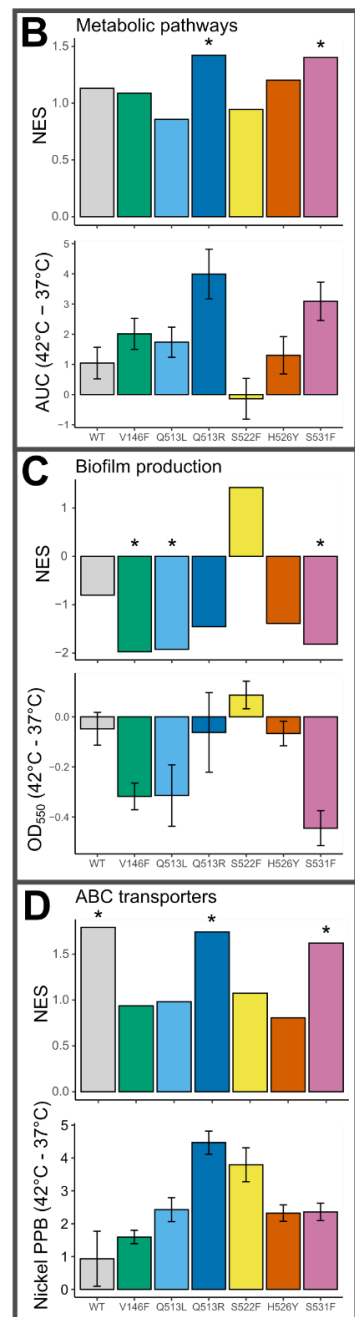
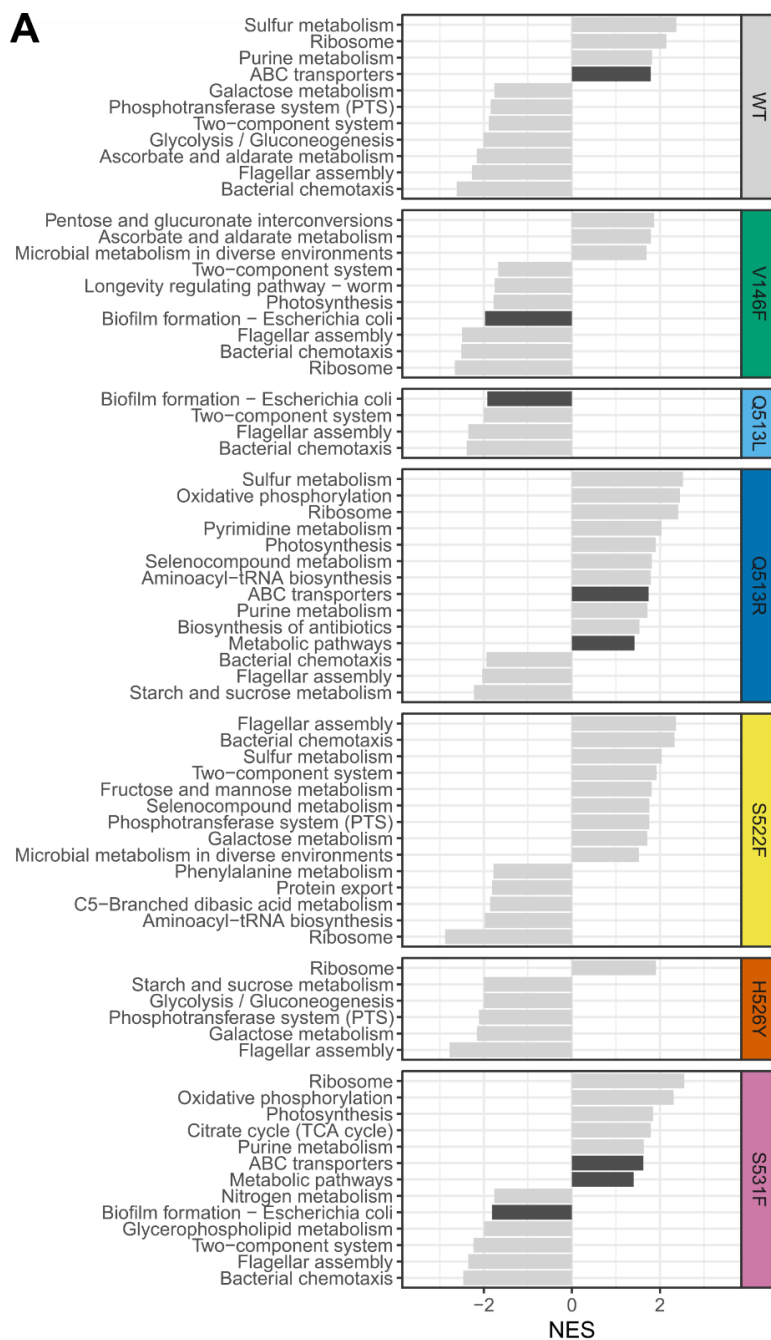
We next explored biofilm production as another measure of organismal-level responses. Genes involved in biofilm formation were significantly enriched among downregulated genes in V146F, Q513L, and S531F, suggesting biofilm formation is suppressed when those mutants are grown at 42°C compared to at 37°C (Figure 3.6A). Indeed, when the level of biofilm was measured *in vitro*, the level of biofilm varied significantly between mutants ($F_{6, 77} = 3.21$, $p = 0.007$) and we found the same 3 mutants showed reduced biofilm at 42°C versus 37°C (Figure 3.6C). In comparison, WT, Q513R, S522F, and H526Y showed little change in biofilm between the 2 temperatures. Interestingly, the only mutant with a modest increase in biofilm production at 42°C was S522F – the same mutant with the only positive enrichment score (NES) for genes involved in biofilm formation, although neither the NES nor the biofilm production data were statistically significant for this mutant.

Not all measurements of organismal level responses were so directly linked to differentially expressed gene sets, however. The KEGG category ‘ABC transporters’ appeared among significantly enriched gene sets in WT, Q513R and S531F (Figure 3.6A).

The ABC transporter family of transmembrane proteins are involved in the ATP-dependent active transport of a wide range of substrates, many of which are essential for bacterial growth (Moussatova *et al.*, 2008). To measure the activity of one particular ABC transporter, we quantitatively assessed the concentration of a substrate metal—nickel—that relies on the activity of the Nik system ABC transporter (Eitinger and Mandrand-Berthelot, 2000). Whilst we found significant variation between mutants in nickel concentrations in Q513R at 42°C versus 37°C ($F_{6, 35} = 7.38$, $p < 0.001$), and seemingly higher levels in S522F, neither WT nor S531F showed similar increases (Figure 3.6D). Further, S522F had the second highest increase in nickel concentrations, where enrichment of ABC transporter genes was not seen among DEGs.

Analysing the consequences of transcriptome-level changes with GSEA revealed targeted dysregulation of sets of functionally related genes that reflected in specific, measurable organismal-level effects. However, relationships between mutational effects at different levels of biological organisation, and how temperature interacts with and influences these effects, remain extremely complex.

Figure 3.6 – Cellular-level effects on transcription alter specific organismal fitness traits. (A) Gene Set Enrichment Analysis of differentially expressed genes for each mutant at 42°C vs 37°C, showing all KEGG categories with significant normalised enrichment scores (NES). Bars highlighted in dark grey are focal categories for experimental analysis (which are shown again in panels B, C and D; highlighted as significant NES scores with *). (B) Top: NES for ‘Metabolic pathways’ KEGG group for each mutant. Bottom: Difference in area under the curve (AUC) from growth curves of each mutant at 42°C minus 37°C (n = 12). (C) Top: NES for ‘Biofilm production’ KEGG group for each mutant. Bottom: Difference in level of biofilm (measured as optical density at 550nm – see *Chapter 3.3.4.2 – Biofilm formation*) at 42°C minus 37°C (n = 6). (D) Top: NES for ‘ABC transporters’ KEGG group for each mutant. Bottom: Difference in cellular nickel parts per billion (PPB) at 42°C minus 37°C (n = 3). Error bars in B, C and D show error propagation (standard error of the mean).



3.5 Discussion

Understanding how mutations affect bacterial phenotype and fitness lies at the heart of understanding microbial evolution and is essential if we are to develop strategies to combat the spread of infectious disease and, critically, the evolution of antimicrobial resistance. Linking mutations to their consequences remains particularly challenging for mutations with complex pleiotropic effects, as a single point mutation can alter many phenotypes across different levels of biological organization (Figure 3.1). The problem is only further magnified as the mutational effects at any of these levels can be altered by the environment, and the influence of the environment on genotype-phenotype-fitness relationships remains poorly understood. In this study, we explored how a change in temperature modulates the consequences of single point mutations in RNAP across several levels of biological organisation – molecular, transcriptomic and organismal. We identified genotype- and temperature-specific variations in the molecular, transcriptomic and organismal traits of *rpoB* mutants. However, our aim was also to identify links between these different levels of biological organization, as they would point towards mechanistic connections necessary to understand complex consequences of mutations. In attempting to link molecular effects to cellular effects, we found correlations between the number of DEGs between the two temperatures and the distances to non-template DNA and to the sigma factor binding site, suggesting a possible mechanism behind transcriptomic dysregulation (Figure 3.4). We also found a link between groups of dysregulated genes that share a common function and the organismal-level consequences of mutations, such as biofilm formation and metabolism (Figure 3.6).

Specific differences in the behaviour of *rpoB* mutants at different temperatures have implications for the evolutionary potential of those strains. We found that 2 mutants exhibited higher levels of growth at 42°C than other mutants, suggesting they may be better adapted to growth at 42°C purely as a result of their single point mutation in RNAP (Figure 3.5). This could mean some mutants would have an evolutionary advantage in an infection model where body temperature is increased due to fever, or in species with different internal body temperatures. More broadly, the distribution of different resistance alleles globally could be influenced by their ability to grow at different temperatures in diverse climates. Similar differences found in biofilm formation suggest that adaptation to specific niches can depend on genotype-by-environment interactions (Figure 3.6C).

These adaptations would be missed simply by measuring bacterial growth rate, highlighting the importance of teasing apart the intermediate phenotypes—i.e. the molecular and cellular level effects—of complex mutations.

We identify several limitations with the present study. Our assumption that the temperature change influences the molecular effects of different mutations equally, so that the rank order is maintained in both temperature environments, may not be accurate. Previous work from our group and others has highlighted the complex genotype-by-environment interactions that influence the behaviour of *rpoB* mutations (Gifford, Moss and MacLean, 2016; Soley *et al.*, 2023). Such interactions may result in the structural effects of some mutations being more pronounced than others when the temperature is shifted. However, this assumption was necessary given the difficulties in accurately predicting temperature effects on molecular structures, with experimental investigations into the consequences of temperature on RNAP structure lacking.

Whilst our findings are constrained by the small number of mutations selected, the characterisation of other *rpoB* mutations in other studies suggests that many of the intermediate phenotypes explored in our work are not exclusive to the mutations we investigated (although the nature of the interactions between them may vary to those uncovered in our work), and in fact highlight that there are many more intermediate phenotypes than those we focused on (Jin and Gross, 1989; Cui *et al.*, 2010; Qi, Preston and Maclean, 2014). Further studies that identify other potential consequences of *rpoB* mutations, and how they influence each other, will improve our understanding of how such mutations alter phenotype and fitness. In particular, we lack molecular understanding of how *rpoB* mutations and temperature alter the functioning of RNAP – the initiation, elongation and termination of transcription. Data from previous studies has revealed how transcriptional efficiency varies between *rpoB* mutants – indeed, it has been proposed that this is responsible for reduced fitness in certain mutants (Reynolds, 2000; Qi, Preston and Maclean, 2014). However, whether the transcriptional efficiency of mutants is altered by temperature in a mutant-specific manner is as yet unknown. Further, neither study observed *increased* transcriptional efficiency in instances where mutants showed improved fitness relative to the wildtype (Reynolds, 2000; Qi, Preston and Maclean, 2014), suggesting that the fitness of mutants may be determined by more complex interactions.

The use of *E. coli* as a model organism, while common in microbial research, also puts limitations on the generalisability of our findings, particularly since rifampicin is not generally used as a treatment for *E. coli* infections. However, some of the *rpoB* mutations we investigated are found in other species including those treated with rifampicin such as *M. tuberculosis*, as are similar effects on transcription (Vogwill, Kojadinovic and Maclean, 2016; Guerillot *et al.*, 2018; Trauner *et al.*, 2021). Further, our approach to identifying the effects of complex mutations in different environments is applicable regardless of the species.

Despite a wealth of research on rifampicin-resistant mutants, it remains impossible to predict the effects of *rpoB* mutations *a priori*. Here, we described some of the different consequences of *rpoB* mutations and the ways in which they might impact one other. This builds upon previous work identifying many of the effects of *rpoB* mutations (Jin and Gross, 1989; Qi, Preston and Maclean, 2014; Cutugno *et al.*, 2020; Yang *et al.*, 2023). Generally, studies focus on one or two effects and lack a holistic approach that takes into account the consequences of *rpoB* mutations on different levels of biological organisation. We highlighted several key intermediate phenotypes, which can be predicted and/or measured experimentally and that could be affected by *rpoB* mutations. We suggest a framework for a bottom-up, holistic approach for investigating the mechanisms through which mutations influence bacterial fitness, and how changes in temperature alter those mechanisms. Such a framework is critical for accurate understanding, and ultimately predicting, the consequences of complex pleiotropic mutations.

4 High-throughput method rapidly characterizes hundreds of novel antibiotic resistance mutations

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4.1 Abstract

A fundamental obstacle to tackling the antimicrobial resistance crisis is identifying mutations that lead to resistance in a given genomic background and environment. We present a high-throughput technique – Quantitative Mutational Scan Sequencing (QMS-Seq) – that enables quantitative comparison of which genes are under antibiotic selection and captures how genetic background influences resistance evolution. We compared four *E. coli* strains exposed to ciprofloxacin, cycloserine, or nitrofurantoin and identified 975 resistance mutations, many in genes and regulatory regions not previously associated with resistance. QMS-Seq revealed that multi-drug and antibiotic-specific resistance are acquired through categorically different types of mutations, and that minor genotypic differences significantly influence evolutionary routes to resistance. By quantifying mutation frequency with single base pair resolution, QMS-Seq informs about the underlying mechanisms of resistance and identifies mutational hotspots within genes. Our method provides a way to rapidly screen for resistance mutations while assessing the impact of multiple confounding factors.

4.2 Introduction

Antimicrobial resistance (AMR) is an ongoing global health crisis limiting our capacity to treat infections. AMR infections lead to five million deaths each year, and this burden is predicted to increase considerably (Murray *et al.*, 2022). AMR often arises through chromosomal mutations that alter susceptibility to one or more antibiotics (Munita and Arias, 2016). Consequently, predicting drug susceptibility from genomic data is becoming vital to antimicrobial stewardship by informing and stratifying treatments (Schürch and van Schaik, 2017). However, the accuracy of such predictions is limited by our existing knowledge of which mutations contribute to AMR and through what mechanisms.

Mapping mutations to AMR phenotypes is typically achieved through sequencing, either of clinical isolates or laboratory strains that have undergone selection for resistance. These approaches have proven powerful tools but are often biased towards mutations that confer high-level resistance, and struggle to discern how AMR evolution varies in different contexts, for example how the evolution of secondary resistance varies between strains with different primary resistance mutations. Smaller-effect mutations, epistatic interactions with genetic background, and environmental factors contribute significantly to overall resistance level, complicating genomic predictions (Wong, 2017). As such, a key unmet challenge is to quantify the potential mutation space for resistance beyond the well-trodden evolutionary paths we currently understand.

To achieve this, we developed Quantitative Mutational Scan-Sequencing (QMS-Seq), an experimental evolution technique that adapts metagenomic sequencing to quickly characterize mutational landscapes for resistance. In QMS-Seq, random mutations are allowed to occur in the absence of selection, using subpopulations to maximize diversity by limiting some founder effect bias – where a beneficial mutation gained in an early selective sweep dominates the population (Figure 4.1A). Pooling the subpopulations creates a heterogeneous mixture mostly containing variants with only a single mutation. This population is spread across many selective plates – here containing the minimum inhibitory concentration (MIC) of an antibiotic. After resistant colonies have grown, they are mixed and sequenced collectively with sufficient depth to detect low-frequency resistance mutations. A custom bioinformatic pipeline (Figure 4.1B) stringently filters for high-quality reads and utilises high specificity/sensitivity software (Wilm *et al.*, 2012) to

call single-nucleotide variants and small indels, while another tool, *breseq* (Deatherage and Barrick, 2014), identifies larger mobilization events for known insertion sequences. QMS-Seq's experimental simplicity makes it ideal for high throughput methodologies, enabling quantification and combinatorial comparison across environmental conditions and genetic backgrounds.

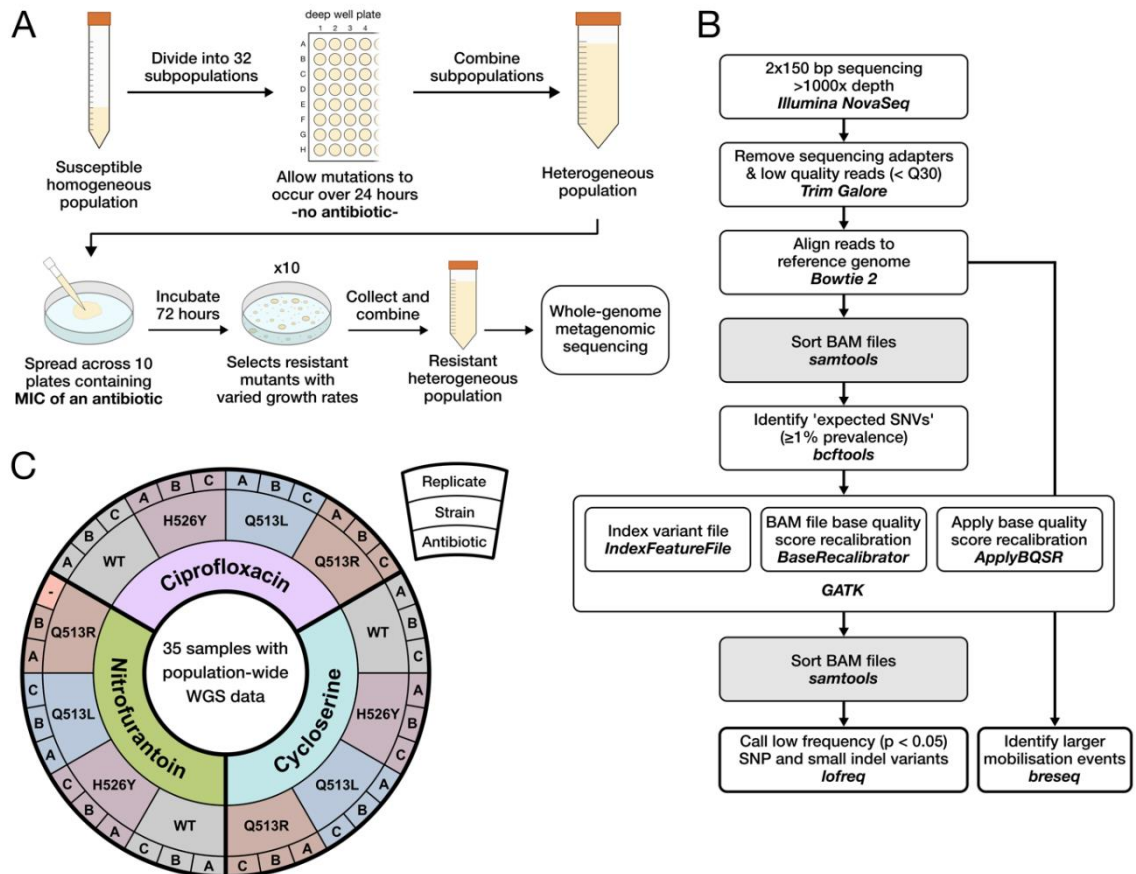


Figure 4.1– Quantitative Mutational Scan-Sequencing, QMS-Seq. (A) Experimental technique. (B) Sequencing and bioinformatics pipeline. (C) Experimental design. We tested three antibiotics and four strains of *E. coli*, performing three replicates for each of the 12 conditions, comprising 36 independent samples. The wildtype (WT) strain is BW25113, which served as ancestor for the three mutant strains. These each carry a different mutation in the *rpoB* gene which provides resistance to rifampicin and significantly changes their transcriptome compared to WT (Soley *et al.*, 2023). One nitrofurantoin/Q513R replicate was excluded from the analysis due to sequencing failure.

We employed QMS-Seq to investigate the mutational landscapes of four *Escherichia coli* strains, three of which have a single mutation in the *rpoB* gene, for resistance to the MIC of three antibiotics with different mechanisms of action (Figure 4.1C). Each *rpoB* mutation confers resistance to the antibiotic rifampicin; utilising these strains will reveal differences in secondary evolution between strains that differ only by a single SNP in an essential gene. This revealed many previously unknown resistance mutations, discriminated between mutations providing multi-drug or antibiotic-specific resistance, offered insights into the mechanisms underpinning resistance, and quantified how antibiotic class and genetic background influence AMR evolution.

4.3 Materials and methods

4.3.1 Experimental strains

The ancestral WT was *Escherichia coli* BW25113 (NZ_CP009273.1) from the Keio collection (Baba *et al.*, 2006) containing a knock-out of the *tolC* gene. The three rifampicin-resistant strains were evolved from this ancestor. *rpoB* mutations were selected for as described in (Soley *et al.*, 2023). Whole genome sequencing was used to confirm that the strains differ by only one point mutation in *rpoB*. The $\Delta tolC$ strain was chosen so only *rpoB* mutations were selected for by rifampicin – TolC is a prominent efflux channel and common target for antibiotic resistance mutations. Its absence from the strains in this study may have increased the relative frequency of alternate resistance mutations and further contributed to the observed diversity. The three rifampicin-resistant strains each carried different amino acid substitutions: H526Y, Q513L or Q513R.

4.3.2 Selecting for antibiotic resistance mutations

We inoculated Mueller-Hinton (MH) media with glycerol stock of each experimental strain, then split the inoculate into three 5 mL tubes to start independent replicate populations. After overnight growth at 37°C we diluted the cultures 1:10,000 in MH and split them into 3 x 32 sub-populations, each 1 mL volume, in a deep 96-well plate. We incubated the cultures for 24 hours at 37°C, allowing random mutations to occur during growth without antibiotic selective pressure. To select resistance mutations, the 3 x 32 subpopulations were then pooled into three independent heterogeneous populations and

spread on selective plates containing the MIC of an antibiotic. We defined MIC as the lowest concentration of a given antibiotic that prevented any colonies from forming after three days of incubation. We determined the MIC of ciprofloxacin (0.005 µg/mL, *Sigma-Aldrich*), cycloserine (100 µg/mL, *Sigma-Aldrich*) and nitrofurantoin (10 µg/mL, *TOKU-E*) using MH agar plates. We determined the MIC for the WT strain and all three *rpoB* mutants separately, but no differences in MIC between strains were identified.

Each of the three replicate populations were spread across 3 x 10 selective MH agar plates for each antibiotic, using 200 µL of culture for each plate. The 30 plates were then incubated for three days to allow resistant mutants even with severe fitness costs to form colonies. After three days, colonies from the 10 plates were collected and recombined into three heterogenous resistant populations to be sequenced (see *Figure 4.1C*). Prior to collection, we manually removed sections of the largest colonies, to increase the likelihood of identifying rarer and low-fitness mutations. QMS-Seq considers how many independent samples a mutation occurs in, rather than how prevalent it is within one sample population, so lowering the relative abundance of high-fitness mutants (which form large colonies) increases the relative abundance of low-fitness mutants, making them easier to detect without biasing later analysis.

To combine colonies from all 10 plates, fresh media was added to the plates and colonies were scraped to resuspend the cells, then tubes were thoroughly vortexed to break up any clumps. We measured optical density at 600 nm to estimate cell count and diluted the populations to 4.0×10^8 cells/mL for storage at -80°C with 15% glycerol. The samples were later thawed, and genomic DNA was extracted using Genomic DNA Purification Kit (*Promega*). We verified the DNA concentration with a NanoDrop before sending for sequencing.

4.3.3 Whole-genome metagenomic sequencing and low-frequency variant calling

We used Azenta Life Science's Illumina NovaSeq 2x150 base pair sequencing package. They performed sequencing to a mean depth of at least 1000x per sample, a value estimated to allow detection of mutations with a frequency of at least 0.5% within the population. For QMS-Seq, raw sequencing reads were first trimmed with *Trim Galore* (v.0.6.7) (Krueger *et al.*, 2023) to remove sequencing adapters and low-quality reads, with a Q score cut-off of Q30. Trimmed reads were aligned to the reference genome (*E. coli*

K-12 BW25113) with *Bowtie2* (v.2.2.5) (Langmead and Salzberg, 2012) using end-to-end alignment and the *–no-mixed* flag. BAM files were sorted with *samtools* (v.1.16.1) (Danecek *et al.*, 2021), and single nucleotide variants (SNVs) with a minimum allele frequency of 1% were identified (these were considered ‘expected SNVs’) using the *bcftools* (v.1.16) *mpileup*, *call* and *view* commands. Prior to variant calling with *lofreq*, BAM files were pre-processed based on GATK best practice protocol as recommended by the *lofreq* documentation. Specifically, the variant file was indexed with *GATK* (v.4.2.6.1) (McKenna *et al.*, 2010) *IndexFeatureFile* command and BAM files underwent base quality score recalibration with the *BaseRecalibrator* and *ApplyBQSR* commands, using the expected SNVs from the previous step as known sites. Recalibrated BAM files were sorted with *samtools*, and low frequency variants were called with *lofreq* (v.2.1.3.1) with a significance threshold of 0.05 and using the *–call-indels* flag.

We have taken several precautions to reduce the possibility of miscalls. We ensured a high-quality alignment with the use of paired end sequencing, stringent quality control of reads, and alignment of paired reads only, while excluding unpaired reads. *lofreq* itself utilizes mapping quality scores and base scores (which were recalibrated to reduce systematic errors in the scores assigned during sequencing) in calling low frequency variants. All samples had very high coverage and sequencing depth. We further checked the sequencing depth at the loci of the 5,037 filtered mutation occurrences – the median depth for all variant sites was 1151x (Figure 4.2).

The pipeline described above can identify SNVs and small insertions or deletions of less than 30 base pairs, at very low frequencies in the population. To identify low frequency mobilization events, which result in larger insertions and deletions, mutation calling was also performed using the *breseq* pipeline (v0.38.1) (Deatherage and Barrick, 2014). Trimming and read quality assessment was first performed on raw FASTQ files using *Cutadapt* (Martin, 2011) and *FastQC* (Andrews, 2010) via the *Trim Galore* wrapper (Krueger *et al.*, 2023), according to default run parameters. Mapping and variant calling was performed using *breseq* polymorphism mode with a ‘polymorphism-frequency-cutoff’ of 0.001, with other run parameters set to default values. Reads were aligned to a modified version of the reference genome for which we have additionally annotated mobile elements (Gifford *et al.*, 2023) as is required for the prediction of mobile element insertion events by *breseq*.

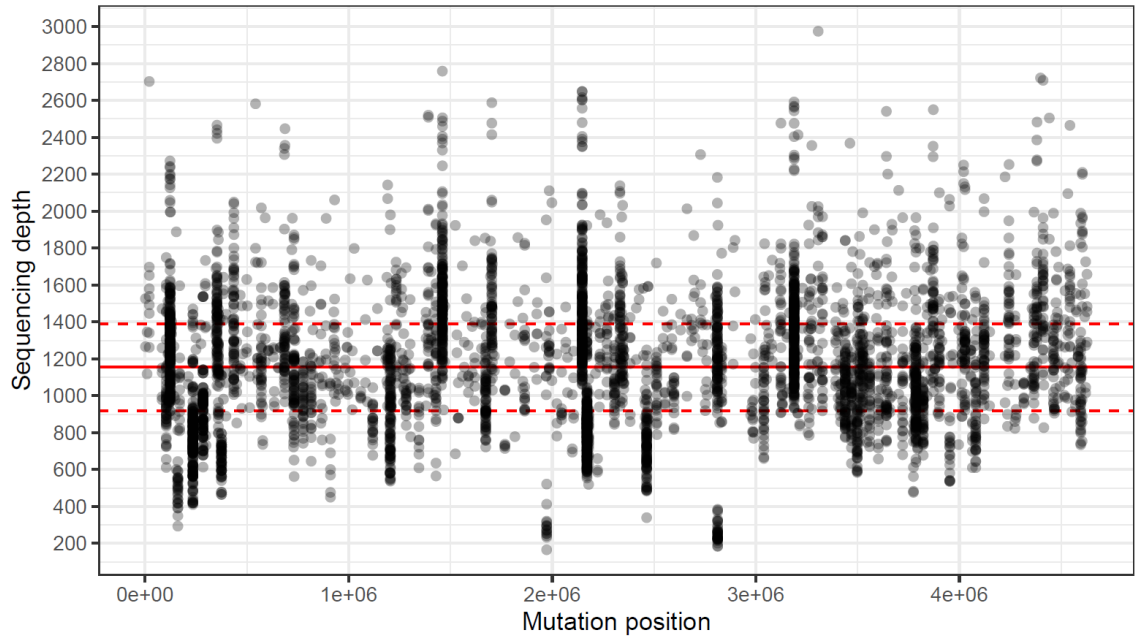


Figure 4.2 – Sequencing depth at the loci of each of the 4,987 mutation occurrences. Solid red line is the median, dashed lines are upper and lower quartiles.

4.3.4 Filtering mutations under clear selective pressure

We adopted a conservative filtering approach in an effort to identify true resistance mutations under clear selective pressure. Assuming a simplified possible mutation space of 1.4×10^7 (the number of bases in the WT genome times three possible mutations at each base), and given an average of 185 mutations occurring per sample, the likelihood of the exact same mutation occurring in 2 or more samples (out of 35) by random chance is:

$$1 - \left(1 - \frac{185}{1.4 \times 10^7}\right)^{35} - 35 \times \frac{185}{1.4 \times 10^7} \times \left(1 - \frac{185}{1.4 \times 10^7}\right)^{34} = 1.04 \times 10^{-7}$$

While the true possible mutation space is limited by essential bases and other factors, this number would not change by orders of magnitude. This calculation also excludes any other types of mutations, such as small insertions or deletions, and hence represents a conservative estimate. As such, we could be confident the 637 mutations which occurred in at least two samples were under strong positive selection.

This criterion was sufficient for intergenic and many genic mutations, however it excluded some genic mutations despite them being clustered in one gene. In certain genes, typically those targeted predominantly by nonsense or frameshift mutations, the *same*

mutation would rarely appear in multiple samples, even if the gene was targeted by many *different* mutations across multiple samples. Hence, if there is insufficient pressure to mutate a specific base (or bases) in the gene over any other, the above criterion fails. To accommodate this, we also included 375 single-sample mutations observed in genes with more mutation occurrences than expected (Figure 4.3).

Our filtering criteria excluded 1,216 of the 2,191 mutations, which reduced the number of targeted genes from over 900 to 278. However, as our criteria generally filtered for mutations that occurred multiple times, the total number of occurrences only decreased from 6,484 to 4,987. This suggests that, while our filtering substantially decreased the observed variety of mutational targets, the filtered mutations still reflected most of the genetic diversity in the samples. To verify that the final 975 mutations we included in our analysis represented strong selective pressure, we compared their distribution throughout the genome to that observed by Foster *et al.* in a mutation accumulation study performed with *E. coli* in the absence of selection (Foster *et al.*, 2013). The gap sizes between mutations in our dataset indicated overwhelming clustering that was absent from the near-random distribution of 1,625 mutations observed by Foster *et al.* (Figure 4.4).

Finally, we estimated the typical number of mutations present in a single genome by performing WGS on 18 individual clones that had been streaked from the glycerol stock of nine different heterogeneous resistant population samples. We chose to examine CIP samples because it leads to DNA damage, and we surmised clones from these samples would have been most likely to acquire additional mutations. Raw sequencing reads were first trimmed with *Trim Galore* with a Q score cut-off of Q20. Variants were then called with *snippy* (v.4.6.0). This confirmed that most clones did not contain more than one mutation (Table 4.1).

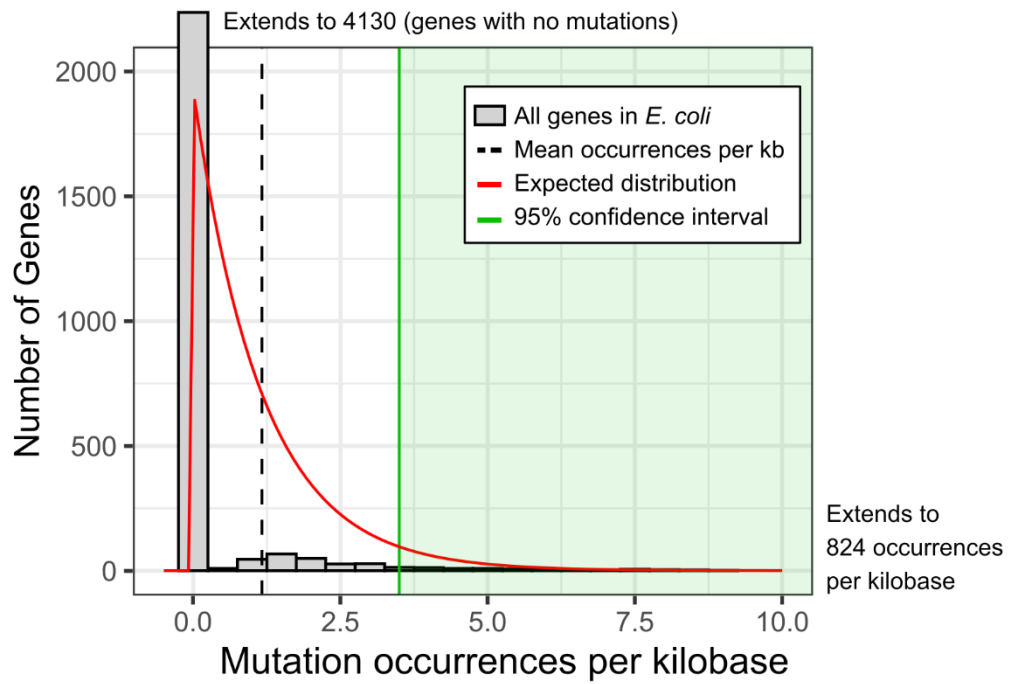


Figure 4.3 – Secondary filtering criterion. In addition to our primary filtering criterion (mutations must appear across two or more independent samples) we also included single-sample mutations in genes where significantly more mutations were observed than expected. To determine this, we calculated the number of mutation occurrences per kilobase for every gene in *E. coli*. We then included single-sample mutations occurring in genes with more mutation occurrences per kilobase than the 95% confidence interval, equivalent to > 3.5 mutation occurrences / kb.

Table 4.1 – Validating the number of mutations per genome by whole-genome sequencing individual clones.

Sample	Clone	Number of mutations	Mutation identity
<i>CIP/WT A</i>	1	1	<i>gyrB</i> (S464Y)
<i>CIP/WT A</i>	2	1	<i>gyrA</i> (D87Y)
<i>CIP/WT B</i>	3	1	<i>gyrA</i> (A119P)
<i>CIP/WT B</i>	4	0	-
<i>CIP/WT C</i>	5	0	-
<i>CIP/WT C</i>	6	1	<i>gyrA</i> (D87Y)
<i>CIP/H526Y A</i>	7	1	<i>gyrA</i> (D87Y)
<i>CIP/H526Y A</i>	8	0	-
<i>CIP/H526Y B</i>	9	0	-
<i>CIP/H526Y B</i>	10	0	-
<i>CIP/H526Y C</i>	11	2	<i>yncD</i> (V242E) 8173_8183del
<i>CIP/H526Y C</i>	12	1	<i>fhuD</i> (p.G264fs)
<i>CIP/Q513R A</i>	13	1	<i>gyrB</i> (L509G)
<i>CIP/Q513R A</i>	14	0	-
<i>CIP/Q513R B</i>	15	1	<i>gyrB</i> (E466A)
<i>CIP/Q513R B</i>	16	1	<i>gyrA</i> (D87N)
<i>CIP/Q513R C</i>	17	1	<i>gyrA</i> (D87Y)
<i>CIP/Q513R C</i>	18	0	-

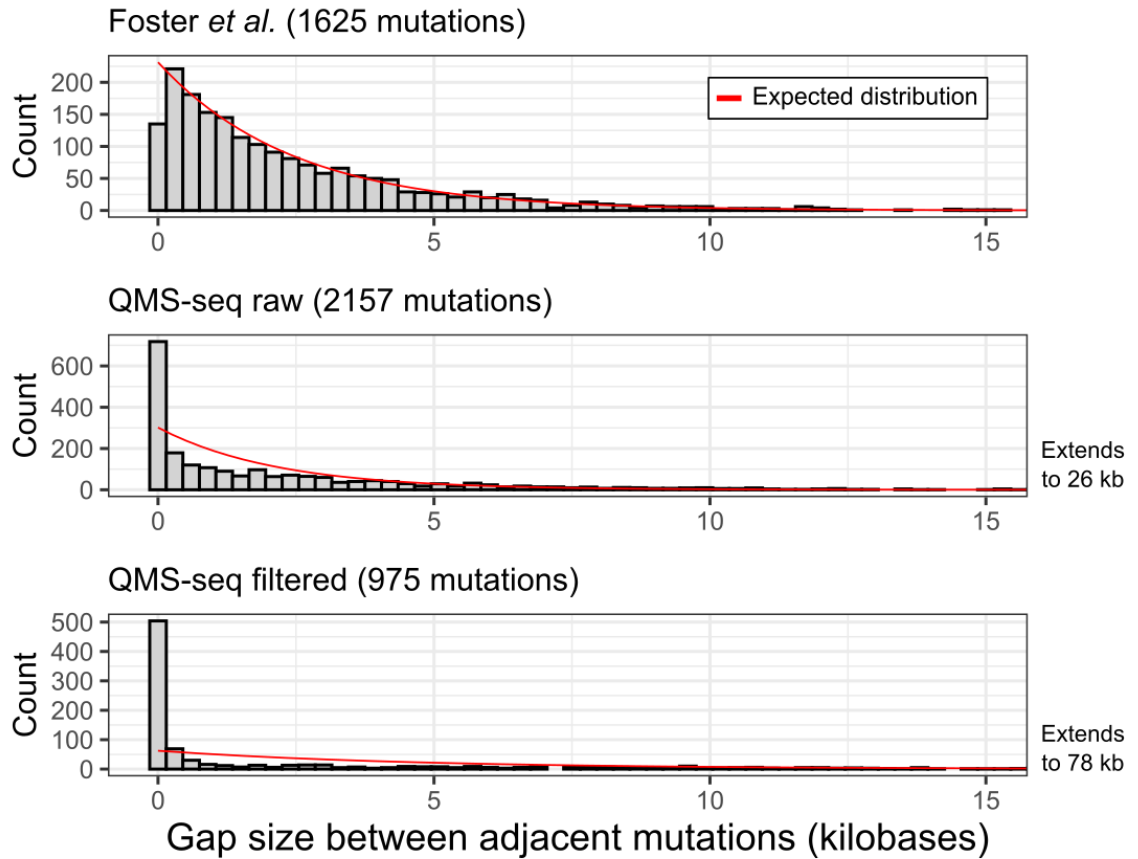


Figure 4.4 – Clustering of QMS-Seq mutations indicates strong selective pressure. Comparing the distribution of mutations throughout the genome between QMS-Seq and a mutation accumulation study performed in the absence of selective pressure using a $\Delta mutL$ strain of *E. coli* MG1655 (Foster *et al.*, 2013). Absent from selection the mutations are spaced randomly, with the gap size between adjacent mutations nearly perfectly matching the expected exponential distribution. In contrast, the spacing of mutations identified by QMS-Seq shows significant clustering, indicative of selective pressure. The filtered QMS-Seq mutations analysed in this study show even greater clustering, suggesting the filtering has successfully discarded most hitchhiker mutations.

4.3.5 Annotating regulatory features in the genome

We mapped intergenic mutations to regulatory features using information from published databases: *regulonDB* (Tierrafría *et al.*, 2022) and (Ju, Li and Liu, 2019), containing positions of known promoters and Rho-dependent/independent terminators in *E. coli*. An additional study was used to confirm the regulatory features of the *sib-ibs* genes (Thomason *et al.*, 2014). In some instances, regulatory annotations overlapped with a coding sequence. Mutations that occurred in such regions were classified as regulatory if they were near (< 20 base pairs) the beginning or end of the gene.

4.3.6 Classifying known resistance and persistence genes

We searched Google Scholar with the following queries: (“*gene name*” + antibiotic resistance + mutation) and (“*gene name*” + antibiotic persistence) for the 278 target genes. For 62 annotated regulatory regions, each gene in the transcriptional unit was inserted for *gene name*. If a publication was found that indicated mutations in the gene were associated with antibiotic resistance it was classed as such, regardless of the antibiotic or specific mutation(s). Antibiotic persistence genes were classified if they had specifically been implicated in persistence or found consistently dysregulated in persister populations. If no relevant publications appeared in the first 20 results, the gene was classed as not having been previously associated with antibiotic susceptibility.

4.3.7 Categorizing the function of target genes

We used *EcoCyc* (Keseler *et al.*, 2011) to annotate every target gene with a known biological role, including those immediately downstream of regulatory targets. To classify the genes into broad functional categories we used a combination of gene ontology (GO) biological process terms and published literature. For ‘membrane transport’, ‘translation’, ‘respiration’, ‘lipid metabolism’, and ‘transposase’ we identified the GO term(s) that were associated with each of these categories and grouped target genes annotated with one of those terms. For transcription factors, which are annotated with GO terms indicating their enzymatic role rather than the downstream pathways they affect, classifications were supplemented with literature. This was also done for the ‘toxin-antitoxin’ category, as there is no relevant GO term. Target genes were classified as ‘other’ if they were annotated with GO biological process terms, but none fell under

the above categories. They were classified as ‘uncharacterized’ if they were not annotated with any GO biological process, and we could not find a description of their function in literature. To construct an equivalent categorization of all the genes in the *E. coli* genome, we cross-referenced our list of GO terms against the GO annotations of the entire *E. coli* genome from *PANTHER* (Thomas *et al.*, 2022), and identified the number of known toxin-antitoxin genes from literature (Yamaguchi and Inouye, 2011).

4.3.8 Identifying matching mutations in global dataset of *Escherichia coli* isolates

To contextualize QMS-Seq mutations against global *E. coli* isolates, we downloaded all *E. coli* genomes from the BV-BRC database (as of April 2023) (Olson *et al.*, 2023). After filtering for high quality data from Illumina platforms and excluding isolates that lacked sufficient metadata, we kept 21,711 genomes for downstream analysis, consisting of 4918 raw sequence files and 16,793 assemblies. Variants were called using *snippy* (v4.6.0) with the same reference genome used above (NZ_CP009273.1), and we counted the number of isolates with sequences matching mutations in our dataset.

4.3.9 Categorizing antibiotic-specific and strain-specific targets

Given the quantity of mutations and their complex distribution across 12 experimental conditions, we chose to be conservative when categorizing mutational targets as ‘multi-drug resistance, MDR’ vs. ‘antibiotic-specific resistance, ASR’. For example, many mutations were identified in three different samples, two from one antibiotic and one from another. For mutations that appeared in only two samples from the same antibiotic, we could not be confident a hypothetical third occurrence would appear in the same antibiotic. Thus, we employed generalized linear modelling in *RStudio* (v2023.09.1) to statistically determine which genes/regulatory features were disproportionately targeted in one antibiotic. Targets with a mutation distribution where the antibiotic variable explained a significant ($P < 0.05$) amount of variation were classified as ASR. Targets that were mutated in at least one sample from *all* antibiotics were classified as MDR. Targets that were mutated in either one or two antibiotics, but with insufficient occurrences to pass our confidence threshold, were classified as ‘unclear’. The same approach was used to identify targets whose mutational distribution varied significantly between strains. We identified mutational distributions with higher-order interactions

between antibiotic and strain, and targets where antibiotic and strain were both significant, but independent. We had the statistical power to identify targets with only strain effects (no antibiotic bias) but observed none. We used one-way ANOVA with multiple comparisons to examine differences in the total number of target genes (Figure 4.8A) between the 12 experimental conditions.

4.3.10 Mutation impact and dispersion analysis

Predicting the impact (high, moderate, or low) of genic mutations on the encoded protein was automated by *SnPEff* (v5.1) (Cingolani *et al.*, 2012) which compares the variant sequences to the reference genome to predict the effect of single-nucleotide polymorphisms or small insertions/deletions. To calculate the average mutation impact for every targeted gene, each impact type was given a numerical value: high = 1, moderate = 0.5, low = 0. An average was taken for all the mutation occurrences in one gene to correspond with the three-color scale in Figure 4.6B, red = 1, blue = 0.5, green = 0. We used Pearson's Chi-squared test with Bonferroni correction and pairwise comparison to examine differences in the distribution of high/moderate/low impact types in MDR vs ASR mutations.

We complemented this analysis by calculating the 'mutation dispersion' for each gene, a metric quantifying how much of the gene's total length mutations were identified across. Mutation dispersion was defined as the distance (in base pairs) between the furthest 5' and furthest 3' mutations, divided by the total length of the gene. When calculating mutation dispersion, we discounted 'outlier' mutations: low occurrence mutations located distantly from a cluster of high occurrence mutations. For each gene we considered every mutation occurrence as a number representing that mutation's position between the first base of the gene and the last. We then calculated the interquartile range of these values and excluded mutations corresponding to outlier values. While still genuine resistance mutations, excluding the outlier mutations from this analysis provides a dispersion value that better reflects the way the gene is most commonly targeted.

4.3.11 Principal component analysis of experimental conditions

We set up a table with 12 columns for each experimental condition and 975 rows for all identified mutations, then gave every mutation a value between 0 and 3 for each condition

corresponding to the number of replicates the mutation was identified in. We used the *factoextra* (v1.0.7) package to perform principal component analysis and visualize the variables distributed along the first two components.

4.3.12 Genome visualization and sequence alignment

We used *Geneious Prime 2023.3.2* (<https://www.geneious.com>) to visualize mutations throughout the reference genome in relation to the positions of coding and regulatory sequences. This software was also used to perform *MUSCLE* (v5.1) sequence alignment for mutational targets with homologous regions, such as the *aceF* functional sites and *sib-ibs* copies.

4.4 Results

4.4.1 Mutational landscape for resistance

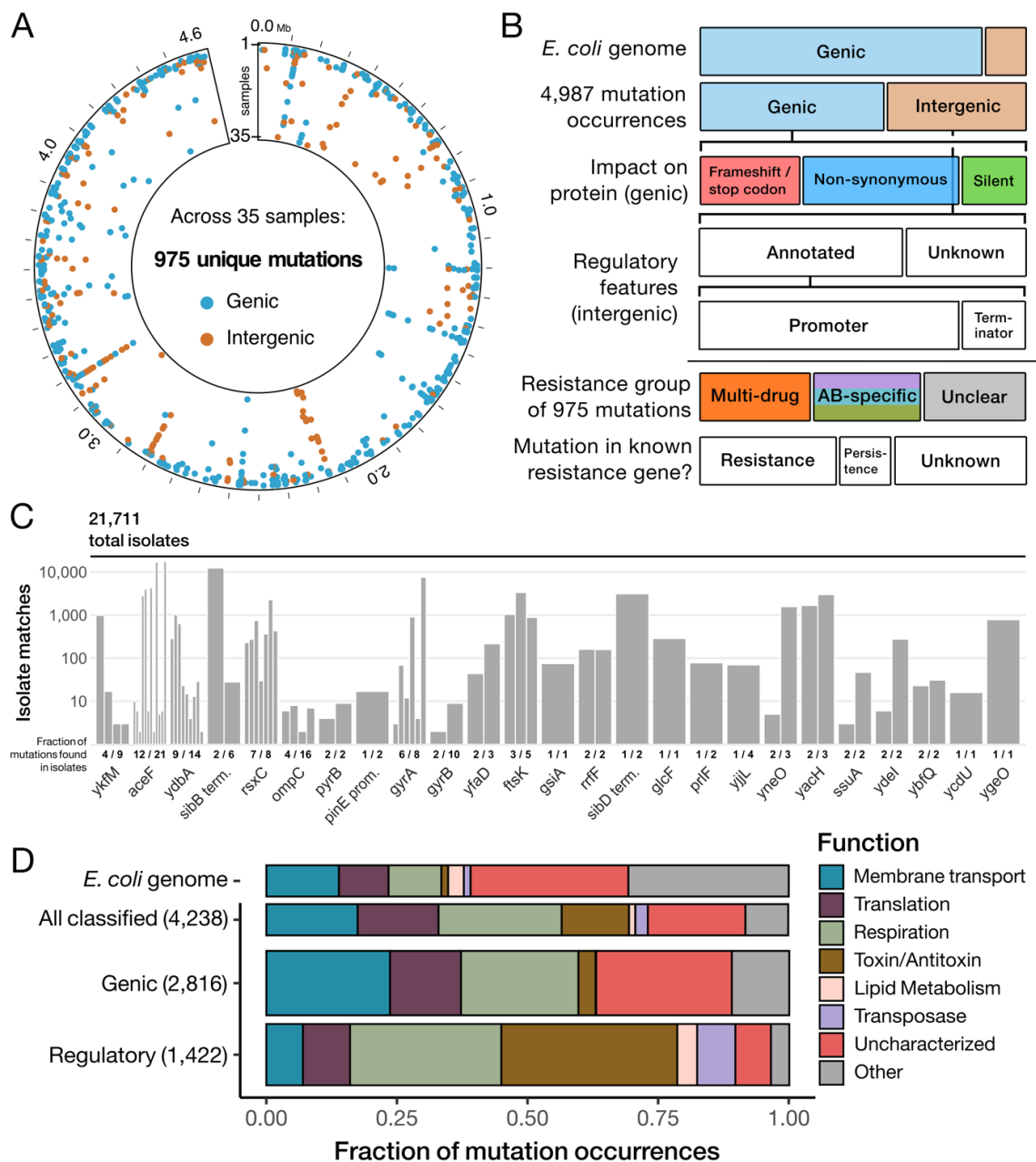
Our variant calling pipeline identified 2,191 unique mutations across the 35 samples, which corresponded to 6,484 mutation occurrences, since many mutations were identified across multiple samples. The mutations were incredibly diverse, targeting over 900 different genes, around 1/5 of the WT genome. We were concerned that our pipeline, designed to pick up very low frequency variants in the population, had identified a host of additional mutations other than those providing antibiotic resistance. This seemed especially possible given the bacteria had three days of growth on the selective plate to accrue secondary mutations in addition to the resistance one. Further, there is a chance that observed mutations are actually the result of sequencing errors (Stoler and Nekrutenko, 2021). In an effort to rectify this, the list of mutations underwent a stringent filtering process (see *Chapter 4.3.4 – Filtering mutations under clear selective pressure*).

After filtering, we observed 975 unique mutations across all conditions, in 278 genes and 62 regulatory features (Figure 4.5A). We quantified the mutation *occurrence* – the number of independent samples each unique mutation was identified in. Interestingly, many more mutations occurred within intergenic regions (43%) than expected given the composition of *E. coli*'s genome (13% intergenic), suggesting that regulatory changes contribute to resistance evolution more than typically thought (Thorpe *et al.*, 2017)

(Figure 4.5B). Nearly half the mutations were in genes that have not previously been associated with resistance to any antibiotic, including 12 of the 30 most targeted genes. This indicates how much of the mutational landscape for resistance remains uncharacterized and emphasizes QMS-Seq's utility in exploring it. Surprisingly, 15% of mutations affected genes known to be involved in antibiotic persistence. Persistence is a transient phenomenon that arises through the dysregulation of such genes (Brauner *et al.*, 2016), yet we found heritable genetic changes, suggesting overlapping mechanisms for resistance and persistence at MIC. To verify the mutations were also relevant to non-laboratory settings, we assessed their prevalence in 21,711 clinical and environmentally isolated *E. coli* genomes from the BV-BRC database (Olson *et al.*, 2023) (Figure 4.5C). Matching variants were found for 68 genes (of 278), with some identified in thousands of isolates including ciprofloxacin's target *gyrA* and the three genes with highest mutation occurrence in our study (*ykfM*, *aceF*, and *ydbA*).

Most of the genes with known functions were involved with three broad biological processes: membrane transport, translation, or cellular respiration (Figure 4.5D), which are all associated with antibiotic resistance and/or persistence (Delcour, 2009; Gomez *et al.*, 2017; Babosan *et al.*, 2022; Lopatkin *et al.*, 2024). The considerable variety of genes in each category suggests altering membrane architecture or core cellular processes is a frequent strategy for acquiring MIC-level resistance. There were notable differences between genic and regulatory mutations – the former much more common in membrane transporters, the latter accounting for 90% of toxin-antitoxin mutations. Toxin-antitoxin systems are thought to be a key driver of persistence (Page and Peti, 2016), indicating that the regulatory mutations we observed may recapitulate differential expression to provide a phenotype that mimics persistence, while being heritable rather than transient. Additionally, we observed mutations in multiple promoters for lipid metabolism genes and different copies of the *insH1* transposase, without equivalent genic mutations. Despite substantial overlap of functional categories between genic and regulatory mutations, the disparities indicate which resistance mechanisms favour differential expression over protein modification.

Figure 4.5 – Summary of resistance mutations identified by QMS-Seq. (A) Circumferential axis is the mutation's location in the *E. coli* genome (4.6×10^6 bp). Radial axis is the mutation occurrence – the number of samples each mutation occurred in (out of 35). Colour indicates if the mutation occurred in a coding sequence (genic) or not (intergenic). (B) Genic mutations were further categorized based on how they affect the amino acid sequence; intergenic mutations based on whether they appear in known regulatory features (Thomason *et al.*, 2014; Ju, Li and Liu, 2019; Tierrafría *et al.*, 2022). Mutations were considered 'multi-drug' if they occurred in samples from each antibiotic, and 'antibiotic-specific' if they were significantly over-represented in one (see *Chapter 4.3.9 – Categorizing antibiotic-specific and strain-specific targets*). For every gene, any previous association with antibiotic resistance or persistence was confirmed through a literature search. (C) The 25 genes with the most isolates possessing matching mutations from the BV-BRC (Olson *et al.*, 2023), a WGS database for global human/animal-sourced *E. coli* isolates. Genes are ordered by their mutation occurrence in our samples. (D) All genic and annotated-intergenic (regulatory) mutation occurrences categorized by the biological function of the gene they affect. Also shown is the same classification of the WT genome (4,553 genes).



4.4.2 Multi-drug vs. antibiotic-specific resistance

QMS-Seq differentiates between multi-drug resistance (MDR) mutations that confer resistance to all three antibiotics and mutations that confer antibiotic-specific resistance (ASR) – see *Chapter 4.3.9 – Categorizing antibiotic-specific and strain-specific targets*. We found categorical differences in both their intragenic positioning and their typical impact on the encoded protein. MDR mutations almost exclusively clustered within a small region of the gene, even for genes with many unique mutations (Figure 4.6A). Furthermore, mutations leading to MDR were less likely to be ‘high-impact’ (frameshift or nonsense) while ‘low-impact’ (synonymous) mutations were significantly more frequent in MDR ($\chi^2 = 608$; $P < 0.001$) (Figure 4.6B), suggesting that MDR is acquired through protein modifications rather than loss-of-function, for example functional site alterations. Conversely, for most of the top ASR targets, we identified high-impact mutations across the entire length of the gene, indicating loss-of-function provided resistance. Supporting this conclusion, over 75% of the insertion element mobilizations that we detected localized within ASR genes. MDR mutations occurred more frequently than ASR mutations and accounted for nearly all regulatory mutations (93%), implying that expression-mediated resistance is less dependent on the antibiotic’s mechanism of action (Figure 4.6C). Therefore, at MIC, the most common mutations are active site modifications and regulatory changes that provide MDR, whereas ASR typically arises from high-impact mutations.

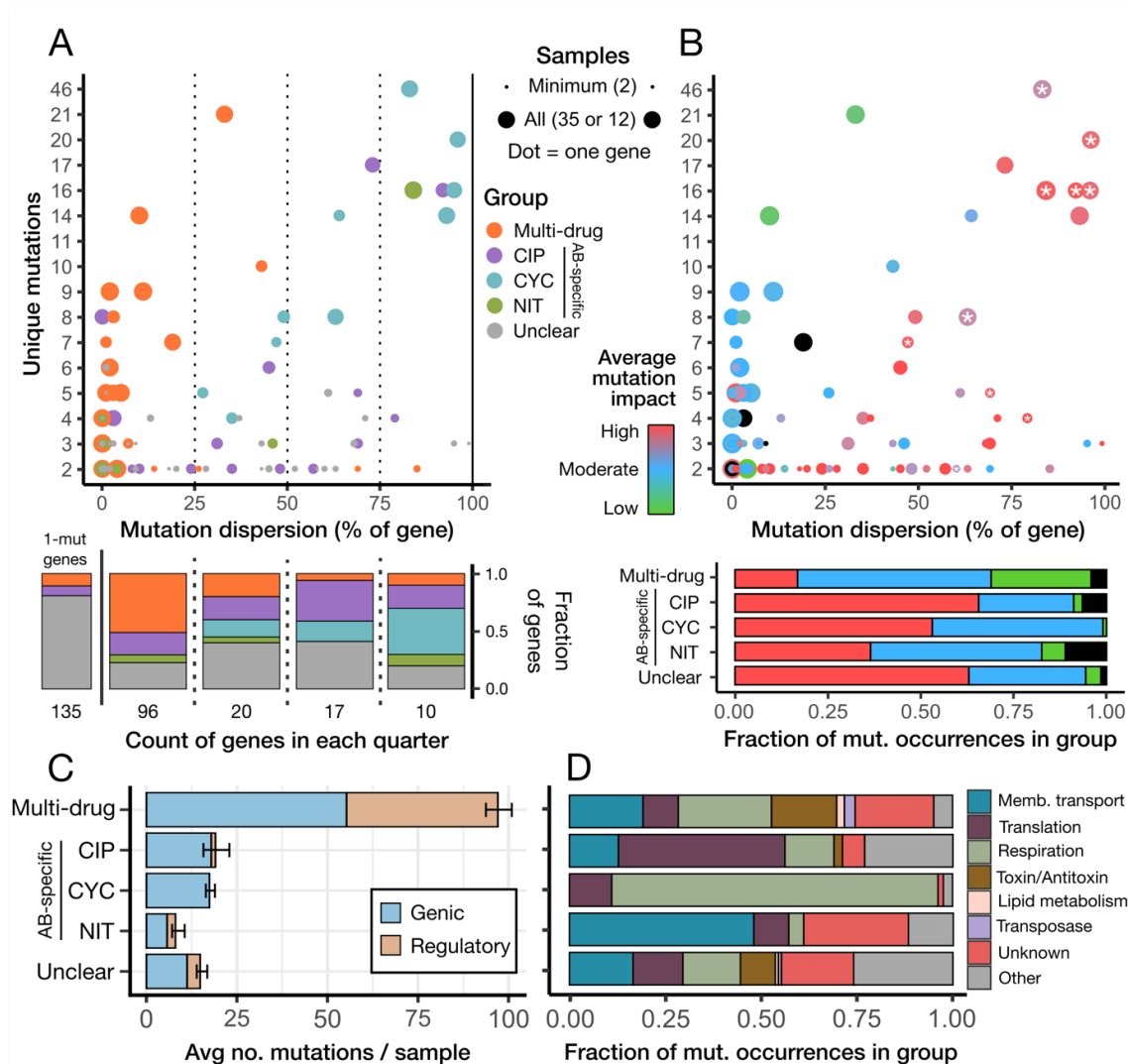


Figure 4.6 – Multi-drug versus antibiotic-specific resistance mutations. (A) Every gene arranged by the number of unique mutations detected in that gene and the percentage of the gene's length the mutations were dispersed across (see *Chapter 4.3.10 – Mutation impact and dispersion analysis*). Colour shows the resistance group classification of the gene. Size indicates the proportion of samples in which mutations were observed, out of a maximum of 12 for antibiotic-specific genes or 35 for multi-drug and unclear genes. The bars below summarize each quarter of the plot; far left bar represents 135 genes with only one unique mutation. (B) The same plot as A, with points colored by average impact of mutations observed in that gene. High: frameshift and nonsense; Moderate: in-frame insertion/deletion and non-synonymous; Low: synonymous. Black indicates a non-coding RNA. Stars indicate there was at least one occurrence of an insertion element mobilization event within the gene. Below is the impact-type distribution of all mutations occurring in each resistance group. (C) The mean number of mutations associated with each resistance group in any one sample. Error bars are standard error of the mean. (D) Proportion of mutations in each resistance group associated with different functional categories.

MDR mutations occurred in all the functional categories we identified, and almost every toxin-antitoxin mutation fell into this group (Figure 4.6D). In contrast, ASR mutations were more likely to target specific functions, with translation favoured by ciprofloxacin (CIP), respiration by cycloserine (CYC), and membrane transport by nitrofurantoin (NIT). As such, antibiotic mechanism of action may explain the observed biases in targeted biological systems, guiding the evolutionary pathways for ASR. Among the top ASR genes were well-documented targets of resistance mutations: *gyrA* and *icd* for CIP (Helling and Kukora, 1971; Jung-Yien *et al.*, 2016), *ispA/B* and *ubiA/E/G/H* for CYC (Baisa, Stabo and Welch, 2013; Melissa *et al.*, 2022), and *ompC* for NIT (Lou *et al.*, 2011). The latter, a membrane porin, is typically considered a target for MDR, rather than being NIT-specific. However, despite mutations in every NIT sample (14/16 of which were nonsense or frameshift mutations), we identified none in CIP or CYC. Interestingly, with CIP's other main resistance target (*gyrB*) we saw a cluster of CIP-specific mutations in the described quinolone-resistance region, but also several mutations outside this region in CYC and NIT (Figure 4.7), pointing to a potential role for *gyrB* in MDR.

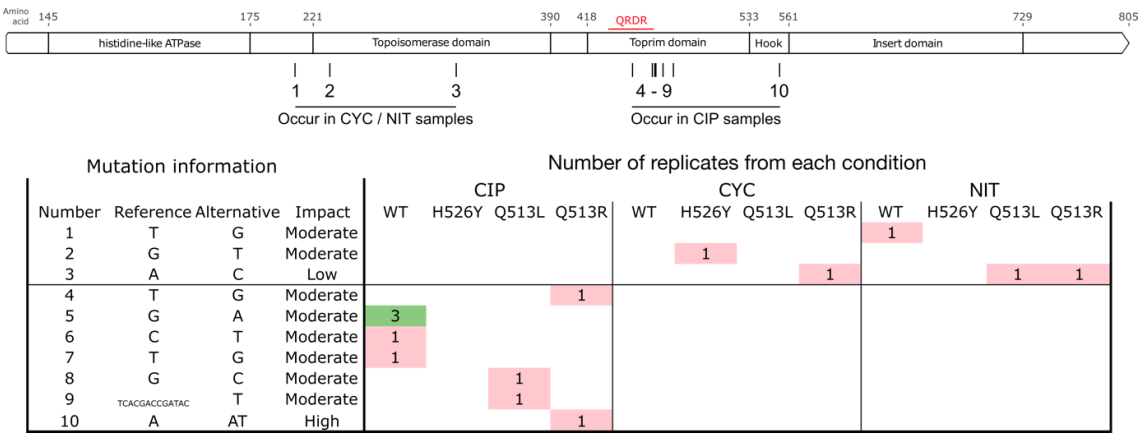


Figure 4.7 – Mutations in *gyrB*. The GyrB amino acid sequence showing locations of known functional sites, and the positions of mutations observed by QMS-Seq. Most are mutations specific to ciprofloxacin and cluster in or near the quinolone-resistance determining region (red). However, three other *gyrB* mutations occur in cycloserine and nitrofurantoin samples, in a different region of the gene.

4.4.3 Genotype influences resistance evolution

We utilized four almost identical *E. coli* strains: the WT ancestor and three descendants, each with a single substitution in the *rpoB* gene encoding the β subunit of RNA polymerase (Figure 4.1C). The minor structural change provides resistance to rifampicin (which targets RNA polymerase) but also affects the transcription of many unrelated genes (Soley *et al.*, 2023). The different expression phenotypes associated with these mutations led us to hypothesize they may shape mutational pathways to secondary antibiotic resistance. Such a phenomenon would improve our ability to predict secondary resistance evolution if the primary resistance mutation is known.

We quantified the differences in mutational landscapes associated with each strain, which showed a clear interaction between CIP and the Q513R strain – the number of unique genes targeted in this condition was significantly higher than any other ($F_{3,8} = 6.2$; $P < 0.05$) (Figure 4.8A). Over 70% of the mutations unique to CIP/Q513R were insertions or deletions, compared to ~25% globally, suggesting a faulty DNA repair system played a role in the effect. In support of this, we identified several mutations exclusive to CIP/Q513R samples in critical genes for two DNA repair pathways, *recO* and *uvrB* (Seeley and Grossman, 1990; Henry and Henrikus, 2021). Principal component analysis of the twelve conditions indicates the influence this effect had on the mutational landscapes: while other conditions cluster by antibiotic, CIP/Q513R is responsible for most of the variance (Figure 4.8B).

At the individual gene level, strain-specific effects were more common in genes that were also specific to an antibiotic, although the mutational distributions of several MDR genes were shaped by an interaction *between* the antibiotics and strains (Figure 4.8C). This was seen clearly in *rsxC*, which inhibits the SoxR MDR system partially responsible for clearing up reactive species (Koo, 2003). Mutations were found across all antibiotics but were more common in NIT, which actively generates reactive species (McOsker and Fitzpatrick, 1994). Curiously, WT was the most targeted strain in NIT but the least targeted across CIP and CYC, while the inverse was true for Q513L. Our data demonstrates how a single nucleotide polymorphism between strains can influence evolutionary trajectories. While the exact relationship between the transcriptomic effects of rifampicin resistance mutations and secondary resistance remains of interest, QMS-

Seq provides a way to evaluate the complicated interactions between antibiotics and genetic background.

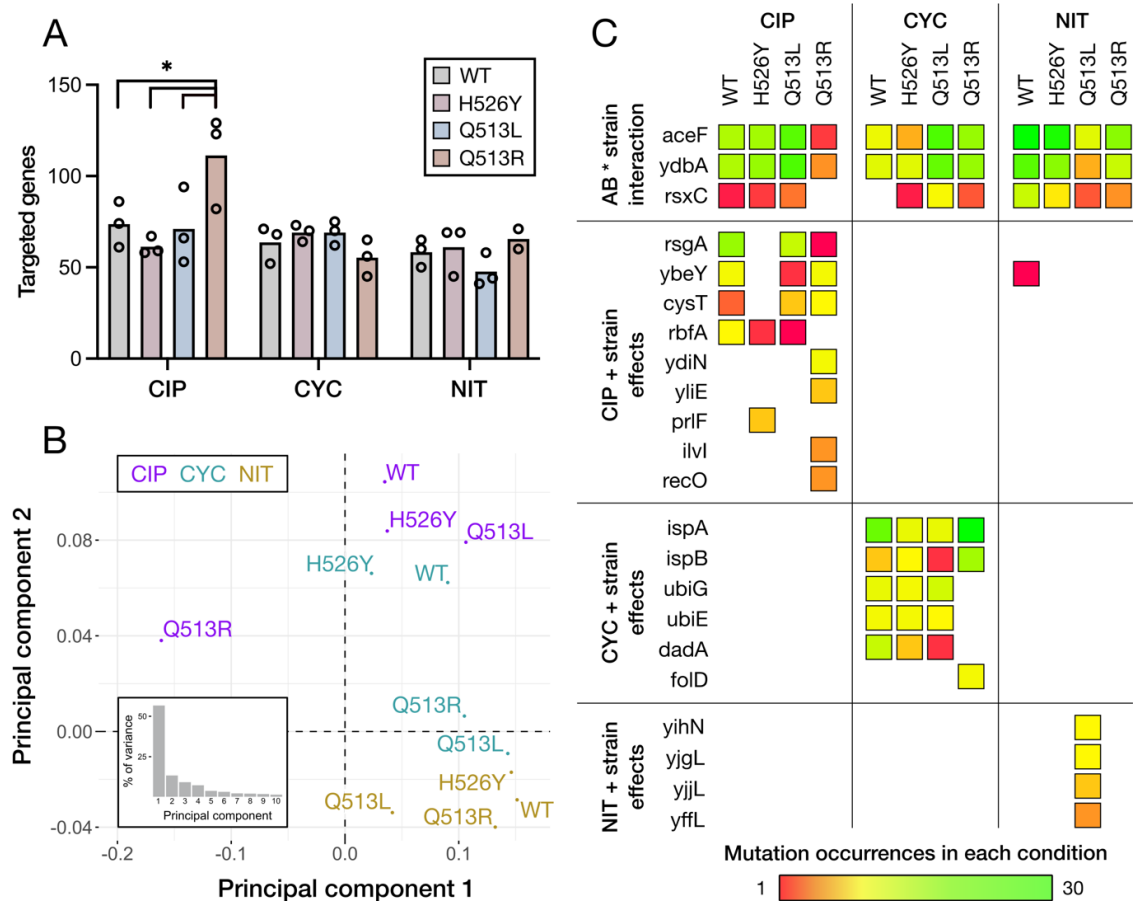


Figure 4.8 – Influence of genetic background on resistance evolution. (A) The mean number of targeted genes (including regulatory targets) in samples from each of the 12 different conditions, with each individual replicate shown. (B) Principal component analysis of the 12 antibiotic/strain combinations, using the number of samples from each condition every mutation was observed in. Inset panel shows the scree plot indicating the percentage of variance described by each principal component. (C) The likelihood of a gene being targeted by mutations can depend on the strain's genetic background. This was observed for MDR genes (top row) as well as ASR genes (bottom three rows). Shows only the top mutational targets with significant strain effects for each category.

4.4.4 Case-studies demonstrate the depth of mechanistic insights afforded by QMS-Seq

4.4.4.1 Top genic target shows synonymous mutations can be under strong site-specific selection

The essential respiratory gene *aceF* is involved in catalysing pyruvate's conversion to acetyl-CoA after glycolysis (John E. Cronan, 2014). Previously found downregulated in persister populations (Sulaiman and Lam, 2020), it was the second highest target for genic mutations in our dataset. All the mutations were found in three homologous regions encoding lipoylation sites on the AceF protein (Paysan-Lafosse *et al.*, 2023) (Figure 4.9A). However, they were not distributed evenly, with only 2 mutation occurrences in site one compared to 64 in site two and 142 in site three – despite the repeat regions previously being considered redundant (Guest *et al.*, 1985). The functional site sequences are 71% identical, yet surprisingly *all* mutations occurred at positions that lack consensus between the three sites, invariably serving to bring one site into better alignment with another (Figure 4.9A). Two-thirds of the mutations were synonymous, suggesting they affect translational efficiency, with the local translation rate potentially altering how the functional sites fold (Rodnina, 2016). Irrespective of mechanism, the mutations in *aceF* exemplify the tight control of essential gene evolution, where marginal changes can give rise to important phenotypes such as antibiotic resistance. This analysis exhibits the resolution QMS-Seq can provide, offering mechanistic insights through the quantification of mutational landscapes, even at the level of individual genes.

4.4.4.2 Duplicates of a toxin-antitoxin system indicate the importance of regulatory mutations

The *sib-ibs* genes form a toxin-antitoxin system with an unclear physiological role (Fozo, 2012). The toxin and antitoxin have fully overlapping sequences but are transcribed from opposite strands, such that the longer antitoxin (a non-coding RNA) suppresses the complementary toxin mRNA. Five copies, in three locations, are present in the WT genome: *sib-ibs A/B* and *D/E* are found in tandem, while *sib-ibs C* is solitary. Fascinatingly, although the *A/B* and *D/E* regions were collectively the top mutational target, no mutations were observed in *sib-ibs C* despite significant homology with its

cousins (Figure 4.9B). Most of the mutations affecting *sib-ibs* copies occurred in annotated regulatory features, although the specific targets varied between copies. These mutations likely increase expression levels of the toxin, either directly or through inhibition of the antitoxin, aligning with previous reports that selective degradation of antitoxins facilitates growth arrest in persister populations (Page and Peti, 2016). Ultimately, the *sib-ibs* illustrate how different gene copies can have distinct evolutionary trajectories under antibiotic selection, either due to differences in sequence/function or their genomic location.

4.4.4.3 Should-be-lethal tRNA mutations provide ciprofloxacin resistance

We observed CIP-specific mutations in the *cysT* tRNA that altered its 3' CCA tail, a conserved feature among tRNAs that is required for aminoacylation (Xiong and Steitz, 2006) (Figure 4.9C). Although unprecedented, this is not entirely without basis; co-treatment of CIP with cysteine (the amino acid for *cysT*) was shown to significantly increase killing of persister cells (Liu *et al.*, 2020). The synergistic relationship between CIP and cysteine appears to be recapitulated in our experiments, except through heritable mutations. However, *cysT* is the only cysteine tRNA in the genome, posing a puzzle as to how disruptions of its 3' CCA are not lethal. *E. coli* carries a 3' CCA repair mechanism (Lizano *et al.*, 2008), which is likely able to rescue enough cysteine tRNA molecules to prevent death, presumably resulting in a variant with reduced translation of nearly every protein. Why this resistance strategy is unique to CIP, and no other tRNAs acquired genic mutations despite the abundance of translation-related targets, remains unclear. Furthermore, *cysT* is a prominent example of evolutionary differences between strains: all CIP samples contained at least one *cysT* mutation, except the three H526Y samples, which had none (Figure 4.8C). Determining how *cysT* plays into the synergistic effect of CIP and cysteine will require further experimentation, but our findings highlight the new avenues for research opened by QMS-Seq's ability to identify rare and low-fitness resistance mutations.

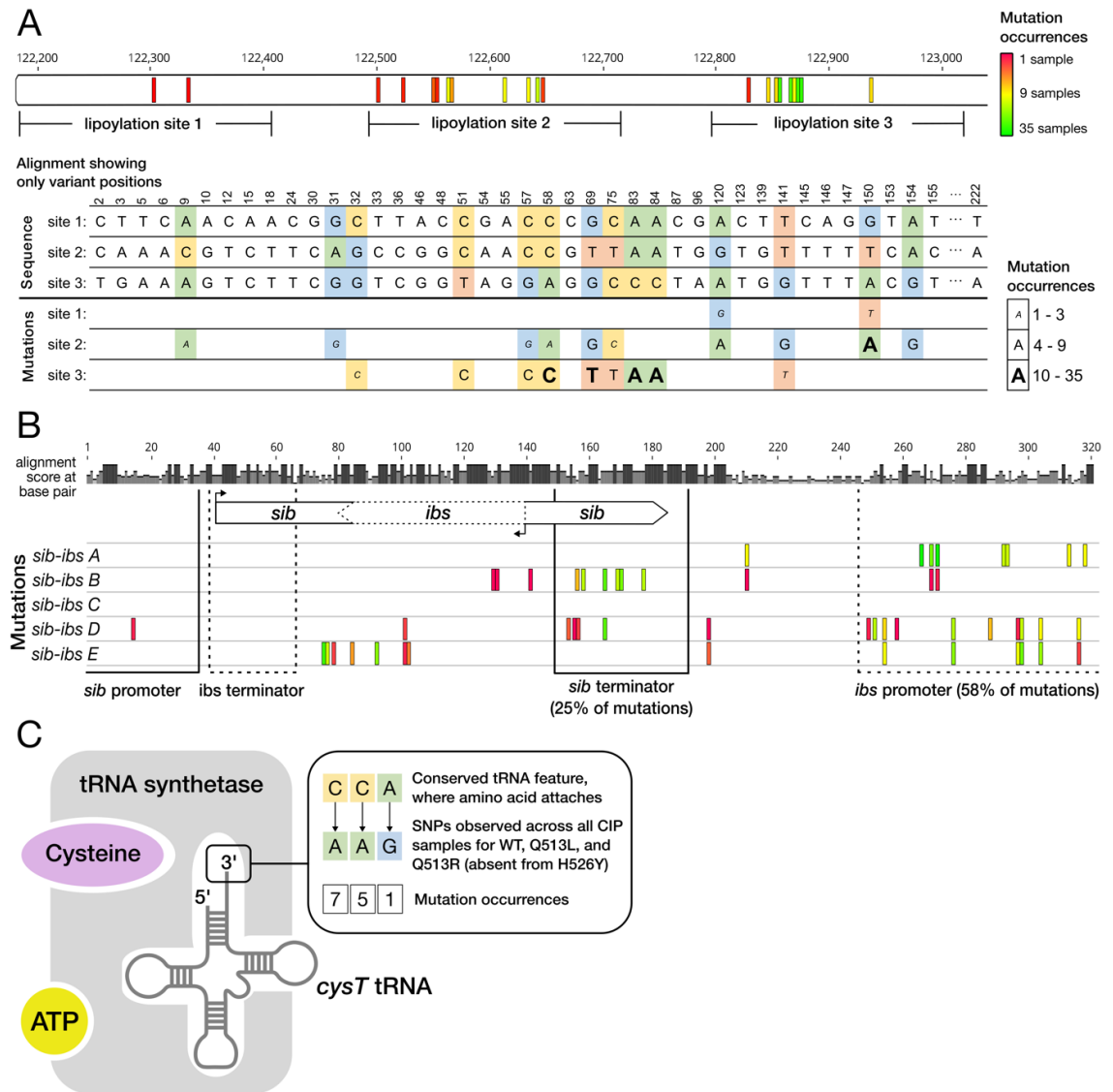


Figure 4.9 – Case studies of notable resistance mutation targets. (A) The first 800 bp of the *aceF* gene (1,893 bp), a region encoding three homologous lipoylation sites. The 21 unique mutations are each colored by how many samples they occurred in. Below is a sequence alignment of the three functional sites, showing only the positions that are *not* conserved between all three sites (variant). Positions where a mutation occurred are highlighted, and the substituted base is shown below, indicating which site the mutation occurred in. 23 variant positions are not displayed because no mutations occurred nearby. (B) Sequence alignment of the five *sib-ibs* copies and their regulatory regions, with mutations colored by their occurrence as in A. The alignment score indicates how many of the copies share consensus at each position. (C) tRNA aminoacylation (where *cysT* is charged with the cysteine amino acid by tRNA synthetase) and the mutations we observed in *cysT*, which in principle should prevent its interaction with cysteine.

4.5 Discussion

We used QMS-Seq to characterize the mutational landscapes for resistance to three different antibiotics, identifying nearly 1,000 mutations across over 300 coding and regulatory sequences. QMS-Seq provides a quantitative metric that captures how mutational effects depend on experimental conditions, enabling high-throughput assessment of the genetic and environmental factors that shape resistance evolution. Through this, QMS-Seq contributes novel insights into microbe biology while also addressing key deficiencies in the effort to monitor and track resistance evolution from genomic data. Its simplicity and broad applicability could help transition from a largely reactive response to the AMR crisis, towards pre-emptive strategies informed by studies examining many strains' evolvability under diverse conditions.

QMS-Seq enabled the discovery of many genes and intergenic regions with previously unknown roles in resistance. This was facilitated by two design choices: (i) minimizing the capacity for high fitness variants to sweep through the population early on by combining replicate parallel cultures, therefore increasing diversity to capture a broader mutational landscape; and (ii) focusing on mutation *occurrence* by considering how often a mutation appears across independent samples instead of how prevalent a mutation is within a population. Because of this latter decision, it is difficult to discern how rare each of the novel mutations identified through QMS-Seq may be. Indeed, WGS performed on individual colonies—to confirm most clones harboured only a single mutation—revealed that the majority of mutations were known SNPs within canonical ciprofloxacin resistance genes *gyrA* and *gyrB* (Table 4.1) (Bhatnagar and Wong, 2019; Perault *et al.*, 2022), and not any of the novel mutations identified with QMS-Seq. However, with a limited sample of 18 individually sequenced clones, further sampling of clones would be required to understand more completely the likelihood of obtaining the mutations identified in this study.

Further contributing to the large diversity of identified mutations is that strains were grown at MIC, where relatively small phenotypic changes can provide resistance. This is in contrast to the majority of clinical isolate and experimental evolution studies that identify resistance mutations with populations exposed to much higher antibiotic concentrations. Using the MIC presumably contributed to the surprising abundance of regulatory and synonymous mutations, sites often thought to be under weak selection

(Thorpe *et al.*, 2017). Such mutations may be enriched during selection at MIC because they are less likely to have a high fitness cost or be lethal (Peris *et al.*, 2010), and the effect they have on gene expression is sufficient to mediate low-level resistance. In known antibiotic persistence genes, which are canonically dysregulated to provide tolerance, regulatory or synonymous mutations may have conferred a permanent persister-like phenotype. The frequency of these mutations corroborates the notion that regulatory and synonymous mutations can significantly affect phenotype, particularly in environments where growth is restricted (Kristofich *et al.*, 2018; Ghosh, N. and Saha, 2020).

The specific experimental design used with QMS-Seq can be tailored to answer a variety of questions about evolution. For example, using only one antibiotic and increasing the number of different strains could classify intraspecies variation in resistance evolution. This would help predict which lineages are most likely to acquire clinical resistance and how, a core goal of AMR surveillance. Similarly, modification of the experimental design could allow researchers to identify the spectrum of compensatory mutations for a range of resistance mutations. Variations to the method could also attempt to mitigate certain limitations of metagenomic sequencing. At present, QMS-Seq is unable to determine if secondary mutations have arisen after a resistance mutation, in the same lineage. Our filtering criteria (see *Chapter 4.3.4 – Filtering mutations under clear selective pressure*) verifies for strong selective pressure and should exclude hitchhiker mutations – but would not exclude subsequent mutations if they were compensatory and under strong positive selection in the given resistance background. This can be ameliorated by decreasing the growth period on selective agar (Figure 4.1A), but this also reduces the capacity to detect rare and costly mutations.

Why is it important to characterise rare and costly mutations? Knowledge of novel resistance mechanisms associated with high fitness costs could help uncover underlying physiological changes that cause reduced fitness, in turn improving our ability to predict the fitness cost of novel mutations (Andersson and Hughes, 2010). Such insights could reveal physiological weaknesses to be exploited. For example, choosing antibiotics for which the resistance mechanism is costly can be a strategy to reduce the evolution of resistance. Thus, identifying costly mechanisms of resistance could guide novel drug discovery. Further, while it is true that costly mutations may be outcompeted by resistance mutations with lower fitness costs, their fitness effects could also be ameliorated through

compensatory evolution (Durão, Balbontín and Gordo, 2018). As such, these rarer resistance mutations are not inherently unimportant by nature of their high cost.

As with any technique reliant on sequencing technologies, there is a likelihood that some mutations identified are in fact a result of sequencing error, which for the NovaSeq has been estimated at 0.109% (Stoler and Nekrutenko, 2021). While attempts were made to stringently filter data and mutations, future work should include specific experiments to validate the identity of resistance mutations beyond their identification with QMS-Seq. While mutation accumulation in our experiments occurred in the absence of antibiotic selection, all samples were then plated on agar containing antibiotic – i.e. QMS-Seq was not performed on any samples grown on antibiotic-free agar plates. Such a control would identify mutations acquired in response to adaptation to growth in our experimental setup, and these could be filtered out of the suspected antibiotic resistant mutation dataset. Future work should also carry out validation of mutations with the susceptibility testing of constructed mutants. This could involve the use of gene knockout strains to test the resistance of nonsense mutations, or the introduction of specific mutations through directed mutagenesis, particularly for synonymous and non-synonymous mutations.

QMS-Seq is experimentally simple, relatively inexpensive, and capable of rapidly identifying resistance mutations across an entire genome while distinguishing between those providing multi-drug vs. antibiotic-specific resistance. It affords sufficient resolution to build a map of which sequence features in those genes are targeted and how, helping to uncover the underlying mechanisms of resistance. Importantly, QMS-Seq evaluates the above in a strain-specific context, enabling researchers to disentangle how resistance evolution depends on the genotype of strains under investigation. In doing so, QMS-Seq can be employed to investigate questions critical to the assaying, monitoring, and stewardship of antimicrobial resistance, as well as provide a deeper view into the nature of microbial evolution.

5 Conclusions

This thesis set out to explore the effects of mutations in *rpoB* on the phenotypes and fitness of *E. coli*, which can largely be summarised with a single word – complex. That such mutations have many interesting and diverse effects on bacterial behaviour has been a subject of research for many decades (Jin and Gross, 1989), and characterising these effects remains important today for the continued clinical use of rifampicin, industrial and research biotechnology applications, and for understanding the evolution of essential molecules. However, the complex effects of *rpoB* mutations remain difficult to predict and the mechanisms underlying these effects are generally poorly understood. It is no wonder given the many factors that influence them. The work in this thesis contributes to the growing knowledge on how *rpoB* mutations influence phenotype and fitness, attempts to build links between molecular differences and the observed transcriptional alterations of each *rpoB* mutant, highlights complex interactions with environmental variables, and reveals how these single point mutations in *rpoB* can influence evolutionary dynamics of secondary resistance. Together, these data are further evidence of how mutations in an essential gene can have unpredictable and wide-reaching implications on the biology of bacteria on a cellular level as well as a population level – important considerations when designing evolutionary-based strategies to mitigate the evolution of antibiotic resistance.

In chapter 2, we see that the fitness effects of several rifampicin resistance mutations are shaped by genotype-by-environment interactions between the specific *rpoB* mutations and the nutrient environment that strains are grown in. This poses a problem for the characterisation of antibiotic resistance mutations if their fitness effects are drastically different depending on the composition of the nutrient environment. Among the effects that were influenced by these GxE interactions was the level of sensitivity to secondary antibiotics, and there were several cases where a mutant went from being sensitive to a second drug, to having low level resistance to it when the nutrient environment was changed. Efforts to exploit these collateral sensitivity relationships to mitigate antimicrobial resistance rely on accurate predictions of the collateral effects of mutations (Andersson *et al.*, 2020). The data presented in this thesis supports recent literature highlighting the complexity of collateral relationships exhibited by different resistance alleles, as well as their environmental-dependency (Nichol *et al.*, 2019; Allen, Pfrunder-Cardozo and Hall, 2021; Liakopoulos *et al.*, 2022; Waller *et al.*, 2023). For such

mitigation approaches to work, it is evident that collateral sensitivities must be validated in a range of environments – in particular clinically relevant environments.

Chapter 3 demonstrates how another environmental variable—temperature—also influences the phenotype and fitness of *rpoB* mutants in a genotype-specific way, exploring how a change in temperature disrupts a handful of mutation effects at different levels of biological organisation, from structural changes at the molecular level to organismal effects on fitness and cell behaviour. Exploring how environmental factors such as temperature influence the evolutionary dynamics of resistant mutants could reveal insights into the success of certain lineages above others, both within individual hosts and at the host population level. Whilst lineage success is determined by a combination of many complex factors (Andersson *et al.*, 2020; Souque, González Ojeda and Baym, 2024), the data presented herein show that differing temperatures, as could be found in different natural environments or in different host species, alter the fitness effects of resistance mutations and so are likely to influence evolutionary dynamics. Understanding the dynamics of resistance evolution in a range of environments is key to designing evolutionary strategies to avoid or combat resistance evolution (Souque, González Ojeda and Baym, 2024). By building on the approach outlined in chapter 3, we can hope to find more explanatory links from genotype to phenotype and fitness, and better predict how these links are influenced by the environment.

Chapter 4 assesses the evolutionary response of *rpoB* mutants to a third environmental factor – secondary antibiotics. The mutational landscape of secondary resistance was shaped by the specific *rpoB* allele, with a single nucleotide variation between strains resulting in evolutionary biases towards alternate mechanisms of resistance. The propensity of an *rpoB* mutant to evolve resistance was also influenced by an interaction with the antibiotics; in other words, the mutant with the most secondary resistance mutations to antibiotic A did not necessarily evolve resistance so readily to antibiotic B. Techniques like QMS-Seq that take advantage of the increased accessibility and affordability of sequencing technologies will play an important role in expanding our knowledge of resistance mechanisms. For example, the rapid identification of resistance mutation spectra could be deployed in the development of novel antimicrobial compounds to quickly identify resistance mechanisms. QMS-Seq could also be adapted to answer further questions regarding resistance evolution, such as the role of ecological

community diversity in shaping the spectrum of resistance mutations (Andersson *et al.*, 2020).

Evolutionary approaches to tackling the AMR crisis rely on an in depth understanding of the factors influencing resistance evolution. Alternative treatment strategies such as those mentioned in chapter 1 will fail if certain resistance alleles confer a fitness advantage where a fitness cost was expected. For example, the success of strategies that exploit collateral sensitivity relies on the repeatability of collateral sensitivity relationships between strains, environments, and clinical vs. *in vitro* experimental models (Andersson *et al.*, 2020). While prior work has shown that these relationships can be consistent between strains with different genetic backgrounds (Podnecky *et al.*, 2018), data from this thesis shows that the difference of a single point mutation, if in an essential gene such as *rpoB*, could result in reversal of collateral interactions. Further, environmental factors can strongly influence the phenotype and fitness of *rpoB* mutants in ways that are difficult to predict – sometimes to the same extent as they are by the specific resistance allele, as the results of this thesis has shown. The phenotypic effects of antibiotic resistance mutations must therefore be considered in the context of the growth environment. Characterisation of resistance mutations should be carried out in multiple environments, especially to identify collateral effects like those observed in chapter 2 (Allen, Pfrunder-Cardozo and Hall, 2021). While it is unfeasible to experimentally measure the effects of every mutation in every environment, a survey of recent literature suggested only a very small percentage of susceptibility testing considers the environment at all[†] so there is scope for improvement. In addition to antibiotic resistance assays, experimental evolution work more broadly would benefit from a greater awareness of the influence of different environments. Further studies, together with data presented in this thesis, will help to improve our mechanistic understanding of mutations such as those in *rpoB*, potentially identify environment-specific responses or common adaptive mechanisms (Utrilla *et al.*, 2016). This in turn will allow better predictive models to be built to more accurately predict mutation effects in different environments *a priori* (Eguchi, Bilollikar and Geiler-Samerotte, 2019; Yi and Dean, 2019).

Throughout this work, many of the complex effects of rifampicin resistance mutations have been explored in *E. coli*. This organism is routinely used in many facets of biological

[†] See Chapter 2.5 – Discussion

research, and evolutionary biology is no exception (John E Cronan, 2014). There are a number of benefits of using *E. coli* as an experimental model: it has a rapid generation time, a well annotated genome, and a huge amount of pre-existing research. However, rifampicin is predominantly used clinically as a treatment for diseases caused by *Mycobacteria*, such as tuberculosis (Rees, Pearson and Waters, 1970; Grobbelaar *et al.*, 2019), and not to treat *E. coli* infection. There is reason to predict that *rpoB* mutations in *M. tuberculosis* will exhibit similarly complex effects as in *E. coli*, and as such the findings of this thesis can inform our understanding of other species. Many of the mutational sites in *rpoB* that confer rifampicin resistance are the same in *M. tuberculosis* (Ma *et al.*, 2021). Further, mutation-specific differences in fitness cost (Mariam *et al.*, 2004) and transcription (De Knecht *et al.*, 2013; Trauner *et al.*, 2021) have been observed much like those present in *E. coli*. However, fundamental differences exist between species. At the molecular level, there are differences in the structure of RNAP (Zenkin *et al.*, 2005); as we have seen in chapter 3, molecular alterations can determine the effects of *rpoB* mutations. At the organismal level, differences in bacterial characteristics such as the cell envelope (Garcia-Vilanova, Chan and Torrelles, 2019) will determine how *M. tuberculosis* mutants respond to environmental variation. Additionally, the unique growth environments that *M. tuberculosis* experiences in the lung, such as complex heterogeneous environments within granulomas, will impose specific constraints on possible genotype-by-environment interactions and the dynamics of evolution on a population level (Torrelles and Schlesinger, 2017; Allué-Guardia, García and Torrelles, 2021). Insights from the data presented in this thesis, particularly surrounding the influence of GxE interactions on the evolutionary dynamics of *rpoB* mutations, could inform predictions of fitness within these *M. tuberculosis*-dependent microenvironments.

While specific focus of this thesis has been on a handful of rifampicin resistance mutations, these findings are applicable outside of this narrow theme. Mutations in *rpoB* can confer resistance to other antibiotics such as ciprofloxacin, and though present in different parts of the gene than rifampicin resistance mutations previous work has shown they can exhibit similar pleiotropic effects (Pietsch *et al.*, 2017). More broadly still, resistance mutations that interfere with other essential cellular processes like translation can arise in response to antibiotics such as streptomycin and tetracycline (Paulander, Maisnier-Patin and Andersson, 2009; Lupien *et al.*, 2015). Such mutations can also result in complex, unpredictable effects and as such are likely to be similarly influenced by

environmental variation (Bershtein *et al.*, 2015). Pleiotropic mutations with complex effects also exist outside the context of antibiotic resistance (Ostrowski, Rozen and Lenski, 2005). From a purely evolutionary perspective, these mutations may drive adaptation in complex molecules and cellular processes, contributing to the evolution of complexity. At this general level, the findings presented here show how a single point mutation can have wide-reaching implications on many aspects of bacterial behaviour and evolution in a highly environment-dependent manner.

One limitation of the data presented here is the short timescales of the experiments. For example, data in chapter 4 highlights how different *rpoB* mutations can constrain evolutionary pathways to secondary resistance. However, strains were only grown for a short period of time to allow a first sweep of selection for secondary resistance mutations. As such, only the immediate effects of *rpoB* mutations on evolution have been assessed. Similarly, in chapter 2 the influence of genotype-by-environment interactions on the evolutionary outcomes of different mutants was predicted based on fitness measurements, but not by measuring actual evolutionary outcomes with long term experimental evolution experiments. Long term evolution studies permit a more robust assessment of evolutionary dynamics (Lenski, 2023), and could reveal whether the constraints on the spectrum of resistance mutations and on the mechanisms of resistance seen in chapter 4 are maintained over many generations. Similar analysis could be applied to data in chapters 2 and 3 – do the fitness advantages that emerge as a result of certain genotype-by-environment interactions improve the evolutionary outcomes of a strain over multiple generations? Conversely, are mutants able to overcome growth deficits when GxE interactions result in higher fitness costs, for example through the evolution of compensatory mutations? The long term evolutionary fate of a mutant depends on many factors not assessed here, such as a changing environment and competition with other mutant strains. Future work could assess the impact of such factors on the evolutionary success of *rpoB* mutants over many generations.

Another important factor in understanding the complexity of mutations is their interactions with other mutations and with the genetic background. Epistatic interactions between *rpoB* mutations and genetic background can determine the magnitude of fitness effects (Vogwill, Kojadinovic and Maclean, 2016; Trauner *et al.*, 2021). In the work presented in this thesis, all mutants have the same genetic background, differing only by the mutation in *rpoB*. While this was a deliberate decision to identify the specific

consequences of each single mutation, future research could explore higher order interactions between genetic background, mutation, and growth environment that may add further complexity to understanding the fitness effects of *rpoB* mutations (Gifford, Toll-Riera and MacLean, 2016). Epistatic interactions also occur between *rpoB* mutations and other mutations in *rpoB*, mutations in other RNAP genes, and other antibiotic resistance mutations (Hall and MacLean, 2011; Durão *et al.*, 2015; Trauner *et al.*, 2021). These interactions may also be altered by different environments. While the effect of epistatic interactions between *rpoB* mutations on fitness tends to be maintained across different environments (Hall and MacLean, 2011), it remains to be seen whether this is true of other mutation effects, such as altered transcription.

Understanding how changes in genotypes alter the phenotype and thus the fitness of an organism remains a fundamental challenge in biology. In the case of antibiotic resistant bacteria, predicting how a novel resistance mutation affects fitness is critical if we are to predict how it disseminates within a bacterial population. The coming together of evolutionary theories with experimental molecular biology has allowed great strides in the building of genotype-phenotype-fitness maps, where the outcome of a mutation on the structure and function of a protein is directly linked to its effect on fitness (Dean and Thornton, 2007; Yi and Dean, 2019). As demonstrated throughout this thesis, the influence of various environmental factors on mutations and their effects can be unpredictable, especially with mutations in essential molecules that result in pleiotropic effects. This points to the need for further characterisation of the complex effects of mutations, and the mechanisms underlying them, in a variety of different environments.

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